Wireless Sensor Networks

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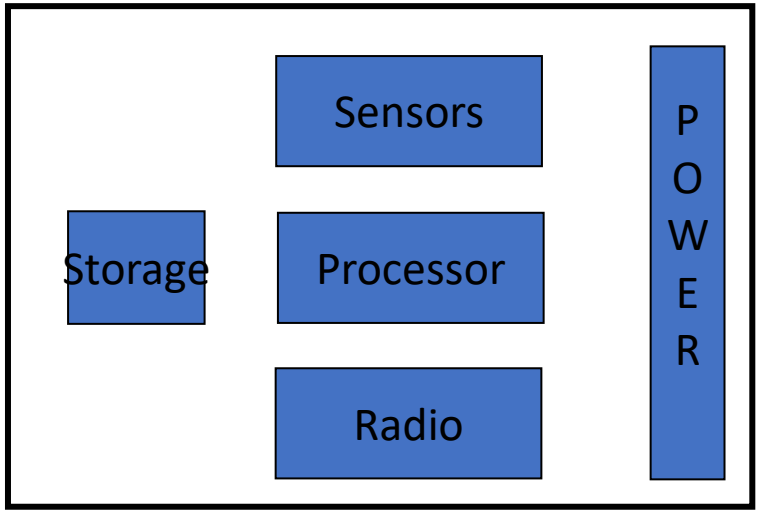
What are wireless sensor networks (WSNs)?

- A wireless sensor network (WSN) is a wireless network using sensors to cooperatively monitor physical or environmental conditions
- Networks of typically small, battery-powered, wireless devices (often MANY, sometimes heterogeneous)
  - On-board processing,
  - Communication, and
  - Sensing capabilities.

Or…

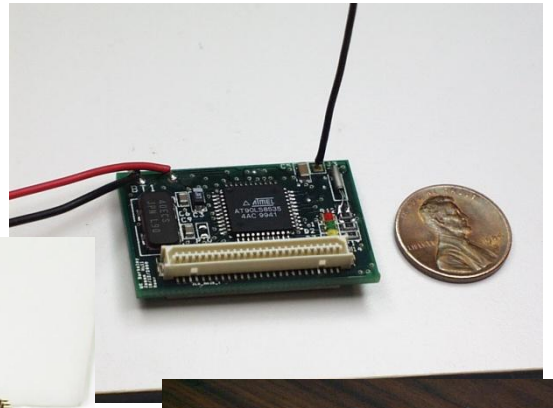
➢ Wireless sensing + Data Networking!

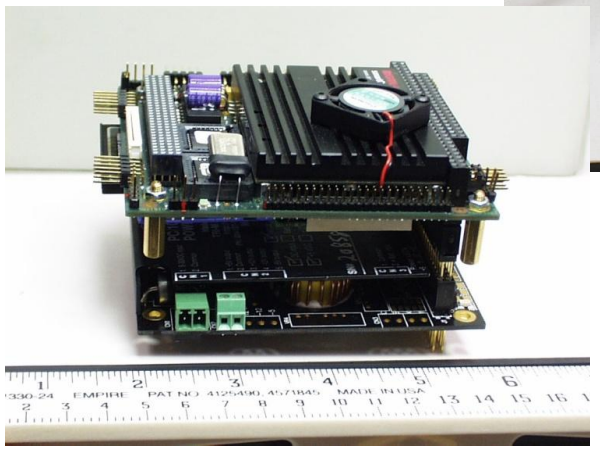
  ➢ Group of sensors linked by wireless media to perform distributed sensing tasks

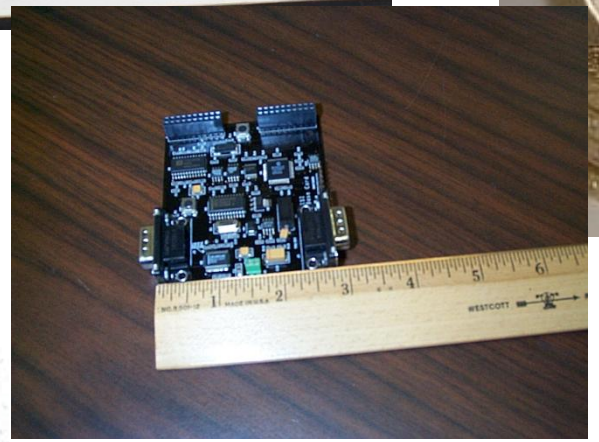


WSN device schematics

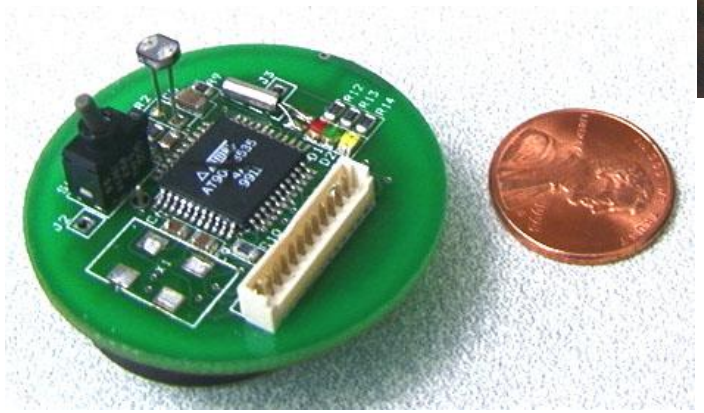
Sensor Nodes and platforms









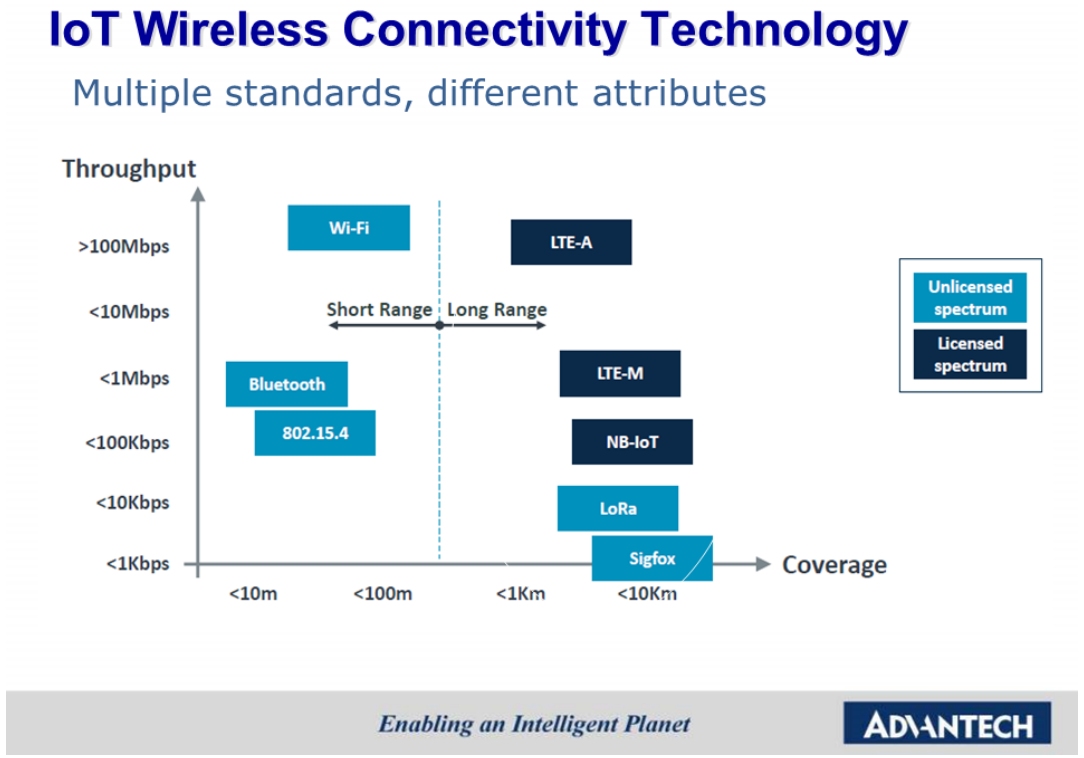






IoT Wireless Connectivity

As with wireless in general, multiple standards with different properties

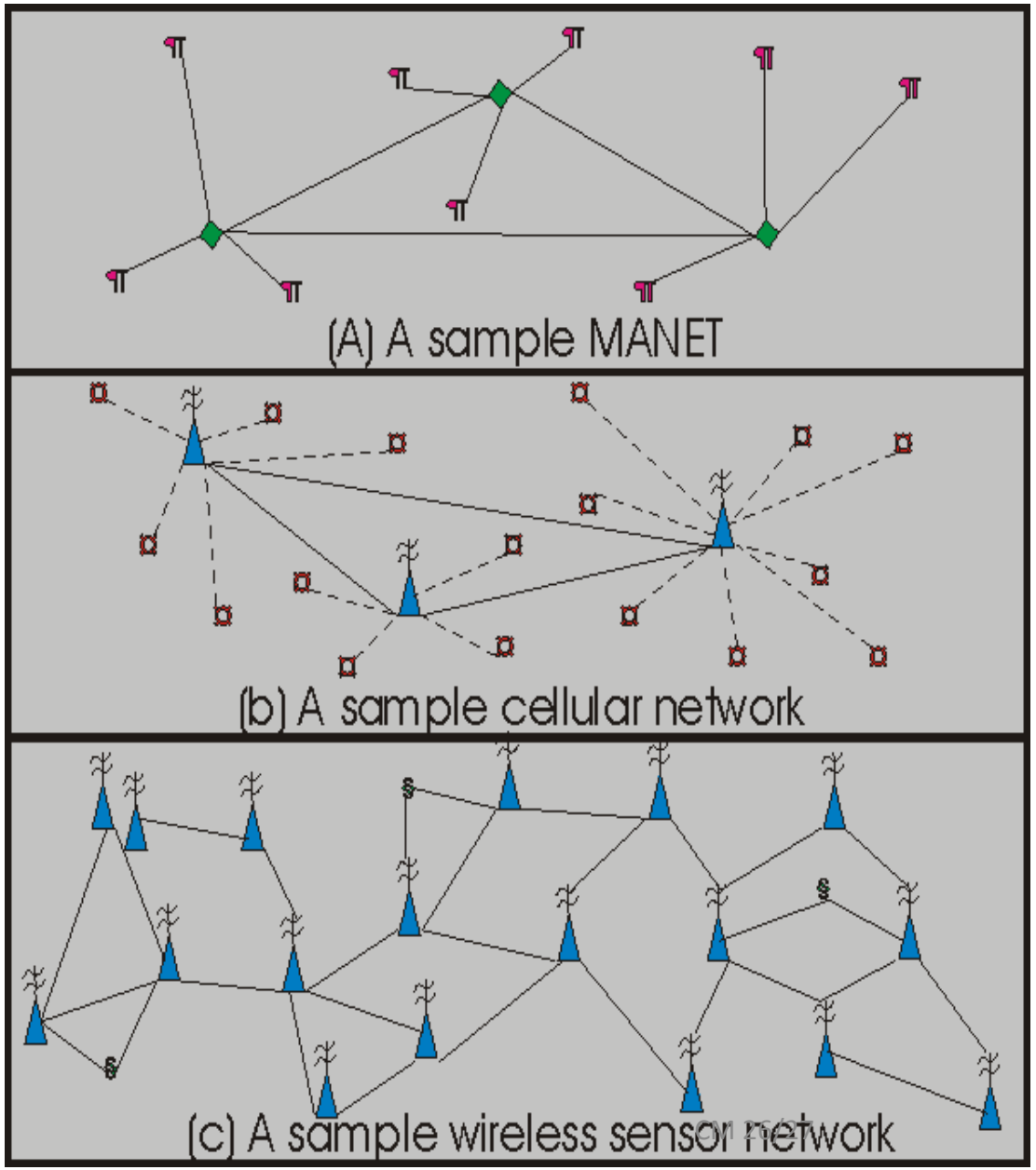


Machine-IoT and Human-IoT are different

- IoT has multiple scenarios, from human-oriented to machine-oriented, and from industrial to forest environments
- WSN need to adapt to these environments.

	Manufacturing IoT	**Consumer IoT**
Goal	Manufacturing-industry Centric	Consumer Centric
Devices	Machines, Sensors, Controllers, Actuators, Smart meters	Consumer devices and Smart appliances
Working Environment	Harsh (vibration, noisy, extremely high/low temperature)	Moderate
Data rate	High (usually)	Low or average
Delay	Delay sensitive	Delay tolerant
Mission	Mission-critical	Non-mission-critical

Types of wireless Networks

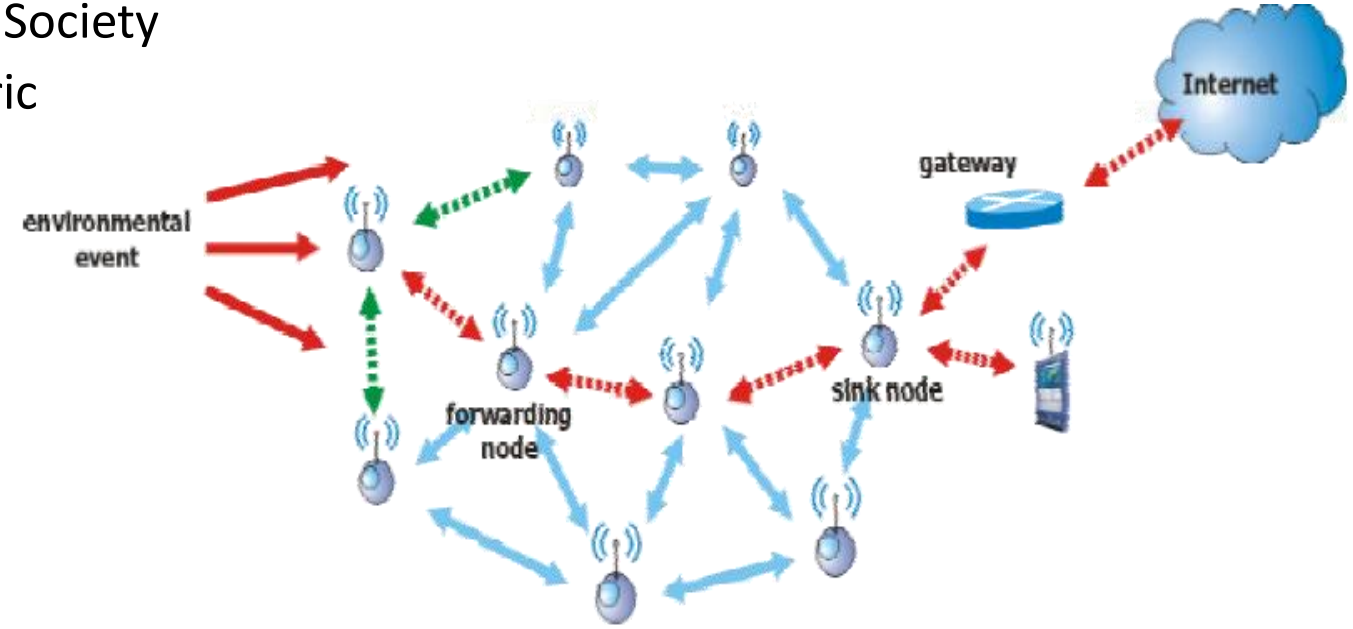


MANET – Mobile Ad-hoc network

WSN can explore the architecture and protocol concepts both of MANETs (mobile ad-hoc networks) and of celular networks.

Wireless Sensor Network

- Focus on:
  - Ubiquitous Computing
  - Ubiquitous Network Society
  - (often) Human-centric
- Ubiquitous
  - Anytime
  - Anyone
  - Anywhere
  - Any Device
  - Affordable
  - All Security
  - Any Information/Service



MAC:
challenges for wireless networking

- MAC is a critical layer for networking
- Traditional problems
  - Fairness
  - Latency
  - Throughput
- For Sensor Networks, more problems are added
  - **Power efficiency**
  - **Scalability**

MAC challenges for WSN

- Sensor networks are deployed in an ad hoc fashion
- Individual nodes remaining largely **inactive for long periods of time…**
- **…**but then becoming **suddenly active** when something is detected.

- These characteristics of sensor networks and applications motivate a MAC that is different from traditional wireless MACs :
  - **Energy conservation** and **self-configuration** are primary goals.
  - Per-node fairness and latency are less important.

Design Challenges

Why are WSNs challenging/unique?

- Typically, severely energy constrained.
  - Limited energy sources (e.g., batteries).
  - Trade-off between performance and lifetime.
- Self-organizing and self-healing.
  - Remote deployments.
- Scalable.
  - Arbitrarily large number of nodes.

Design Challenges

- Heterogeneity.
  - Devices with varied capabilities.
  - Different sensors.
  - Hierarchical deployments.
- Adaptability.
  - Adjust to operating conditions and changes in application requirements.
- Security and privacy.
  - Potentially sensitive information.
  - Hostile environments.

Sensor Network MAC Protocols

- The major sources of energy wastage are:
  - Collisions – *interfering packets*
  - Overhearing – *hearing more than required from a packet*
  - Control packet overhead – *control versus data*
  - Idle listening – *hearing for nothing*

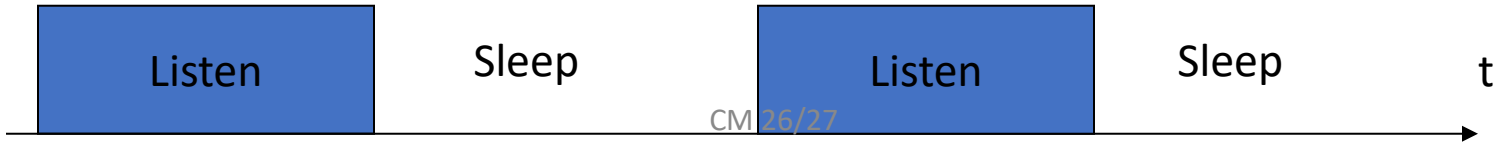
Typical solutions in wireless MACs

- Carrier Sensing
  - Only during low traffic load.
- Contention
  - RTS-CTS only during high traffic load.
- Backoff
  - Backoff in application layer is desired other than in MAC layer.

Achieving good scalability and collision avoidance capability is necessary.

Solution example: Sensor-MAC (S-MAC)

- S-MAC is a medium-access control (MAC) protocol designed for wireless sensor networks.
  - Explores typical solutions also found in many other sensor MACs.
  - **Nodes periodically sleep, and sleep during other nodes' transmissions**
    - Nearby nodes form virtual clusters to synchronize their wake-up and sleep periods
  - Trades **energy efficiency for lower throughput and higher latency**
    - Message passing is used to reduce the contention latency and control overhead



802.15.4 and Zigbee

What is ZigBee?

- Technological Standard Created for Control and Sensor Networks
  - Based on the IEEE 802.15.4 Standard
  - Centered in small radios
- Created by the ZigBee Alliance
  - 200+ members
- History
  - *May 2003: IEEE 802.15.4 completed*
  - December 2004: ZigBee specification ratified
  - June 2005: public availability

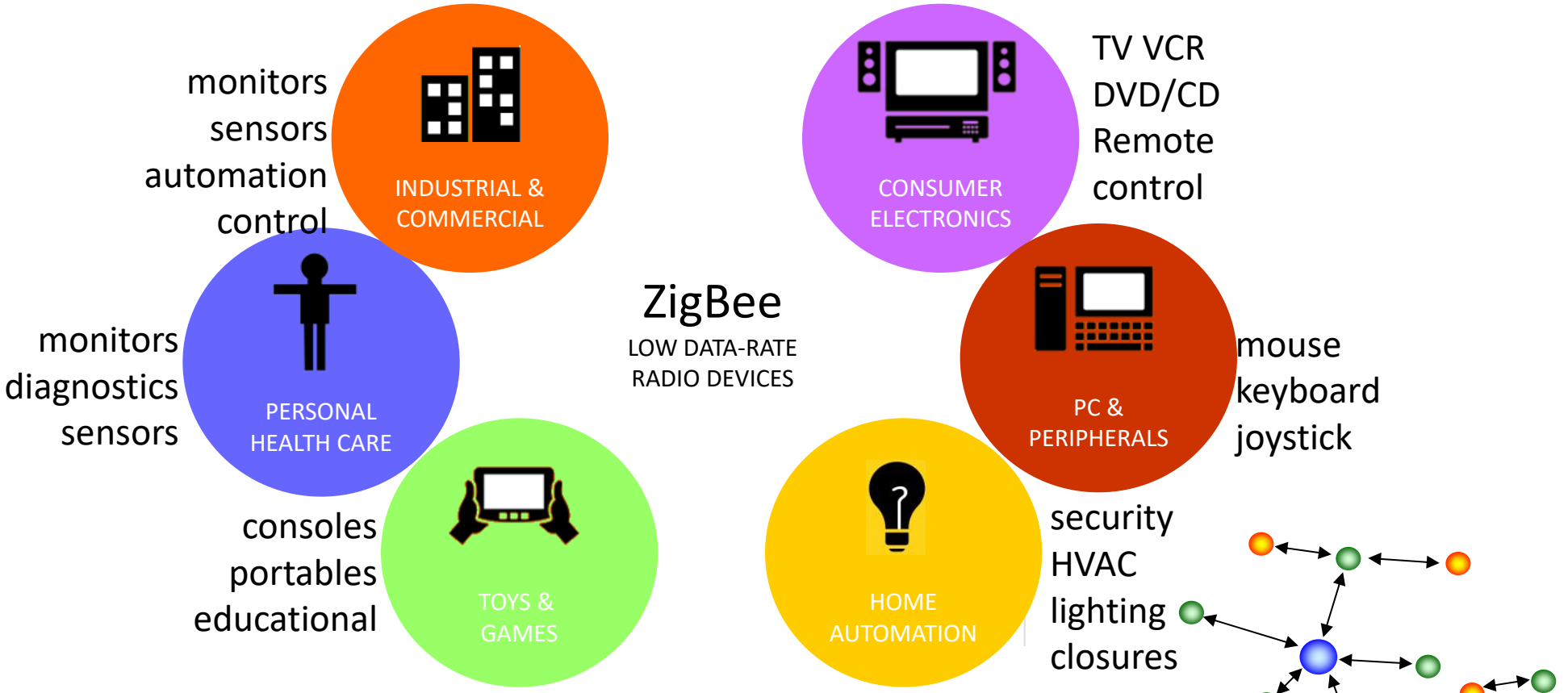
What Does ZigBee Do?

- Designed for wireless controls and sensors
  - Operates in Personal Area Networks (PAN's) and device-to-device networks
  - Connectivity between small packet devices
  - Examples: control of lights, switches, thermostats, appliances, etc.

Zigbee?

- Named for erratic, zig-zagging patterns of bees between flowers
- Symbolizes communication between nodes in a mesh network
- Network components "seen as analogous" to queen bee, drones, worker bees

ZigBee network applications



- Just everything you can imagine for wireless sensor nodes or in general short range communications

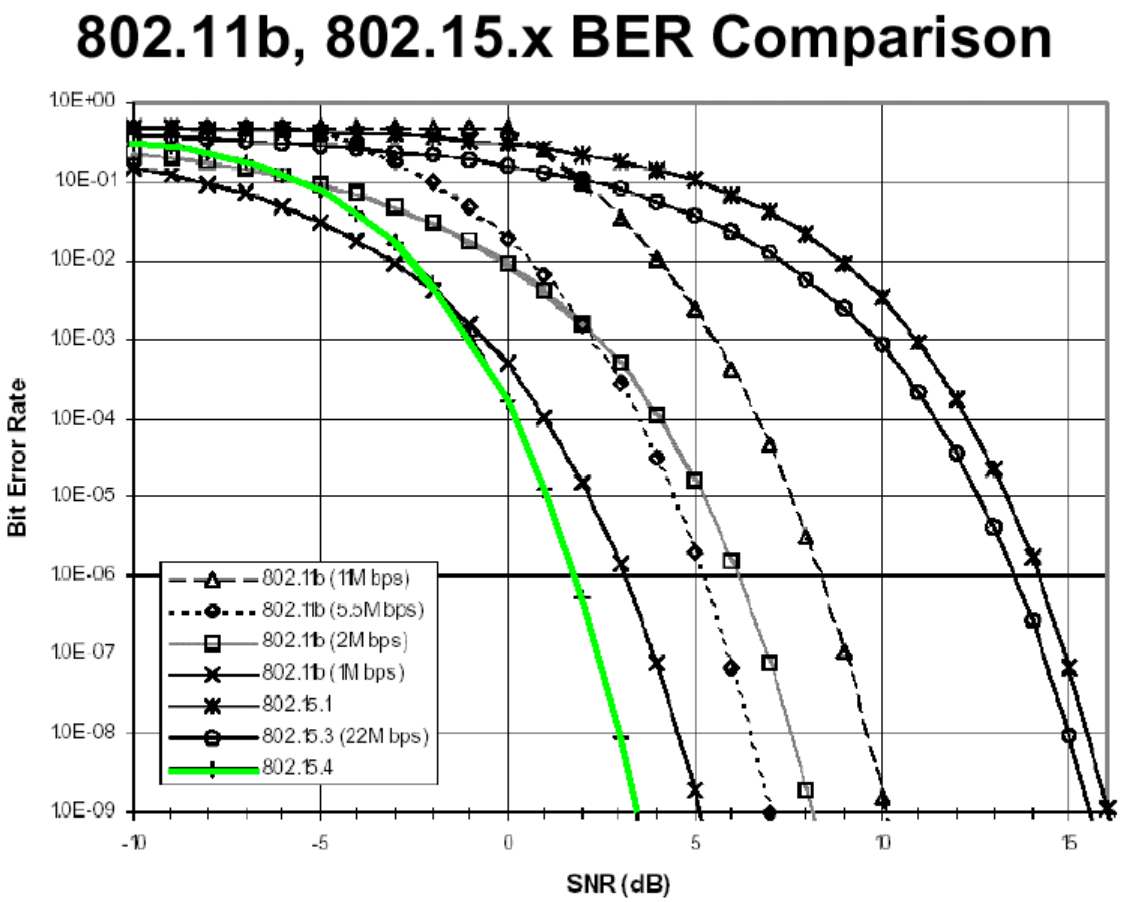
ZigBee and Other Wireless Technologies

Market Name	ZigBee™	---	Wi-Fi™	Bluetooth™
Standard	802.15.4	GSM/GPRS CDMA/1xRTT	802.11b	802.15.1
Application Focus	Monitoring & Control	Wide Area Voice & Data	Web, Email, Video	Cable Replacement
System Resources	4KB - 32KB	16MB+	1MB+	250KB+
Battery Life (days)	100 - 1,000+	1-7	.5 - 5	1 - 7
Network Size	Unlimited (2^{64})	1	32	7
Bandwidth (KB/s)	20 - 250	64 - 128+	11,000+	720
Transmission Range (meters)	1 - 100+	1,000+	1 - 100	1 - 10+
Success Metrics	Reliability, Power, Cost	Reach, Quality	Speed, Flexibility	Cost, Convenience

Source: http://www.zigbee.org/en/about/faq.asp

Why do we need another “WPAN” standard?

- Power consumption
  - ZigBee: 10mA <==> BT: 100mA
- Production costs
  - ZigBee: 1.1 $ <==> BT: 3 $
- Development costs
  - Codesize ZB/codesize BT = ½
- Bit-error-rate (BER)
  - Zigbee is more noise-resilient
- Sensitivity
- flexibility
  - No. of supported nodes
  - ZigBee: 65536 (in a mesh) <==> BT: 7
- Security
- Latency requirements
- Range
  - ZigBee: up to 75 m in LOS condition <==> BT: 10 m

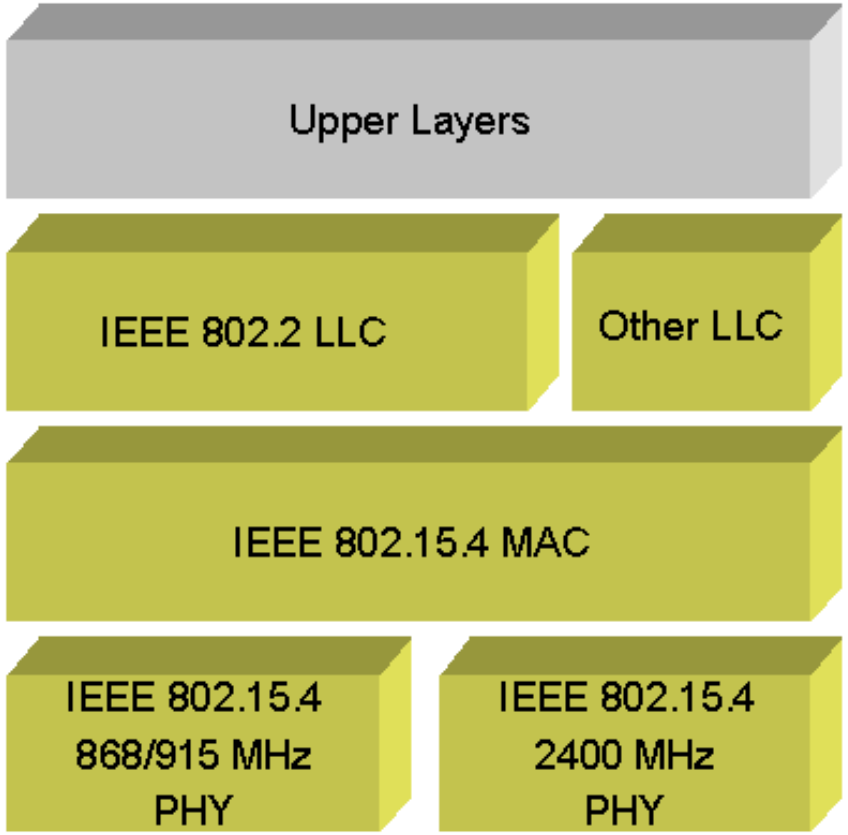


ZigBee/IEEE 802.15.4 features

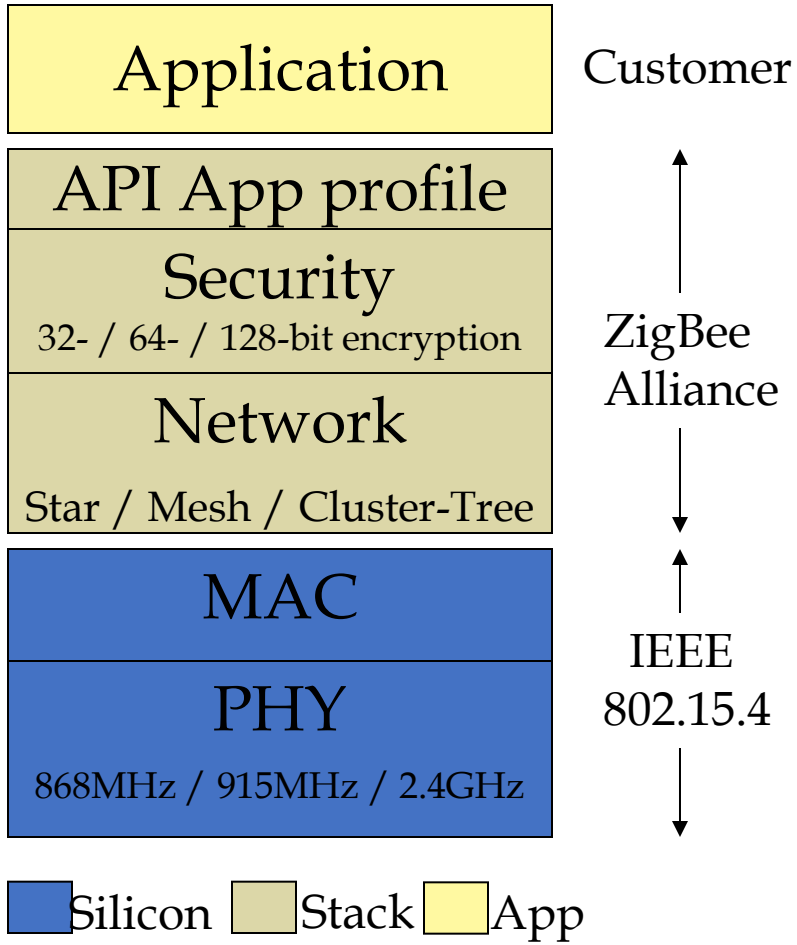
- Low power consumption
- Low cost
- Small packet
- Low offered message throughput
- Supports large network orders (<= 65k nodes)
- Low to no QoS guarantees
- Flexible protocol design suitable for many applications

IEEE 802.15.4 - Overview

- Low Rate WPAN (LR-WPAN)
  - E.g. Sensor networks
- Simple and low cost
  - Fully handshake protocol
- Low power consumption
  - Years on lifetime using standard batteri
- Different topologies
  - Star, peer-to-peer, combined
- Data rates: 20-250 kbps
  - Low latency support
- Operates at different frequencies
  - 868 Mhz, 915 Mhz, 2.4 GHz



IEEE 802.15.4 & ZigBee In Context



ZigBee Alliance

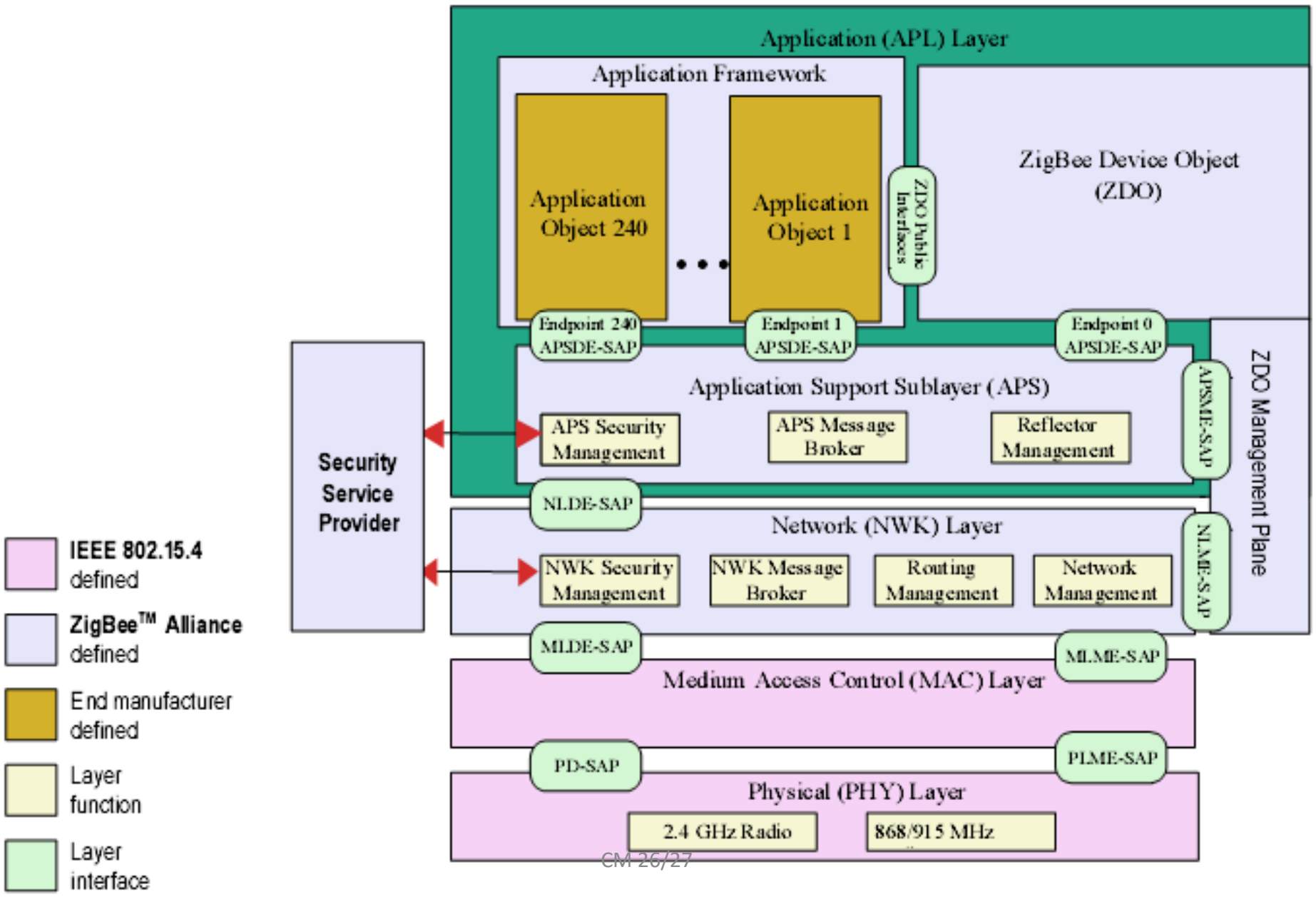
- "the software"
- Network, Security & Application layers
- Brand management

IEEE 802.15.4

- "the hardware"
- Physical & Media Access Control layers

Source: http://www.zigbee.org/resources/documents/IWAS_presentation_Mar04_Designing_with_802154_and_zigbee.ppt

Protocol Stack



How ZigBee Works

- Topology
  - Star
  - Cluster Tree
  - Mesh
- Network coordinator, routers, end devices
- 2 or more devices form a PAN/WSN

How ZigBee Works

- States of operation
  - Active
  - Sleep
- Devices
  - Full Function Devices (FFD's)
  - Reduced Function Devices (RFD's)
- Modes of operation
  - Beacon
  - Non-beacon
- Traffic types
  - Intermittent
  - Repetitive
  - Periodic

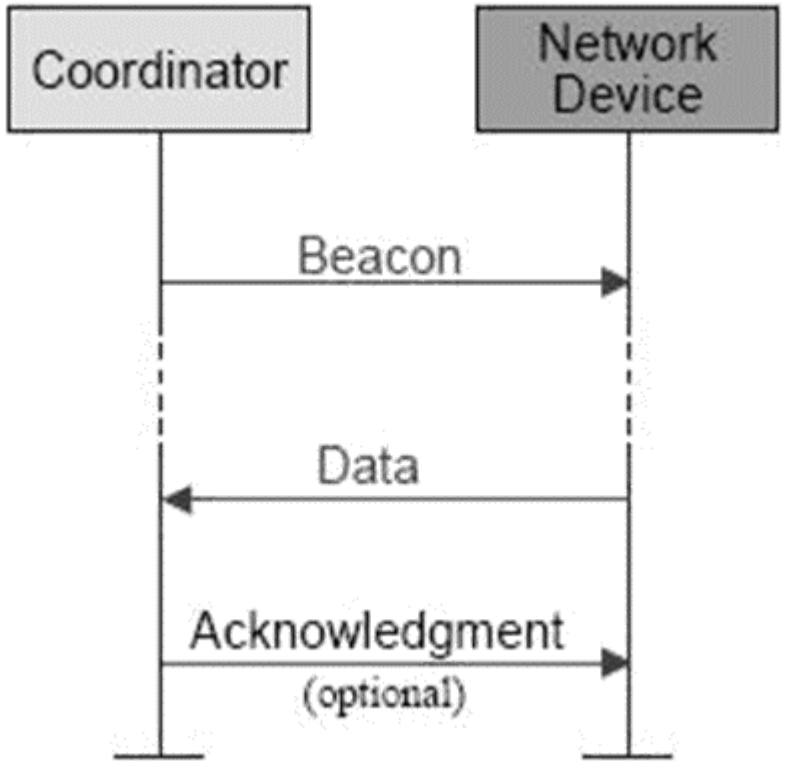
Traffic-Types

- Data is periodic
  - application dictates rate (e.g. sensors)
- Data is intermittent
  - application or stimulus dictates rate (optimum power savings), e.g. light switch
- Data is repetitive (fixed rate a priori)
  - device gets guaranteed time slot (e.g. heart monitor)

Traffic-Modes

Beacon mode:

- beacon sent periodically
- Coordinator and end device can go to power save
- Lowest energy consumption
- Precise timing needed
- Beacon period (ms-m)

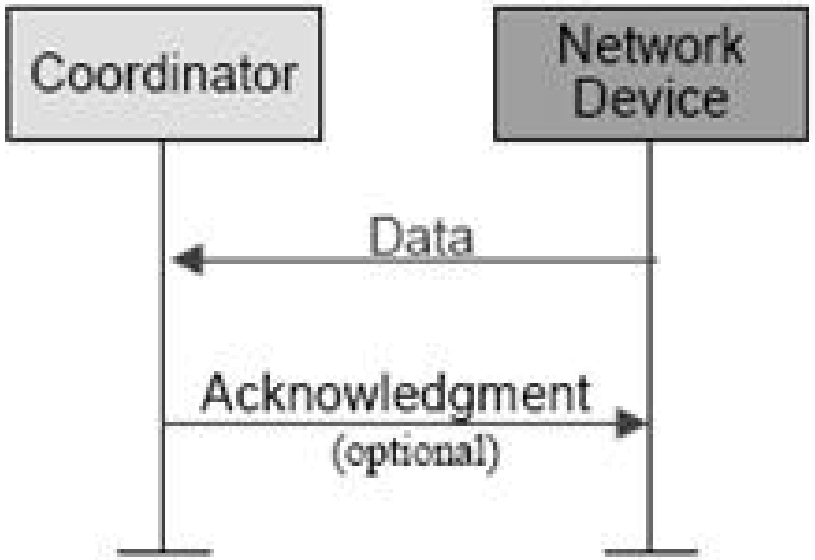


Traffic-Modes

Non-Beacon mode:

- coordinator/routers have to stay awake

  (robust power supply needed)
- heterogeneous network
- asymmetric power



ZigBee Node-Types

ZigBee Coordinator (ZBC) (IEEE 802.15.4 FFD)

- only one in a network
- initiates network
- stores information about the network
- all devices communicate with the ZBC
- routing functionality
- bridge to other networks

ZigBee Router (ZBR) (IEEE 802.15.4 FFD)

- optional component
- routes between nodes, network backbone
- extends network coverage
- manages local address allocation/de-allocation

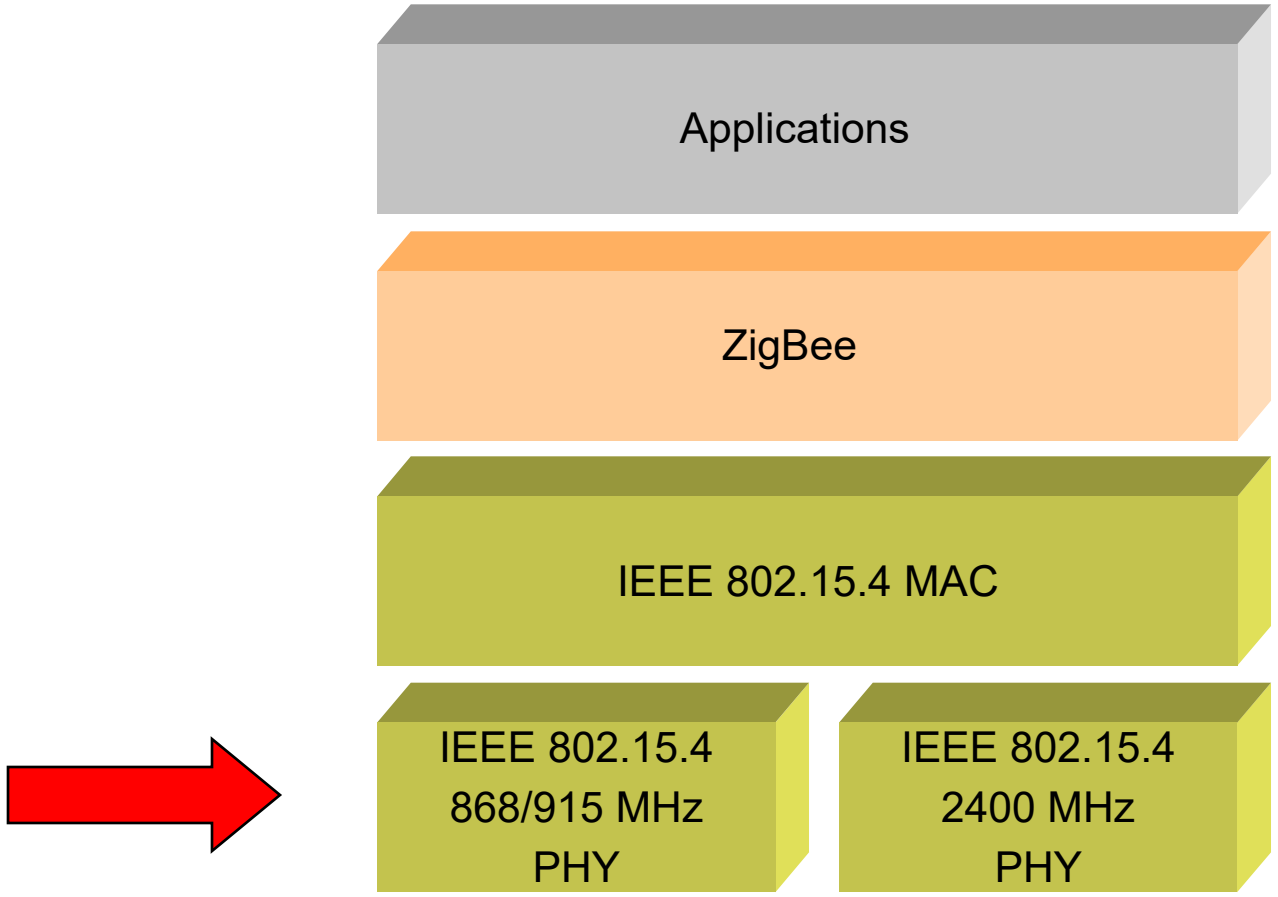
ZigBee End Device (ZBE) (IEEE 802.15.4 RFD)

- optimized for low power consumption
- cheapest device type
  - sensor would be deployed here



Remember: FFD – Full Function Device
RFD – Reduced Function Device

802.15.4 / ZigBee Architecture



- Packet generation
- Packet reception
- Data transparency
- Power Management

IEEE 802.15.4 basics

- 802.15.4 is a simple packet data protocol for lightweight wireless networks
  - Channel Access is via **Carrier Sense Multiple Access with collision avoidance** and optional time slotting
  - Message acknowledgement and an optional beacon structure
  - Multi-level security
  - Works well for
    - Long battery life, selectable latency for controllers, sensors, remote monitoring and portable electronics
  - Configured for maximum battery life, has the potential to last as long as the shelf life of most batteries

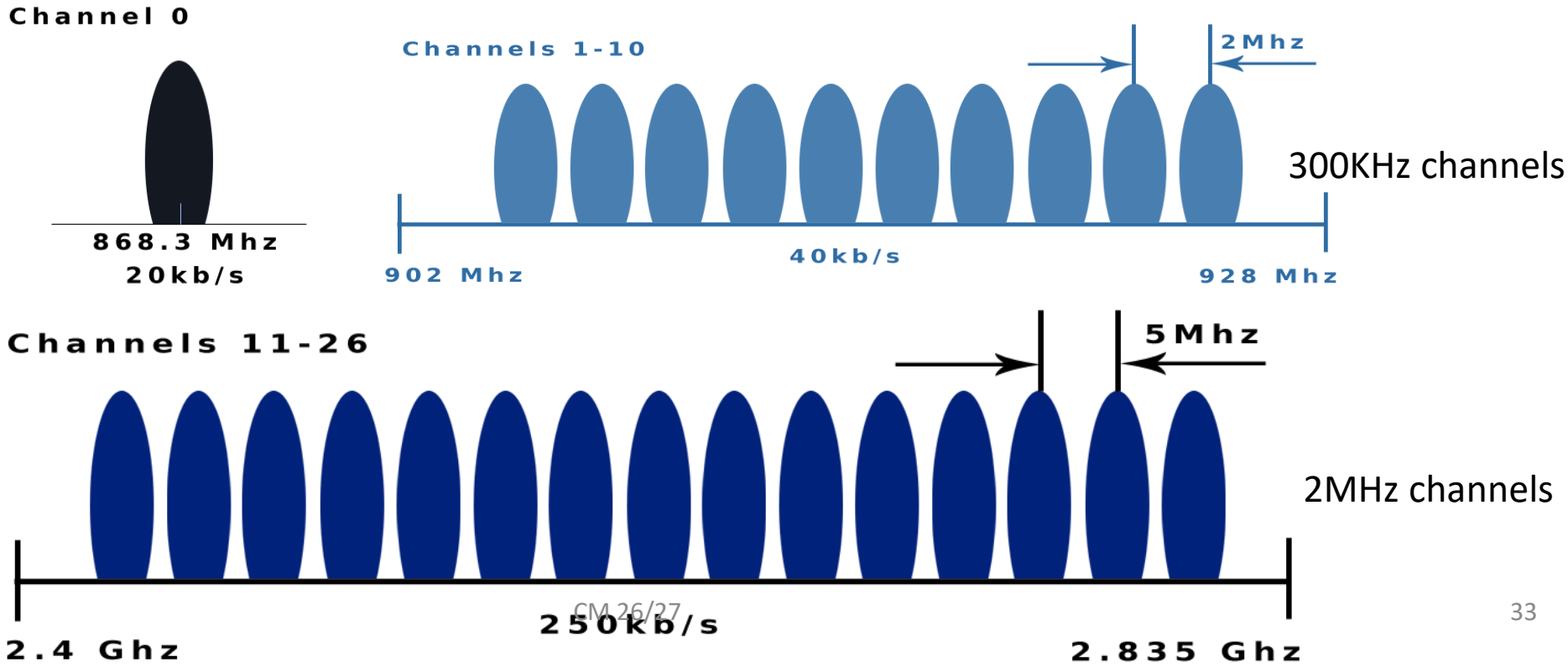
802.15.4 General characteristics

- Data rates of 250 kbps , 20 kbps and 40kpbs.
- Star or Peer-to-Peer operation.
- Support for low latency devices.
- CSMA-CA channel access, with CCA detection
  - Clear Channel Assessment
- Dynamic device addressing.
- Fully handshaked protocol for transfer reliability.
- Low power consumption.
- 16 channels in the 2.4GHz ISM band
- 10 channels in the 915MHz ISM band
- one channel in the European 868MHz band.
- Extremely low duty-cycle (<0.1%)
  - i.e. % of time actively transmitting/receiving in regards to a total operation cycle

802.15.4 frequency bands

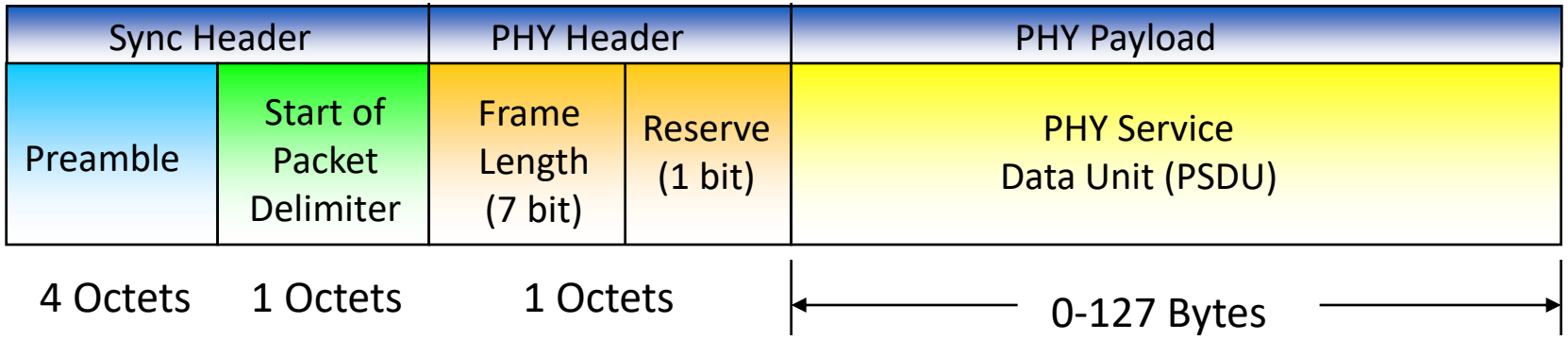
Operates in Unlicensed Bands

- **ISM 2.4 GHz Global Band at 250kbps**
- **868 MHz European Band at 20kbps**
- **915 MHz North American Band at 40kbps**

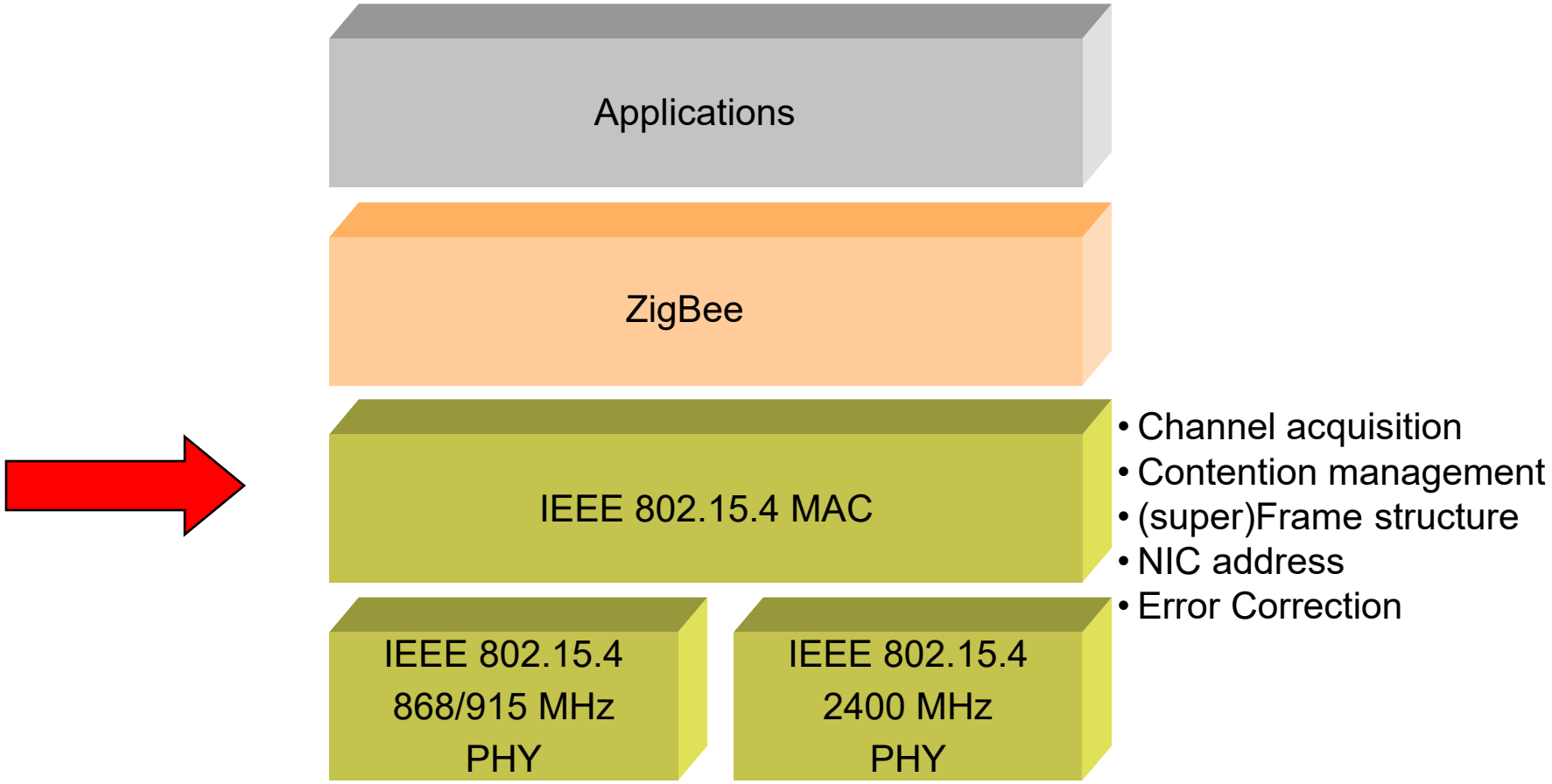


PHY frame structure

- PHY packet fields
  - Preamble (32 bits) – synchronization
  - Start of packet delimiter (8 bits) – shall be formatted as “11100101”
  - PHY header (8 bits) –PSDU length (Physical Layer Service Data Unit)
  - PSDU (0 to 127 bytes) – data field



802.15.4 Architecture (MAC)



IEEE 802.15.4 MAC Design Drivers

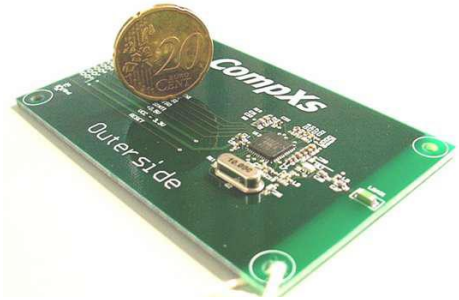
- Extremely low cost
- Ease of implementation
- Reliable data transfer
- Short range operation
- Very low power consumption

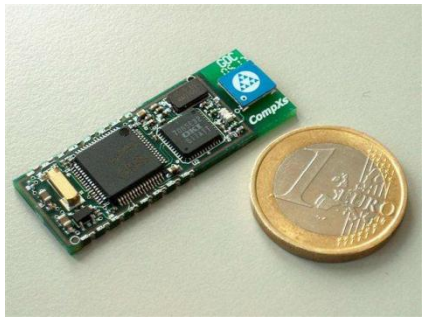
Simple but flexible protocol

IEEE 802.15.4 MAC Overview

Device Classes

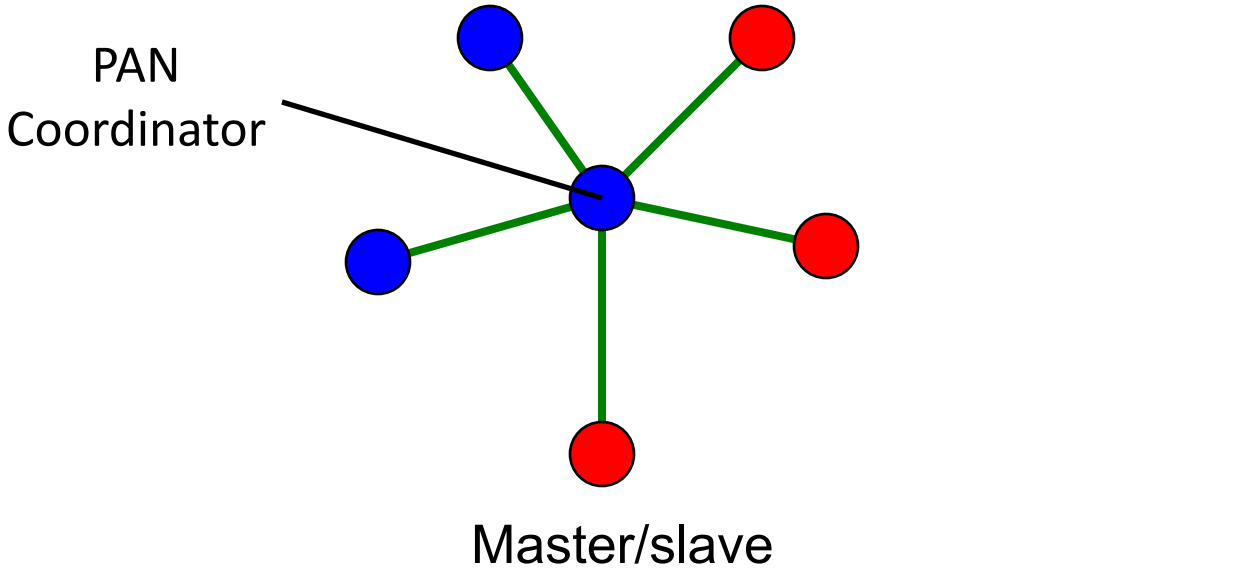
- Full function device (FFD)
  - Any topology
  - Network coordinator capable
  - Talks to any other device
  - The FFD can operate in three modes serving
    - Device
    - Coordinator
    - PAN coordinator
- Reduced function device (RFD)
  - Limited to star topology
  - Talks only to a network coordinator
    - Cannot become a network coordinator
  - Very simple implementation





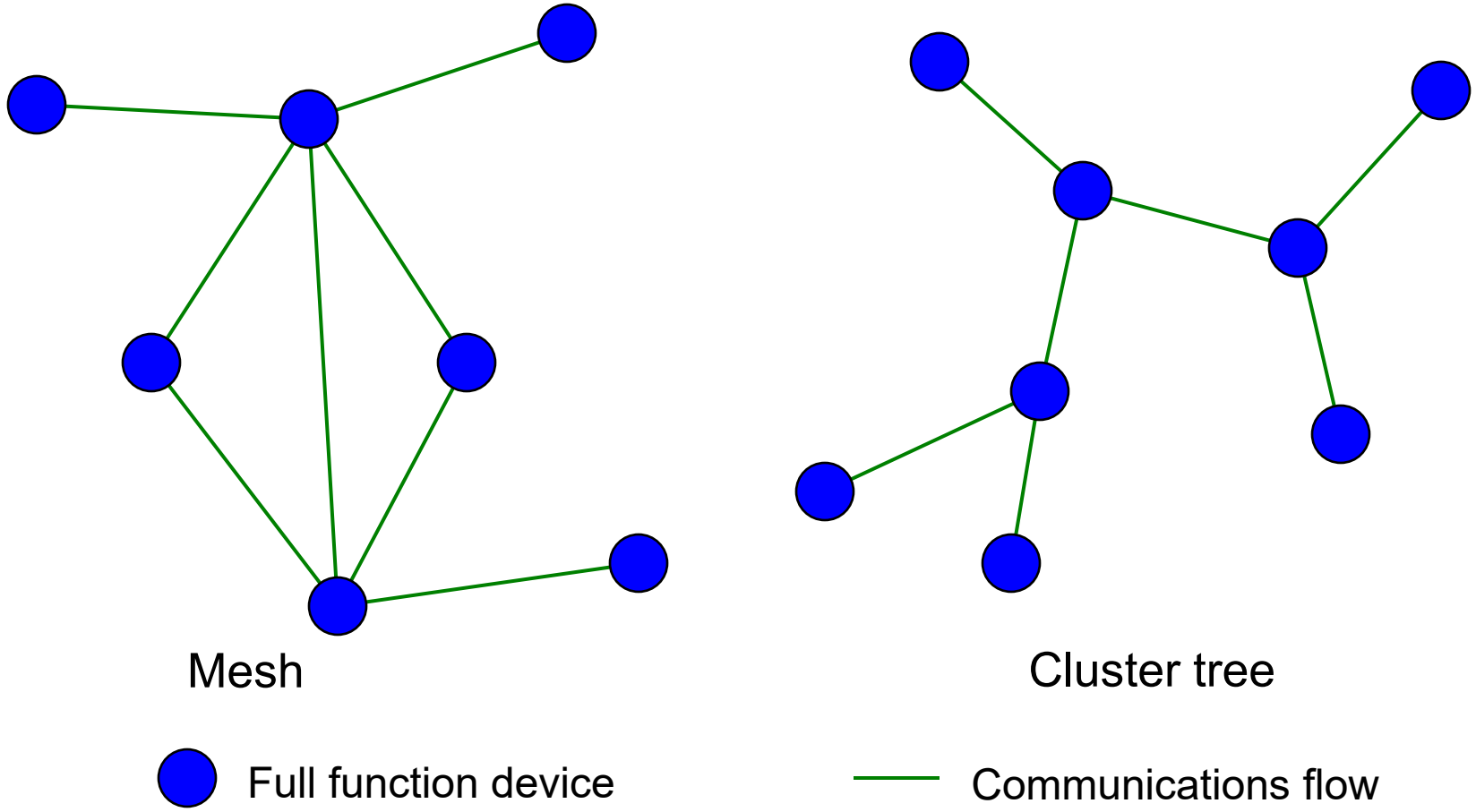
IEEE 802.15.4 MAC Overview

Star Topology



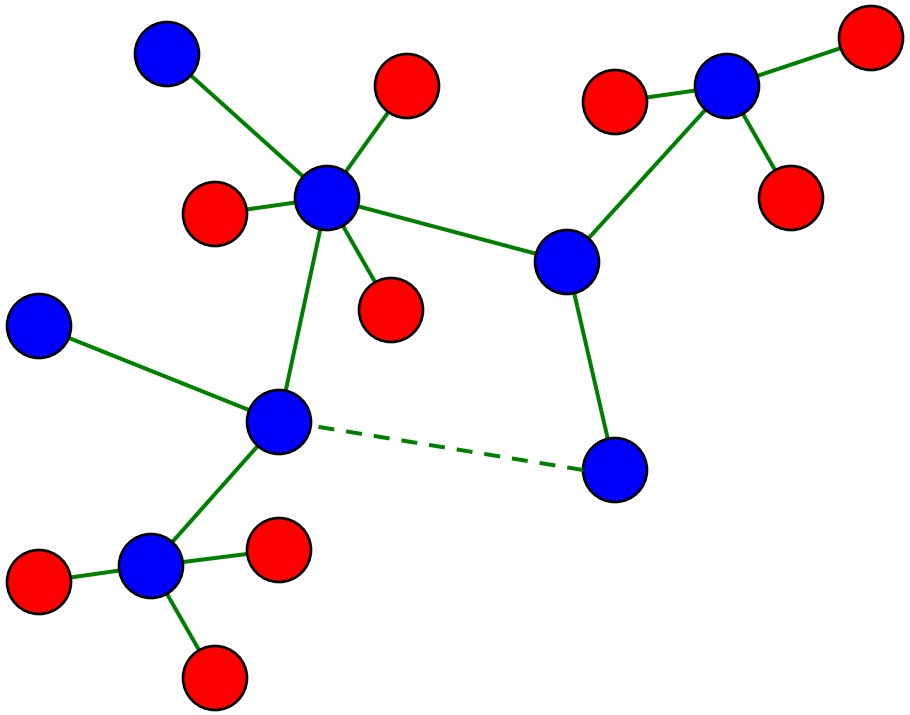
IEEE 802.15.4 MAC Overview

Mesh (Peer-Peer) and cluster tree topologies



IEEE 802.15.4 MAC Overview

Combined Topology



Clustered stars - for example, cluster nodes exist between rooms of a hotel and each room has a star network for control.

May have a mesh structure in some cases as well

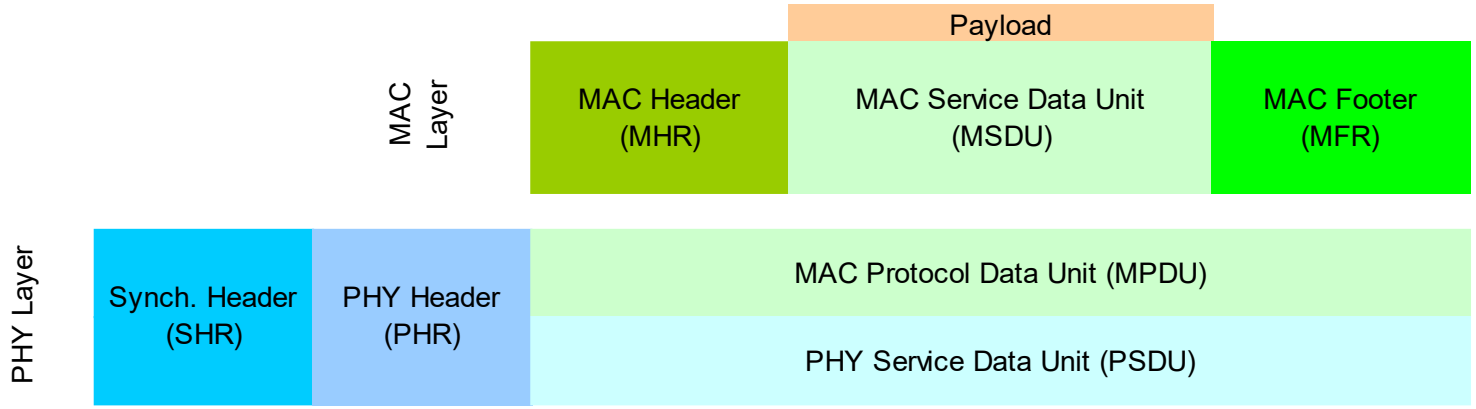
Full function device

Reduced function device

Communications flow

IEEE 802.15.4 MAC Overview

General Frame Structure



4 Types of MAC Frames:

- Data Frame
- Beacon Frame
- Acknowledgment Frame
- MAC Command Frame

MAC layer

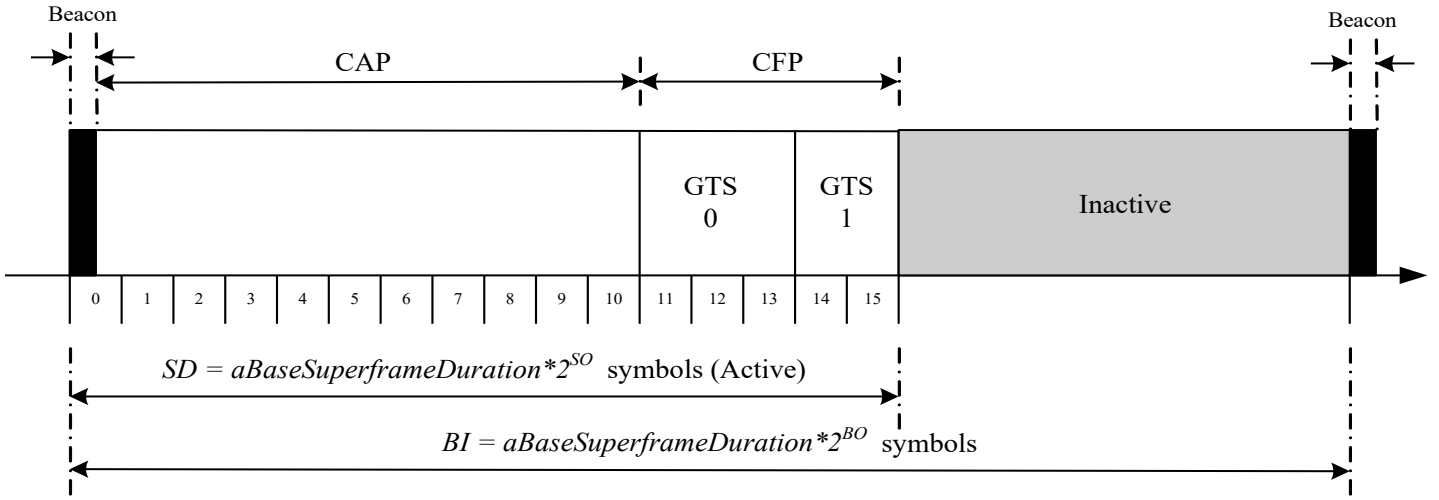
Managing PANs

- Channel scanning (Energy Detection, active, passive, orphan – verifies if it still has a parent)
- PAN ID conflict detection and resolution
- Starting a PAN
- Sending beacons
- Device discovery, association/disassociation
- Synchronization (beacon/nonbeacon)
- Orphaned device realignment

Transfer handling

- Transaction based (indirect transmission)
  - Beacon indication
  - Polling
- Transmission, Reception, Rejection, Retransmission
  - Acknowledged / Not acknowledged
- GTS management
  - Allocation/deallocation/Reallocation
  - Usage

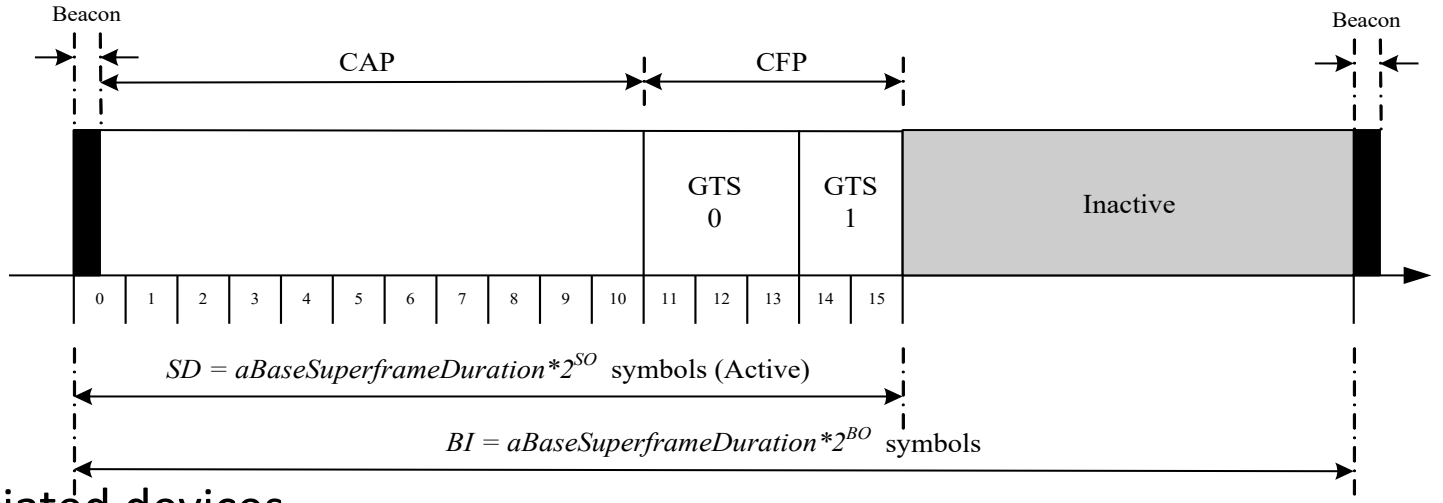
Superframe



- A coordinator in a PAN can optionally bound channel time using a SuperFrame structure
  - bound by beacon frames (i.e., beacon marks the start of a superframe)
- A superframe is divided into two parts
  - Inactive: all devices sleep (including the coordinator
  - Active:
    - Active period will be divided into 16 slots
    - 16 slots can further be divided into two parts
      - Contention access period
      - Contention free period

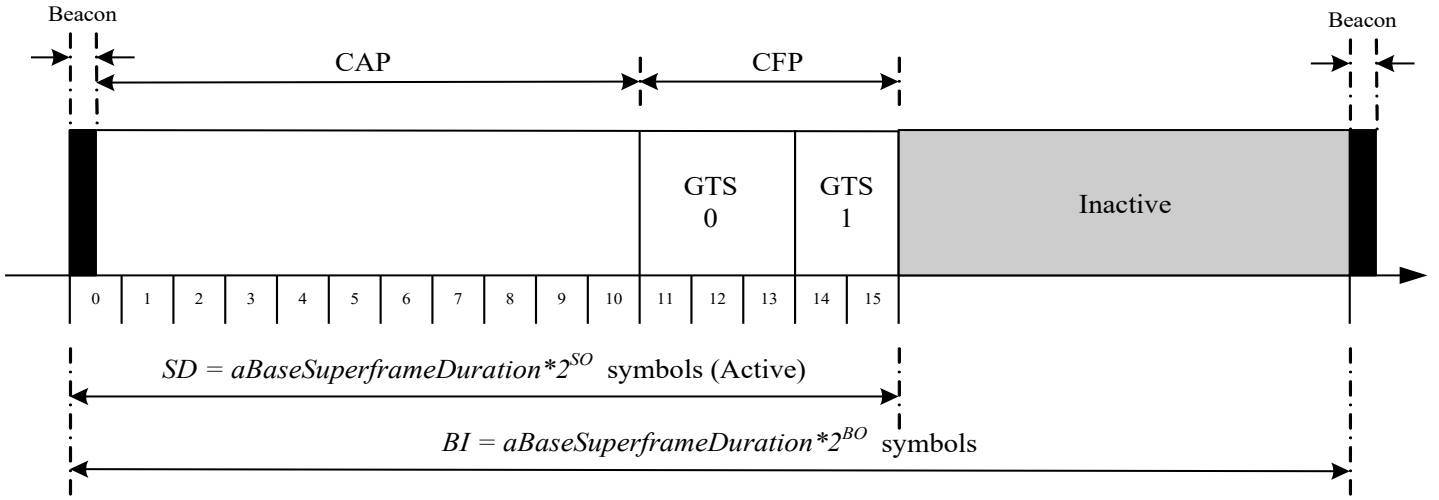
CAP – Contention Access Period
CFP – Contention Free Period
SD – Superframe Duration
BI – Beacon Interval
BO – Beacon Order
SO – Superframe Order

Superframe



- Beacons are used for
  - starting superframes
  - synchronizing with associated devices
  - announcing the existence of a PAN
  - informing pending data in coordinators
- In a beacon enabled network,
  - Devices use the slotted CSMA/CA mechanism to contend for the usage of channels
  - FFDs which require fixed rates of transmissions can ask for *guarantee time slots (GTS)* from the coordinator

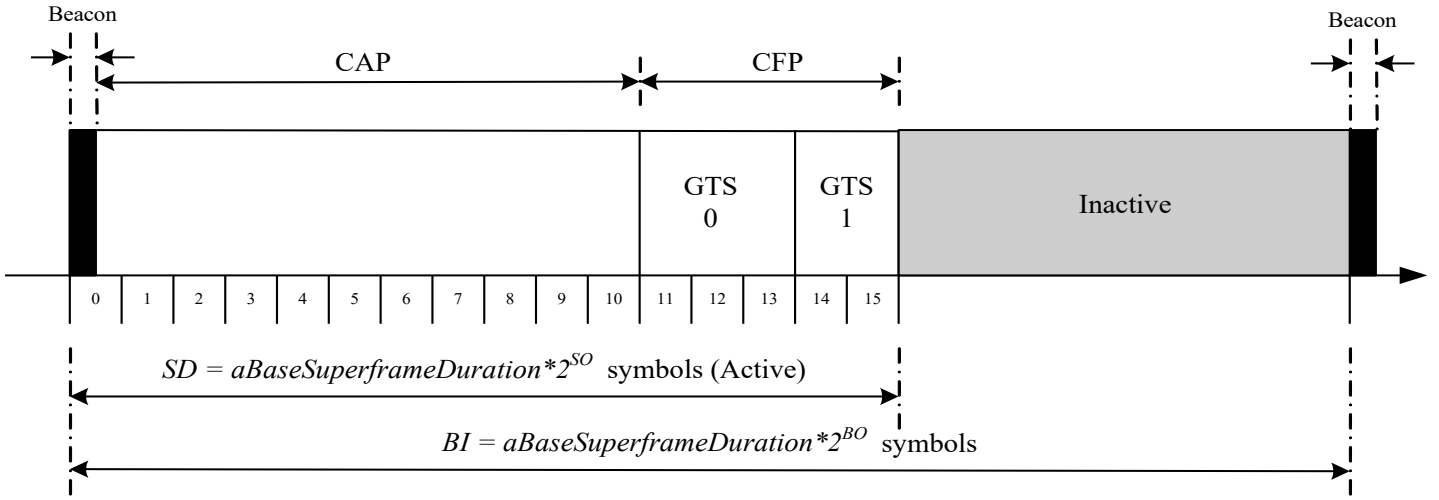
Superframe



- The structure of superframes is controlled by two parameters: *beacon order (BO)* and *superframe order (SO)*
  - BO decides the length of a superframe
  - SO decides the length of the active portion in a superframe

- For channels 11 to 26, the length of a superframe can range from 15.36 *msec* to 215.7 *sec*.
  - which means very low duty cycle
- Remember: Duty Cycle
  - **Duty Cycle indicates the fraction of time a resource is busy.**
  - When a single device transmits on a channel for 2 *time units* every 10 *time units*, this device has a duty cycle of 20%.

Superframe

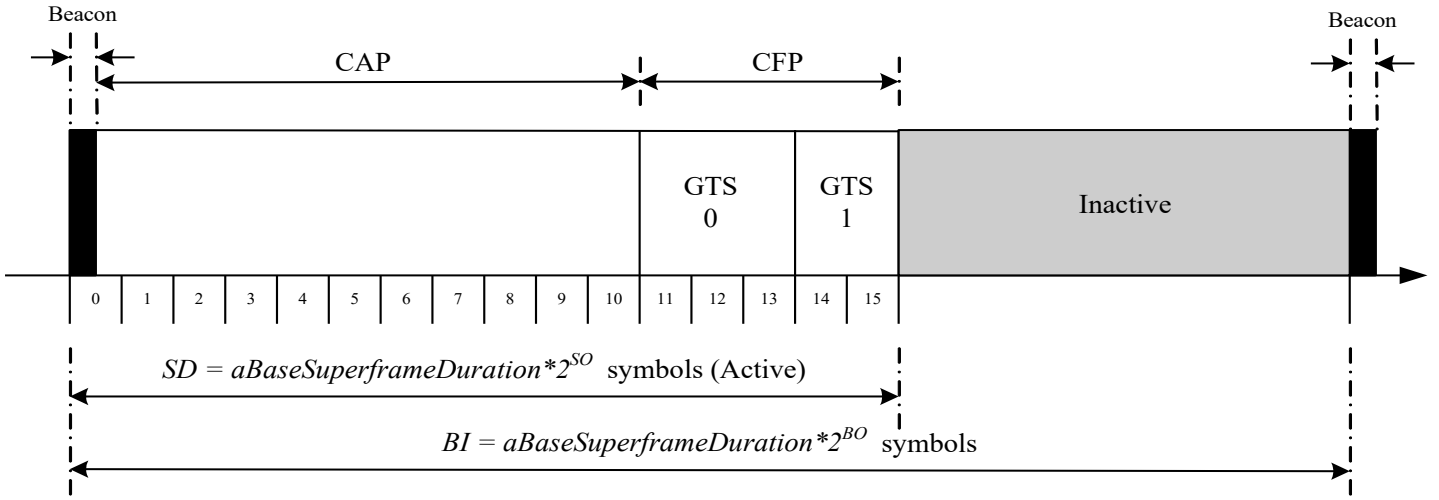


- Each device will be active for $2^{-(BO-SO)}$ portion of the time, and sleep for $1-2^{-(BO-SO)}$ portion of the time
- In IEEE 802.15.4, devices' duty cycle follow the specification

BO-SO	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**≧10**
Duty cycle (%)	**100**	**50**	**25**	**12**	**6.25**	**3.125**	**1.56**	**0.78**	**0.39**	**0.195**	**< 0.1**

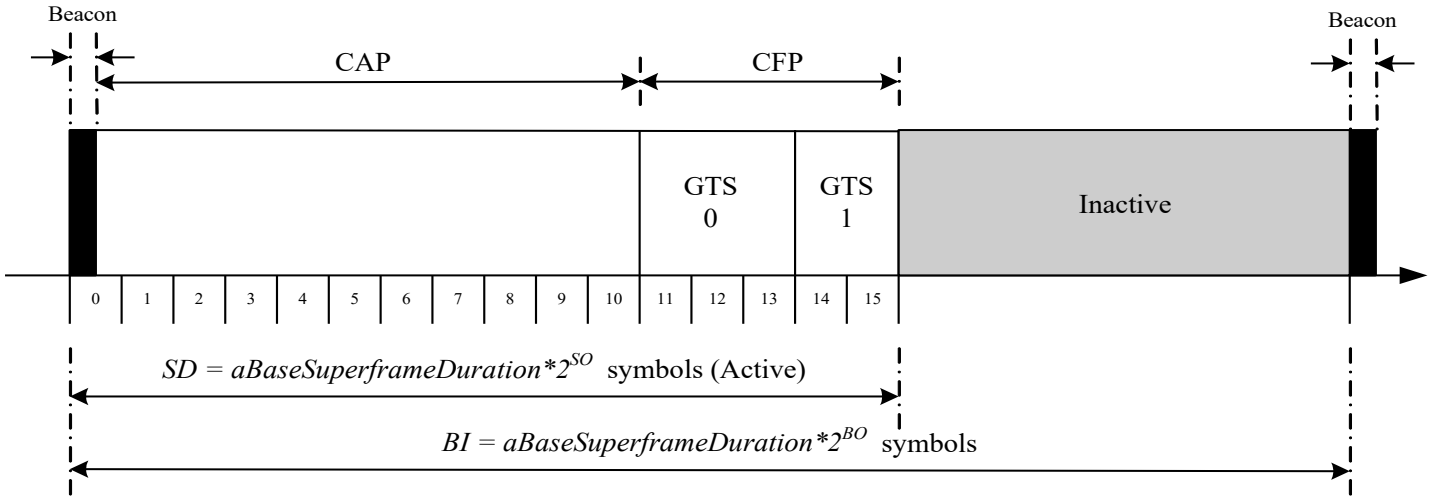
BO – Beacon Order
SO – Superframe Order

GTS concepts



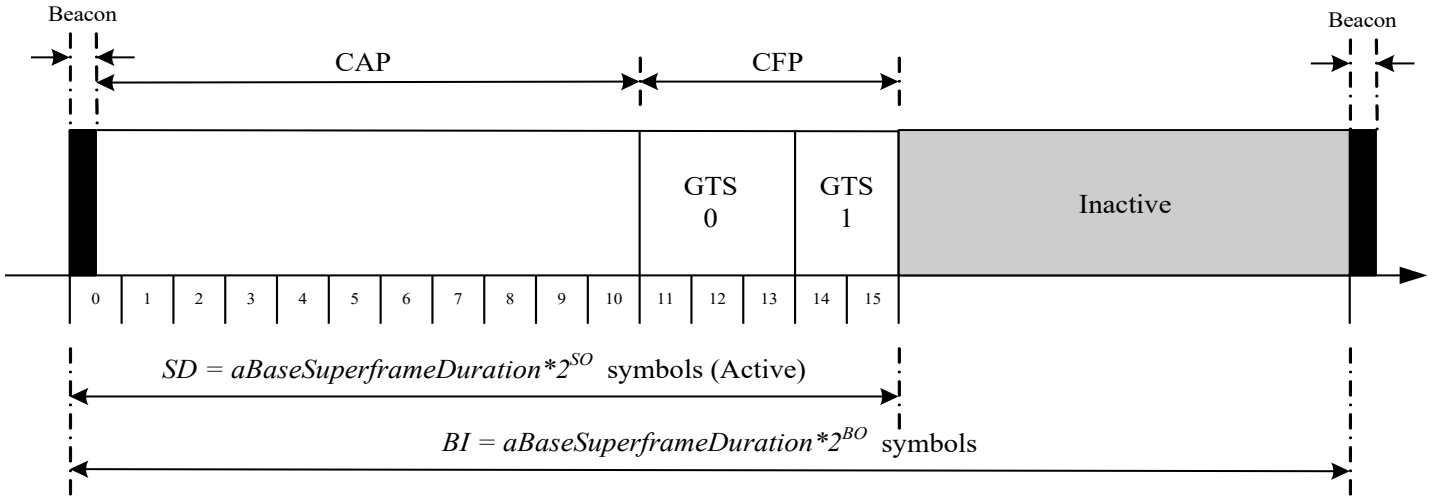
- A guaranteed time slot (GTS) allows a device to operate on the channel within a portion of the superframe
- A GTS shall only be allocated by the PAN coordinator
- The PAN coordinator can allocate up to seven GTSs at the same time
- The PAN coordinator decides whether to allocate GTS based on:
  - Requirements of the GTS request
  - The current available capacity in the superframe

GTS concepts



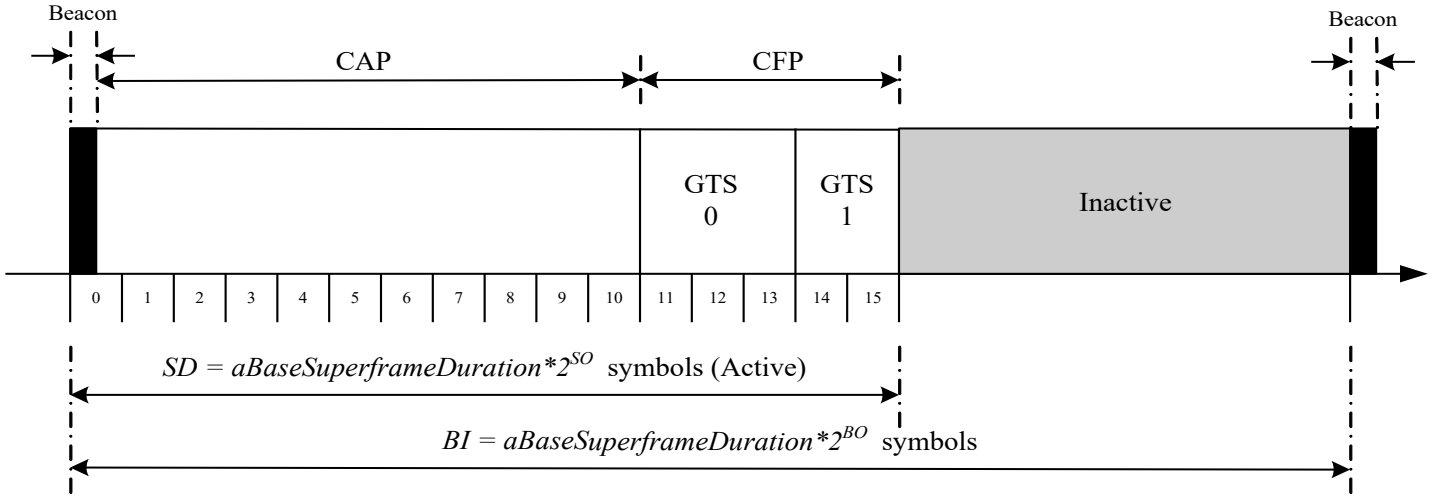
- A GTS can be **deallocated**
  - At any time at the discretion of the PAN coordinator or
  - By the device that originally requested the GTS
- A data frame transmitted in an allocated GTS shall use only short addressing
  - 16-bit instead of 64-bit: reduces overhead
- The PAN coordinator shall be able to store the info of devices that are necessary for GTS, including starting slot, length, direction and associated device address

GTS concepts



- Before GTS starts, the GTS direction shall be specified as either transmit or receive
- Each device may request one **transmit** GTS and/or one **receive** GTS
- A device shall only attempt to allocate and use a GTS if it is currently tracking the beacon
- If a device loses synchronization with the PAN coordinator, all its GTS allocations shall be lost
- The use of GTSs by an RFD is optional

Channel access mechanism



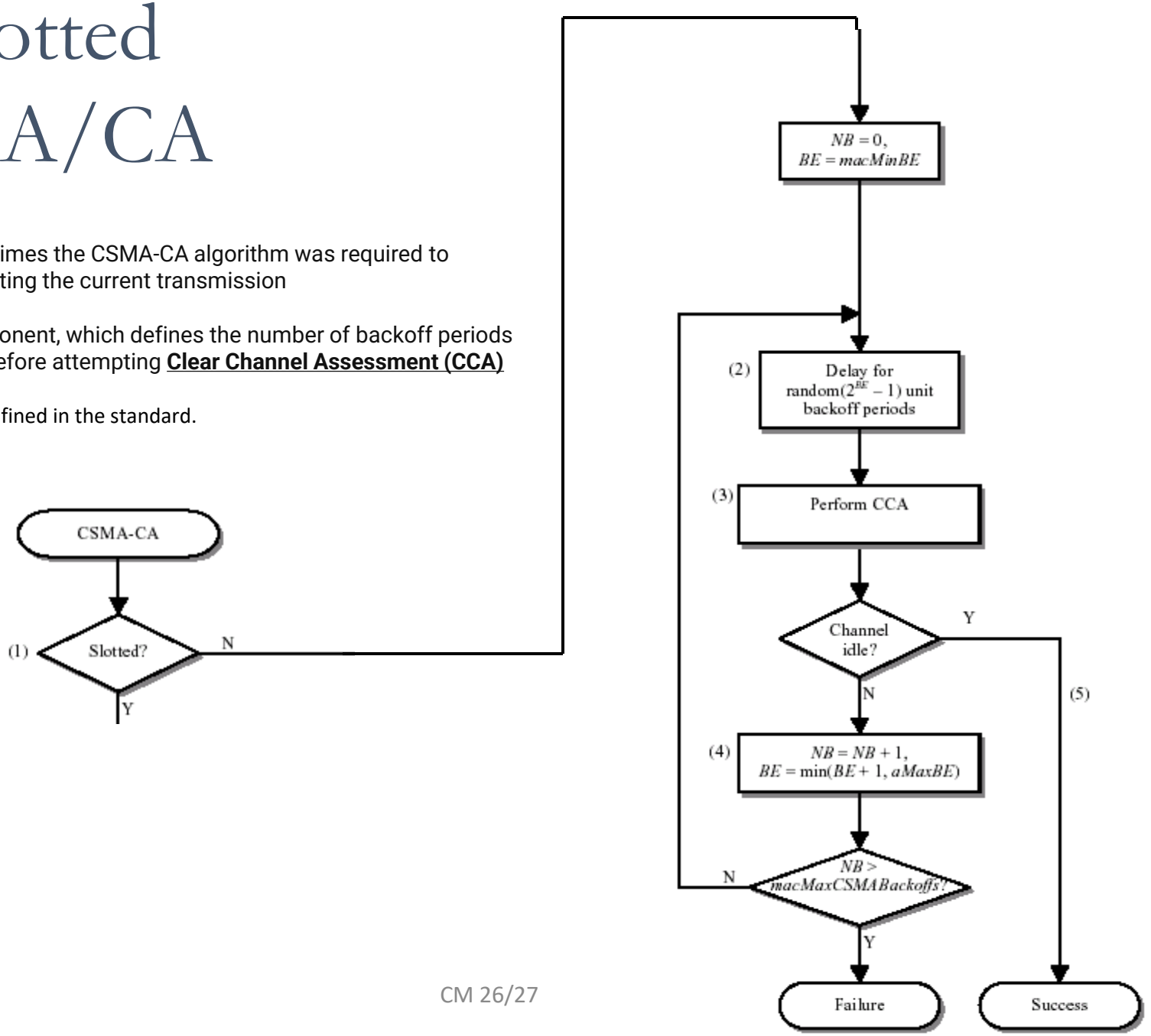
- Two type channel access mechanism:
  - In non-beacon-enabled networks → unslotted CSMA/CA channel access mechanism
  - In beacon-enabled networks → slotted CSMA/CA channel access mechanism

Unslotted CSMA/CA

NB is the number of times the CSMA-CA algorithm was required to backoff while attempting the current transmission

BE is the backoff exponent, which defines the number of backoff periods a node should wait before attempting **Clear Channel Assessment (CCA)**

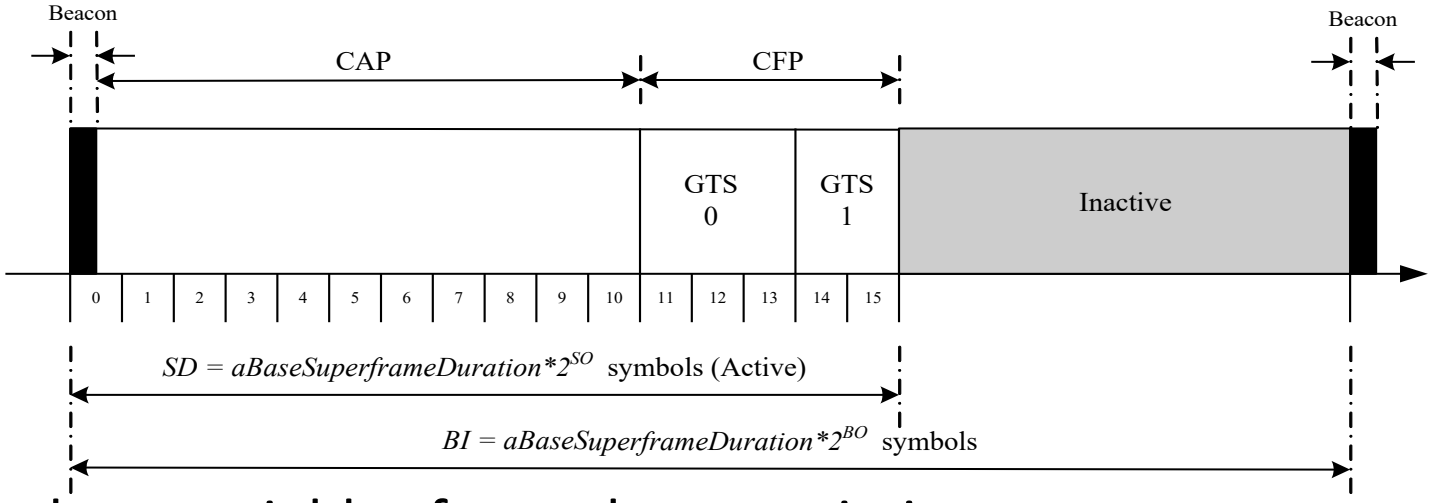
MacMinBE constant defined in the standard.



CSMA/CA algorithm

- In slotted CSMA/CA
  - The **backoff period boundaries** of every device in the PAN shall be **aligned** with the **superframe slot boundaries** of the PAN coordinator
    - i.e. the start of first backoff period of each device is aligned with the start of the beacon transmission
      - All nodes share a common, synchronized timeline driven by the coordinator's beacon.
      - The moment the beacon is sent marks the zero-point for the first backoff period, ensuring no node is out of step with another.
  - The MAC sublayer shall ensure that the PHY layer commences **all of its transmissions on the boundary of a backoff period**
    - A node cannot just start transmitting the moment it finishes counting down its backoff random delay
      - It has to wait for the exact boundary of the next backoff slot.
      - This keeps channel access orderly, aligned, and synchronized across the entire network, allowing devices to properly perform Clear Channel Assessment (CCA) at discrete time intervals.

CSMA/CA algorithm



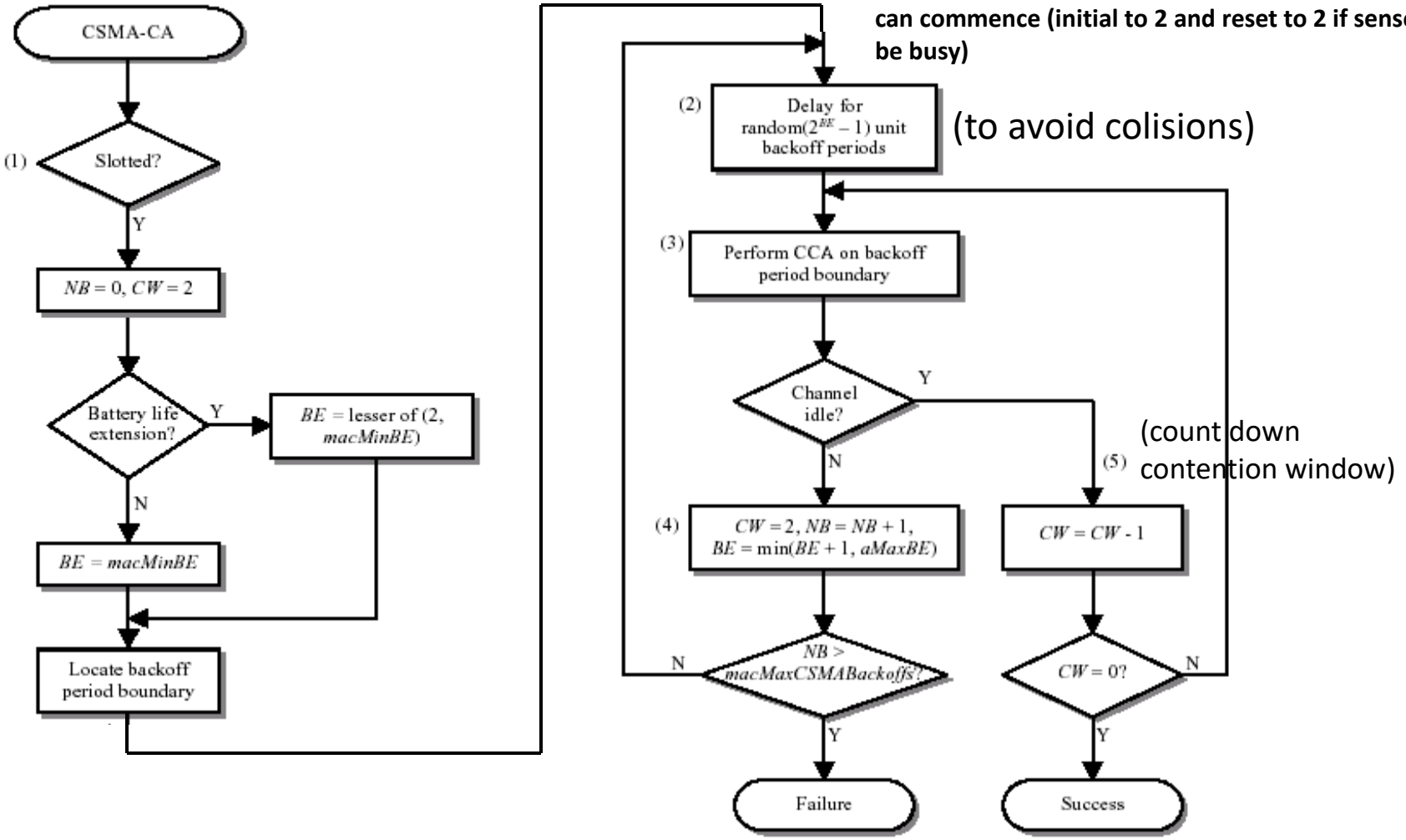
- Each device shall maintain three variables for each transmission attempt
  - NB: number of time the CSMA/CA algorithm was required to backoff while attempting the current transmission
  - **CW: contention window length, the number of backoff periods that needs to be clear of channel activity before transmission can commence (initial to 2 and reset to 2 if sensed channel to be busy)**
  - BE: the backoff exponent which is related to how many backoff periods a device shall wait before attempting to assess a channel

Slotted CSMA/CA

NB is the number of times the CSMA-CA algorithm was required to backoff while attempting the current transmission

BE is the backoff exponent, which defines the number of backoff periods a node should wait before attempting **Clear Channel Assessment (CCA)**

CW: contention window length, the number of backoff periods that needs to be clear of channel activity before transmission can commence (initial to 2 and reset to 2 if sensed channel to be busy)



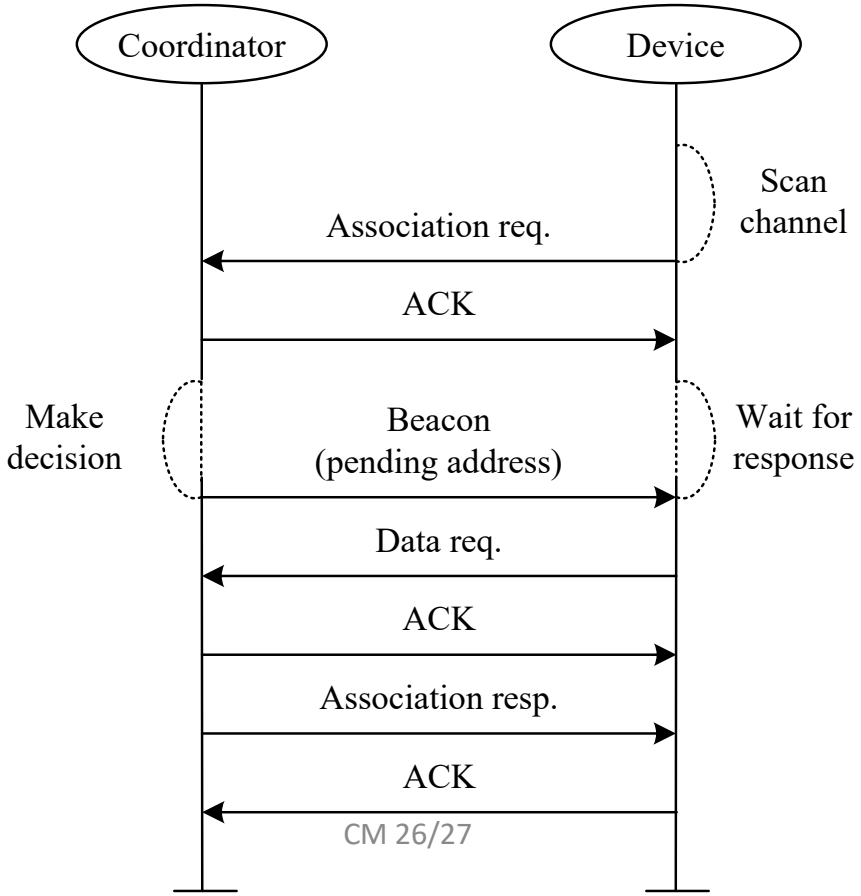
This ensures performing two CCA operations to prevent potential collisions of acknowledgement frames. If the channel is again sensed as idle (CW = 0), the node attempts to transmit.

Slotted vs. non-slotted CSMA/CA

Feature	Slotted CSMA/CA	Unslotted CSMA/CD
Used in	Beacon-enabled networks	Non-beacon-enabled networks
Synchronization	Required (no beacon slots)	Not required
Time Alignment	Access only at slot boundaries	Random access anytime
Power Saving	High (sleep during inactive periods)	Low (devices always listening)
Collision Probability	Lower	Higher
Complexity	Higher (timing and superframe management)	Simpler

Association procedures

- A device becomes a member of a PAN by associating with its coordinator
- Procedures



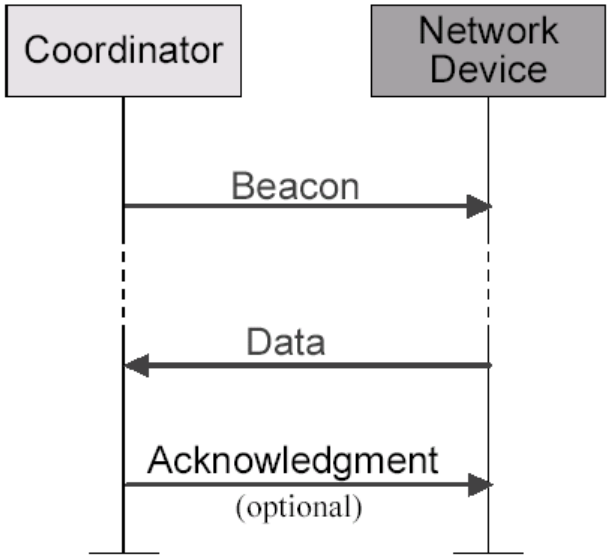
Association procedures

- In IEEE 802.15.4, **association results** are announced in an indirect fashion
- A coordinator responds to association requests by appending devices' long addresses (64 bit) in beacon frames
  - Long ID = device's fixed ID (obtained from factory)
  - E.g.: 00:12:4B:00:11:22:33:44 (EUI-64)
- Devices need to send a data request to the coordinator to acquire the association result

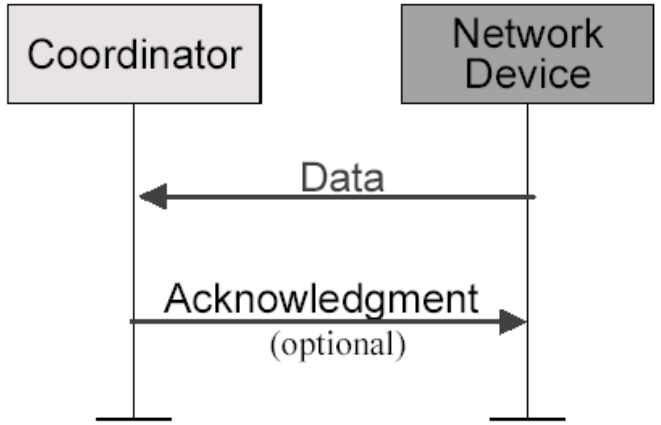
- After associating to a coordinator, a device will be assigned a 16-bit *short address*.
  - Short ID = only valid within a particular PAN
  - E.g.: 0x42AF

Data transfer model (device to coordinator)

- Data transferred from device to coordinator
  - In a beacon-enabled network, the device finds the beacon to synchronize to the superframe structure. Then uses slotted CSMA/CA to transmit its data.
  - In a non beacon-enable network, device simply transmits its data using unslotted CSMA/CA



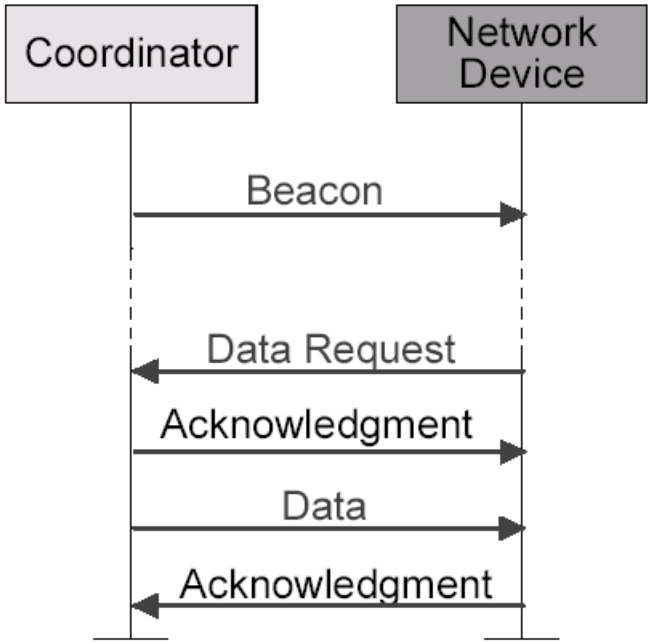
Communication to a coordinator
In a beacon-enabled network



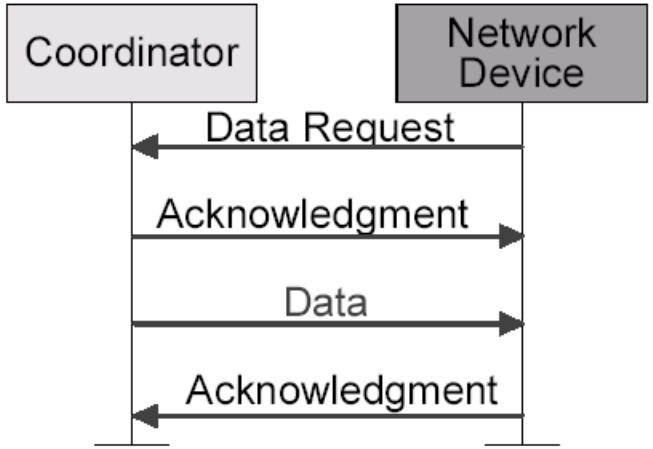
Communication to a coordinator
In a non beacon-enabled network

Data transfer model (coordinator to device)

- Data transferred from coordinator to device
  - In a beacon-enable network, the coordinator indicates in the beacon that the data is pending. Device periodically listens to the beacon and transmits a MAC command request using slotted CSMA/CA if necessary.
  - In a non-beacon-enable network, a device transmits a MAC command request using unslotted CSMA/CA. If the coordinator has its pending data, the coordinator transmits data frame using unslotted CSMA/CA. Otherwise, coordinator transmits a data frame with zero length payload.

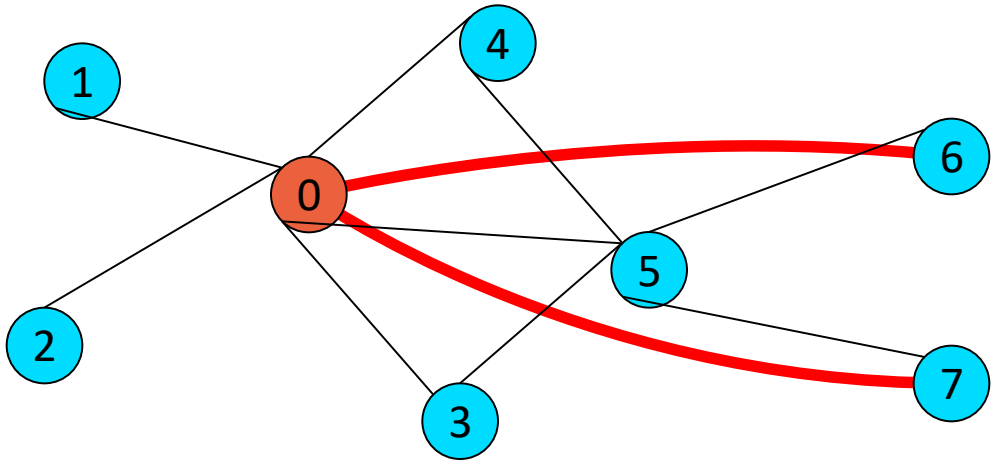


Communication from a coordinator
In a beacon-enabled network



Communication from a coordinator
in a non beacon-enabled network

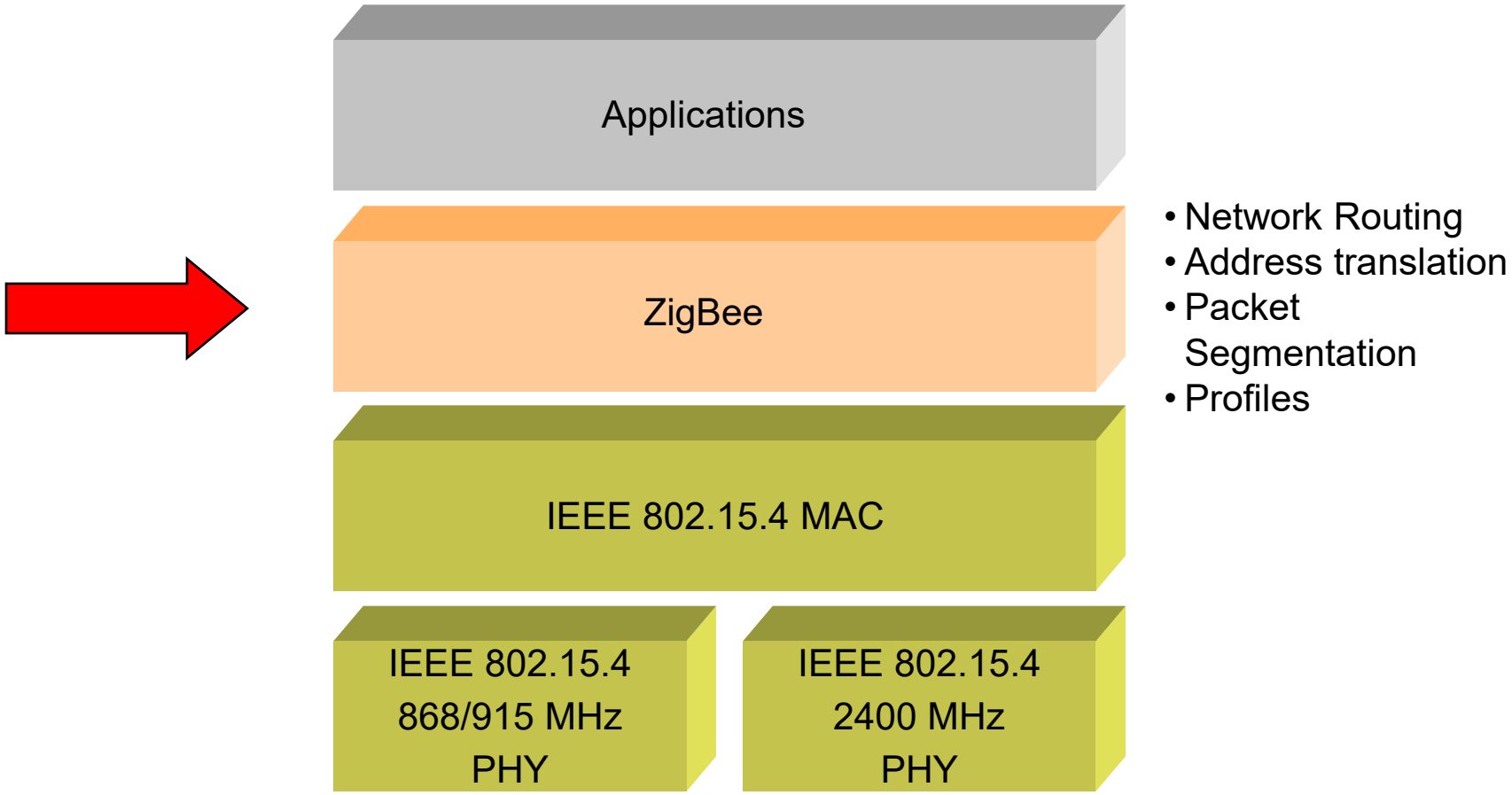
MAC layer



How 6 and 7 connect to coordinator 0?

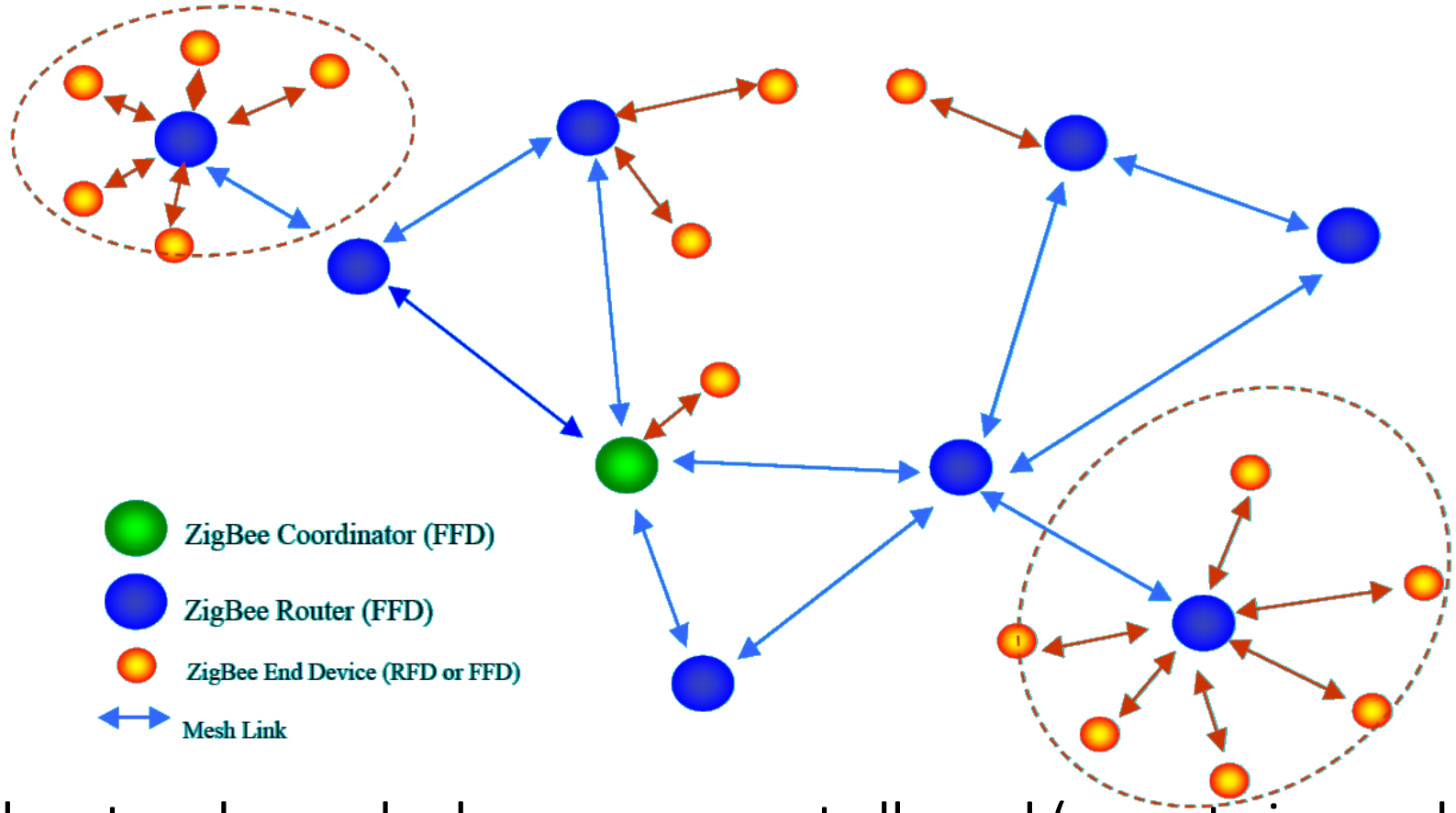
Routing (NWK Layer)

802.15.4 Architecture



- Network Routing
- Address translation
- Packet Segmentation
- Profiles

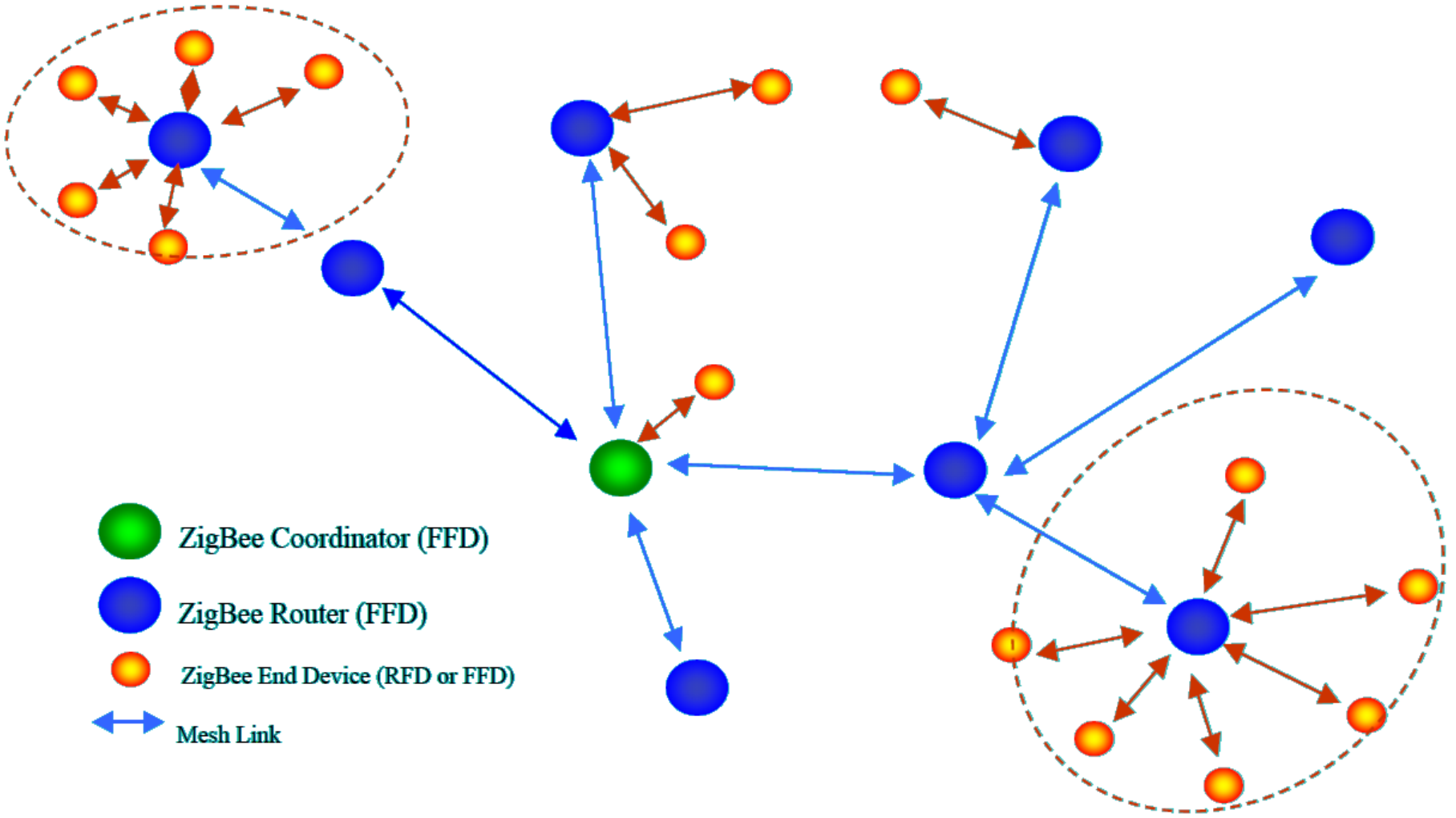
Combined topologies: Mesh Topologies



In a mesh network, regular beacons are not allowed (operate in non-beacon mode).

- Devices in a mesh network can only communicate with each other by peer-to-peer transmissions
- Why? Impossible to maintain inter-router beacon periodicity

Combined Topologies: Tree



In a tree network, the coordinator and routers can announce beacons.

More on Device addressing

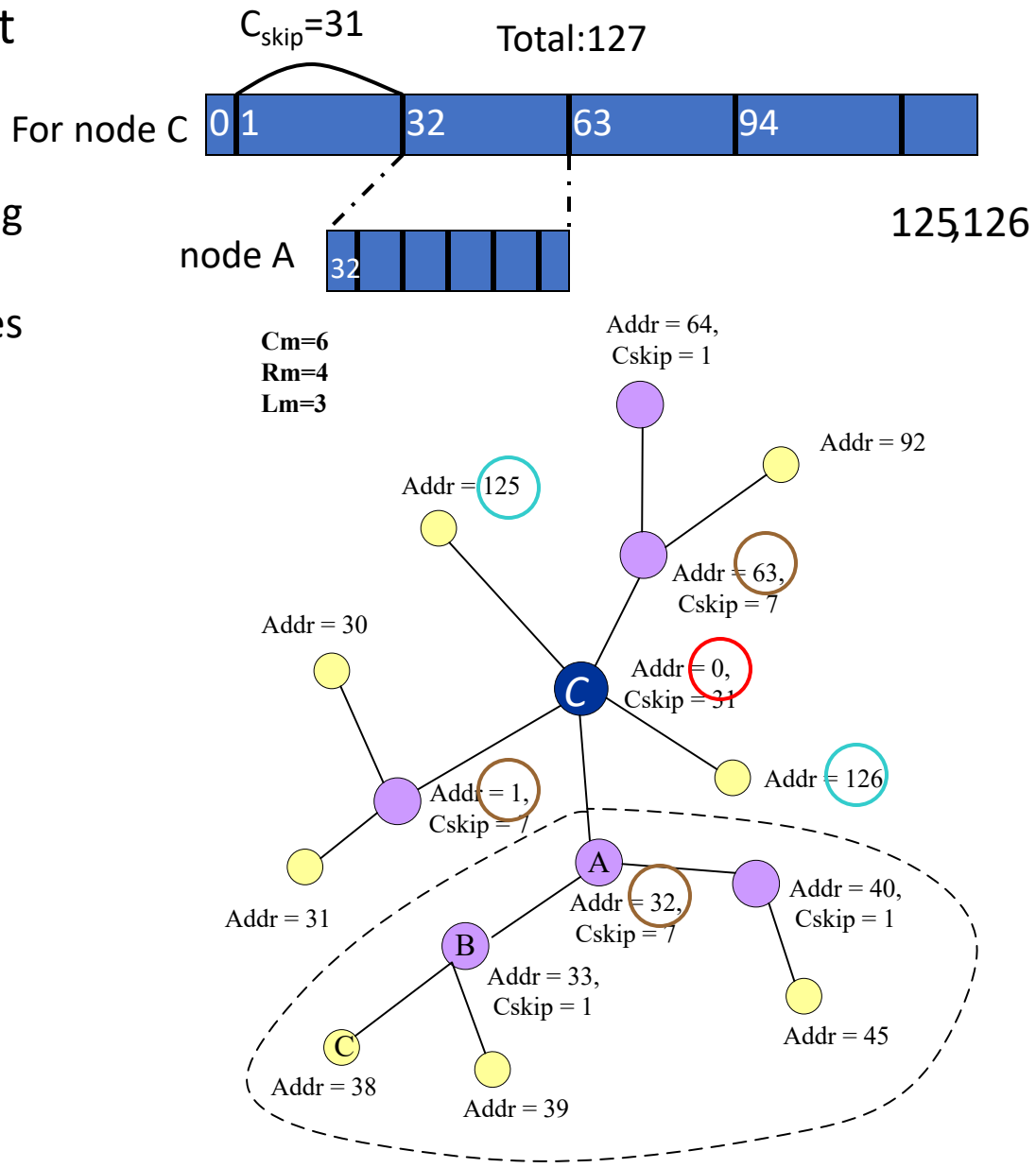
- Two or more devices communicating on the same physical channel constitute a WPAN which includes at least one FFD (PAN coordinator)
- Each independent PAN will select a unique PAN identifier
- All devices operating on a network shall have unique 64-bit extended address (IEEE 802.15.4). This address can be used for direct communication in the PAN
- The network address can use a 16-bit short address, which is allocated to the child routers by the PAN coordinator when the device associates
- 256 sub addresses may be allocated for subunits

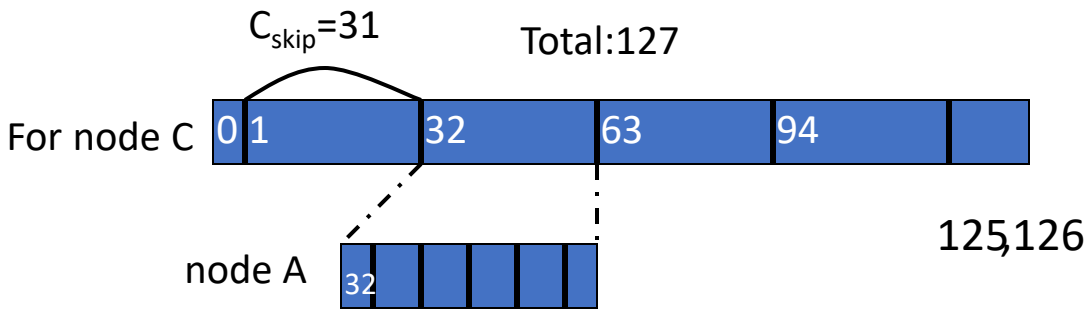
Address assignment in a ZigBee network

- In ZigBee, network addresses are assigned to devices by a **distributed address assignment scheme**
- The ZigBee coordinator determines three network parameters to set the allocations
  - the maximum number of children (C_m) of a ZigBee router
  - the maximum number of child routers (R_m) of a parent node
  - the depth of the network (L_m)
- A parent device utilizes C_m, R_m, and L_m to compute a parameter called C_{skip}
  - which is used to compute the size of its children's address pools

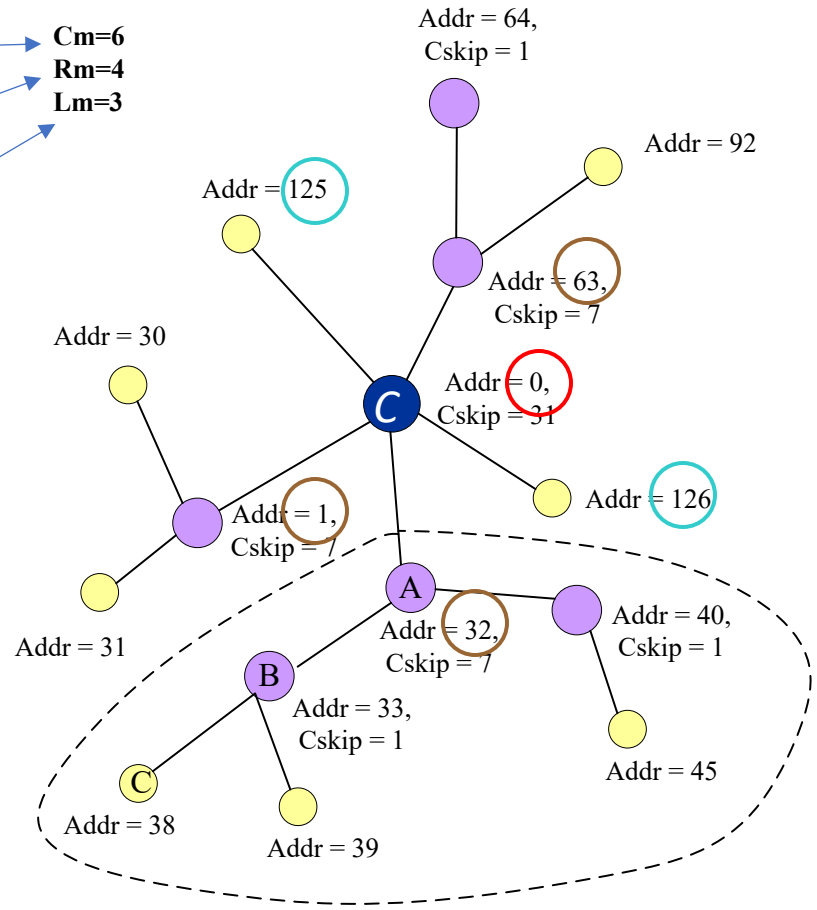
$$Cskip(d) = \begin{cases} 1 + Cm \cdot (Lm - d - 1), & \text{if } Rm = 1 \quad \cdots\cdots\cdots\cdots (a) \\ \dfrac{1 + Cm - Rm - Cm \cdot Rm^{Lm-d-1}}{1 - Rm}, & \text{Otherwise} \quad \cdots\cdots\cdots\cdots (b) \end{cases}$$

- Each router is allocated a block of 16-bit addresses that it can assign to its own children. Why:
  - No two nodes are able to assign overlapping addresses
  - Routing can be inferred from address ranges (tree routing)
- If a parent node at depth *d* has an address A_{parent},
  - the *n*th child router is assigned to address $A_{parent}+(n\text{-}1)\times C_{skip}(d)+1$
  - *n*th child end device is assigned to address $A_{parent}+R_m\times C_{skip}(d)+n$

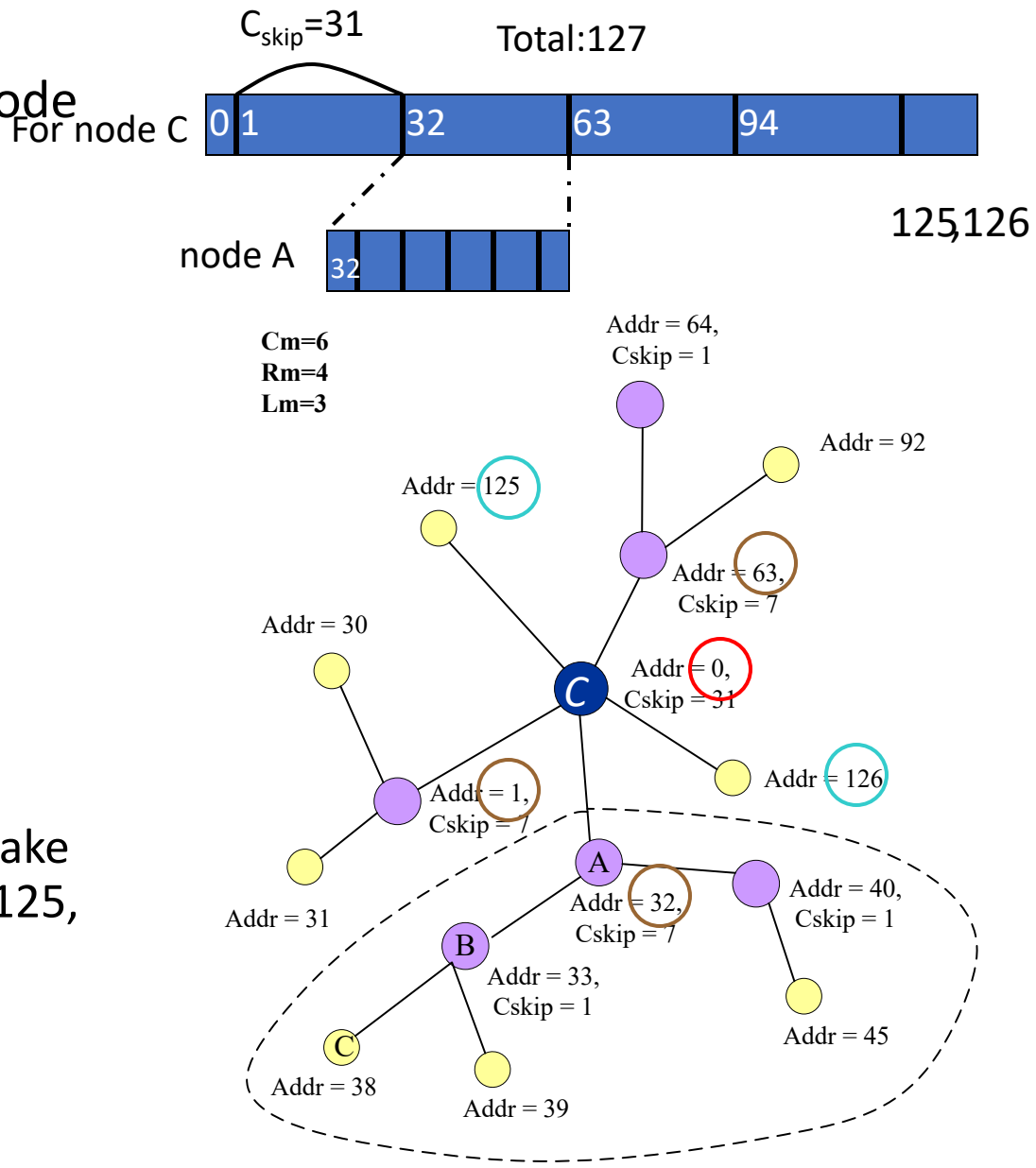




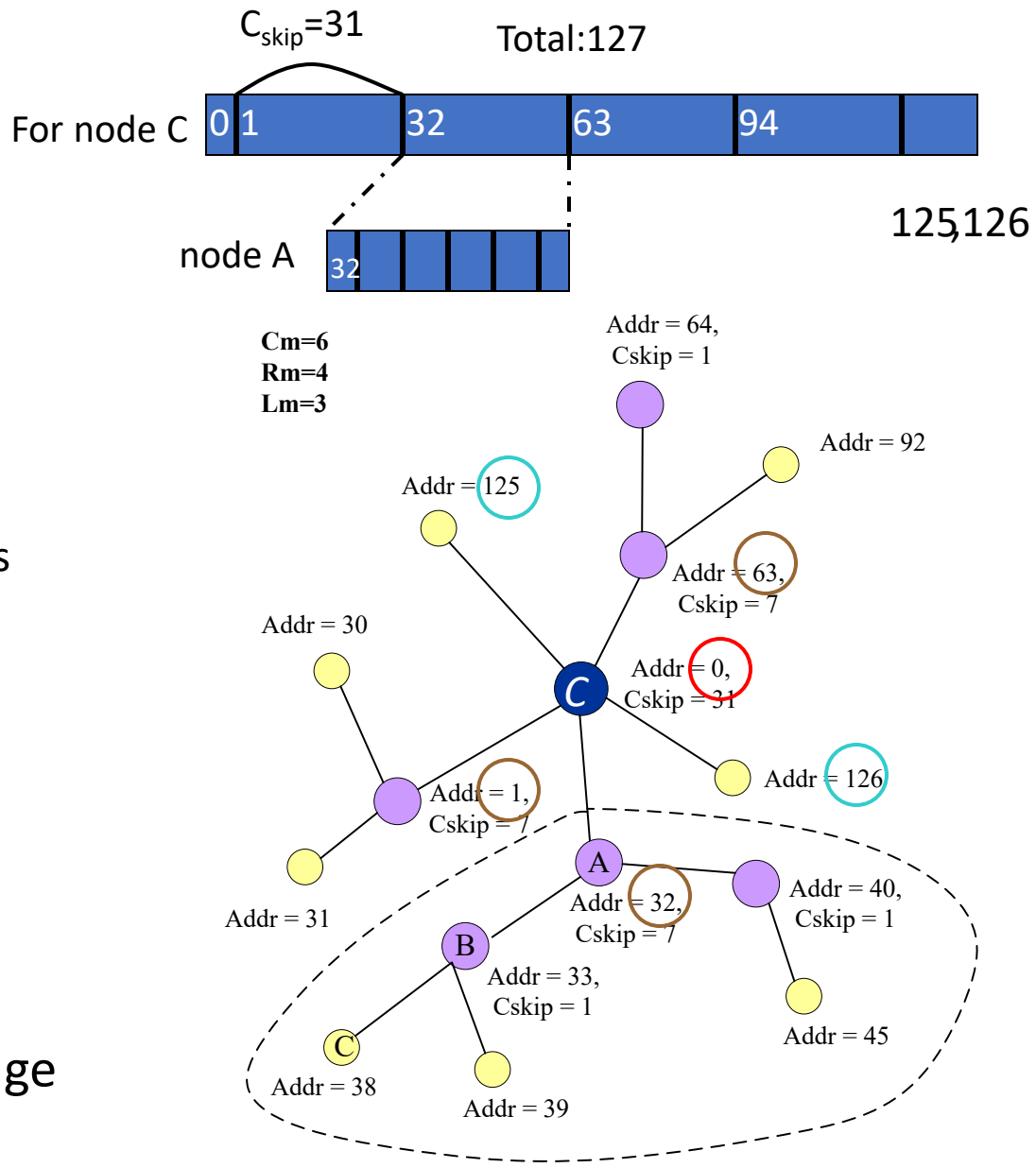
- CM = 6
  - Maximum number of children per parent
- RM = 4
  - Maximum number of children that are routers
- LM = 3
  - Maximum network depth



- At the top, you have the coordinator (node C) with **address 0**
  - Total address space available = 0-127 (128 total)
  - For node C, Cskip = 31
    - Each child router of C gets a block of 31 addresses for itself and its descendants
- So:
  - Router 1
    - Start address: 1 End address **(Start + Cskip -1)** : 31
  - Router 2
    - Start address: 32 End address: 62
  - Router 3
    - Start address: 63 End address: 93
  - Router 4
    - Start address: 94 End address: 124
- Any **end device** under C (non-routers) can take addresses near the end of the total space (125, 126)



- Let's look at node A more closely
- Node A
  - 2nd router child of the coordinator
  - Computes it's own Cskip, based on its depth
  - Remember: Cskip (node C) = 31
    - Router1 : addresses 1 → 31
    - Router 2 (Node A): addresses 32→ 62
- Node A has Cskip=7 ((31-1)/4)
  - Router children get blocks of size 7
  - End devices get the next free single address
- Inside A
  - B is a child router of A, assigned address 33
    - Due to Cskip=7, B's range is 33-39
  - C has address 38
    - Cskip for B is 1 (lowers level), so it can assign one address per child
- Hierarchy: C → A → B
- Each uses Cskip to determine its subrange



ZigBee routing protocols

- In a tree network
  - Utilize the address assignment to obtain the routing paths
- In a mesh network
  - Two options
    - Reactive routing: if having routing capacity
    - Use tree routing: if do not have routing capacity

- Note:
  - ZigBee coordinators and routers are said to have *routing capacity* if they have routing table capacities and route discovery table capacities

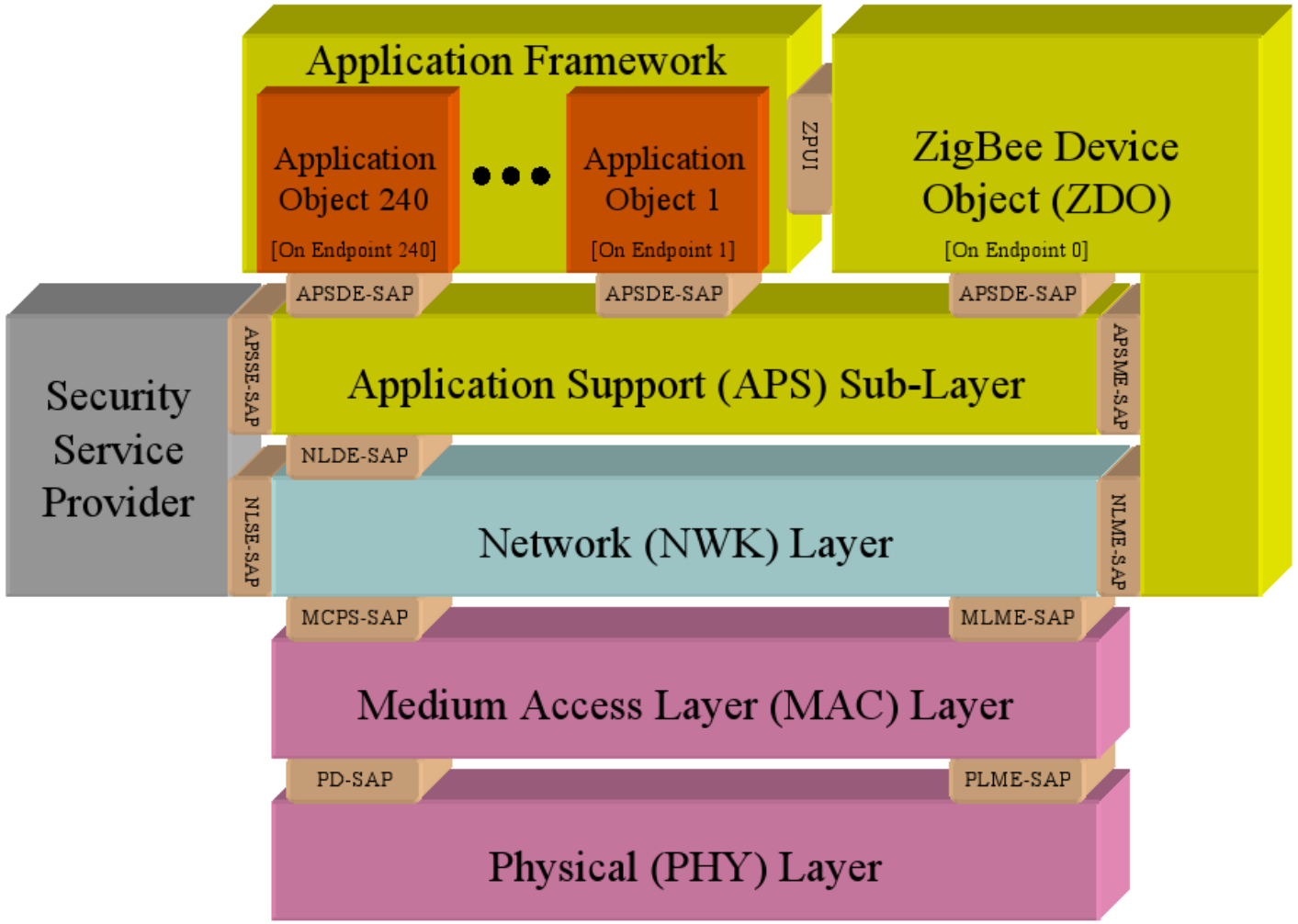
Summary of ZigBee network layer

- Pros and cons of different kinds of ZigBee network topologies

	Pros	Cons
Star	1. Easy to synchronize 2. Support low power operation 3. Low latency	1. Small scale
Tree	1. Low routing cost 2. Can form superframes to support sleep mode 3. Allow multihop communication	1. Route reconstruction is costly 2. Latency may be quite long
Mesh	1. Robust multihop communication 2. Network is more flexible 3. Lower latency	1. Cannot form superframes (and thus cannot support sleep mode) 2. Route discovery is costly 3. Needs storage for routing table

Application Level

CM 26/27



ZigBee defined Objects (ZDO):

- provides common function for applications
- Initializes APS, NWK-Layer and Security Service Specification
- offers services like device-/service-descovery, binding and security management
- assembles information about the network
- for ZBC/ZBR -> e.g. binding table

Command	Addressing	
	Request	Response
End device bind	Unicast to ZC	Unicast
Bind	Unicast to ZC or Src	Unicast
Unbind	Unicast to ZC or Src	Unicast

Profiles:

Definition of ZigBee-Profiles

- describes a common language for exchanging data
- defines the offered services
- device interoperatbility across different manufacturers
- Standard profiles available from the ZigBee Alliance
- **profiles contain device descriptions**
- unique identifier (licensed by the ZigBee Alliance)

ZigBee and BLE

- Business comparison:
  - ZigBee is older. It has gone through some iterations
  - ZigBee has market mindshare, but not a lot of shipments yet.
  - Market barriers: connectivity – ZigBee is not in PCs or mobile phones yet.
- Technical comparison:
  - Zigbee is low power; Bluetooth LE is even lower. Detailed analysis depends on specific applications and design detail, no to mention chip geometry.
  - ZigBee stack is light; the Bluetooth LE/GATT stack is even simpler
    - Remember: GATT – Generic ATTribute profile
- Going forward:
  - ZigBee has a lead on developing applications and presence
  - Bluetooth low energy has improved technology, and a commanding presence in several existing markets: mobile phones, automobiles, consumer electronics, PC industry
  - Replacing “classic Bluetooth ” with “dual mode” devices will bootstrap this market quickly

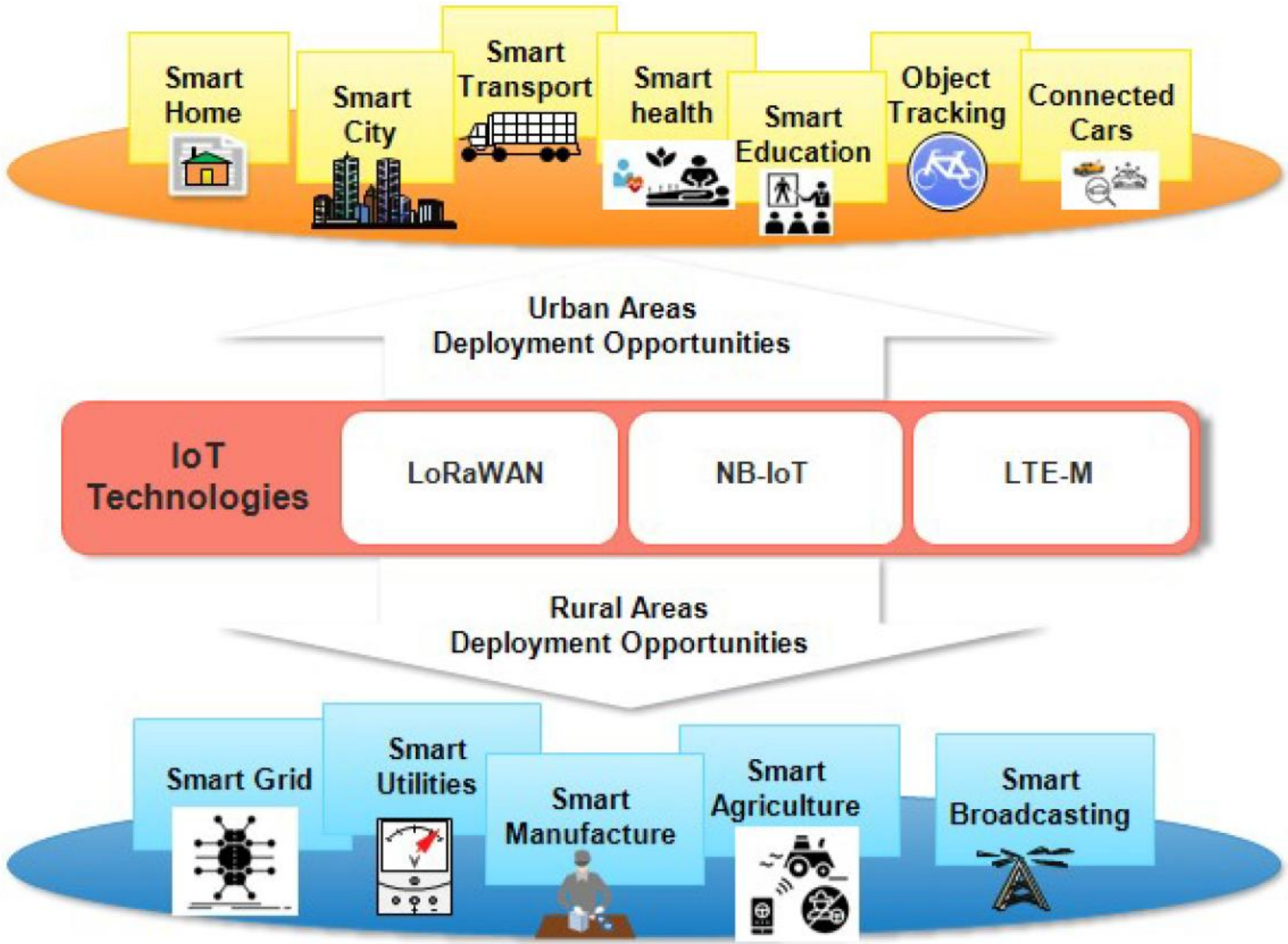
Wide Area Wireless Sensor Networks

WWSN

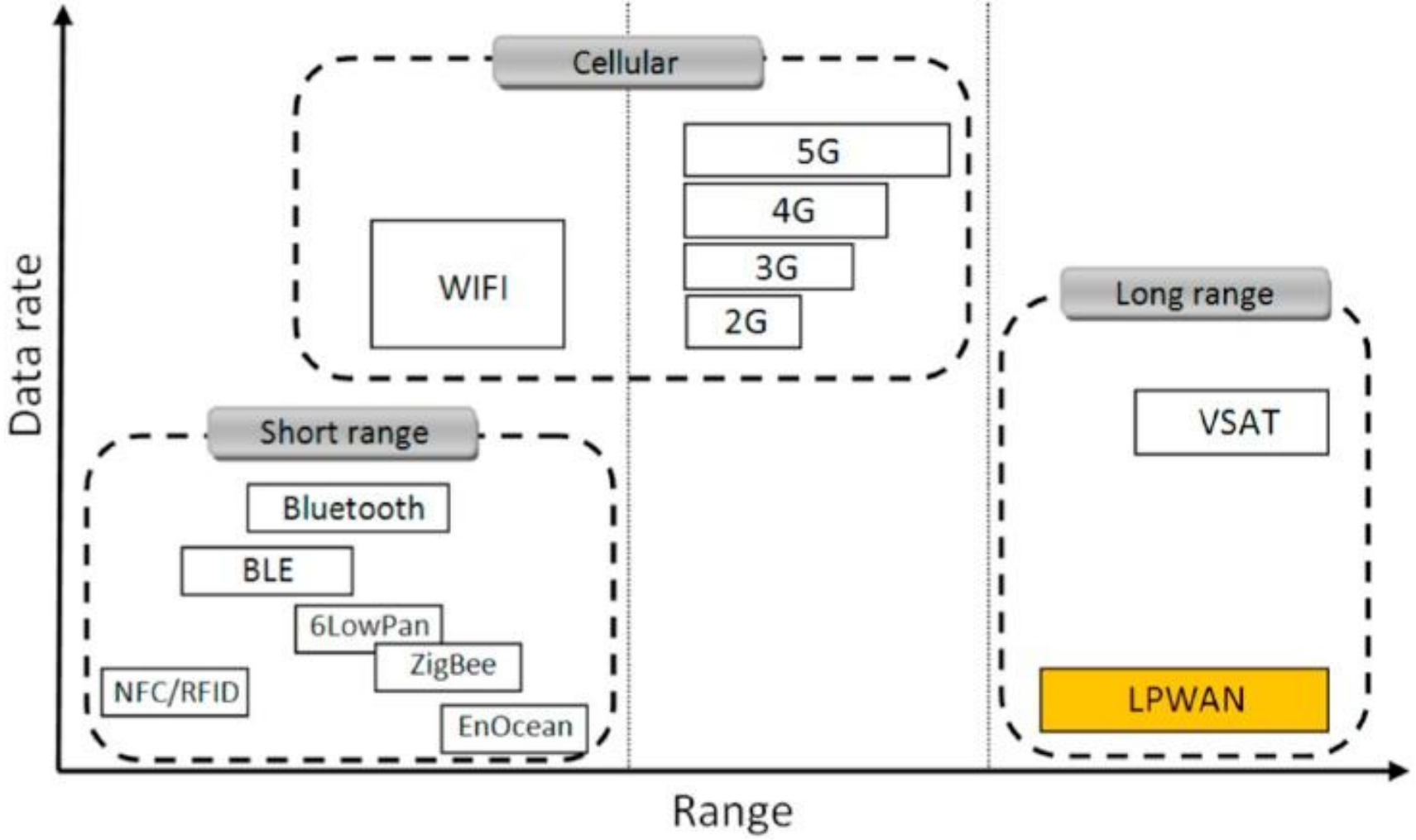
What is this?

- WWSN – wide area wireless sensor networks
- LPWSN – low power wireless sensor networks
- Technologies for sensor networks in wide areas
  - either for low power, or for geography
  - Typically: Sigfox, LoRa, cellular (LTE-M, NB-IoT)

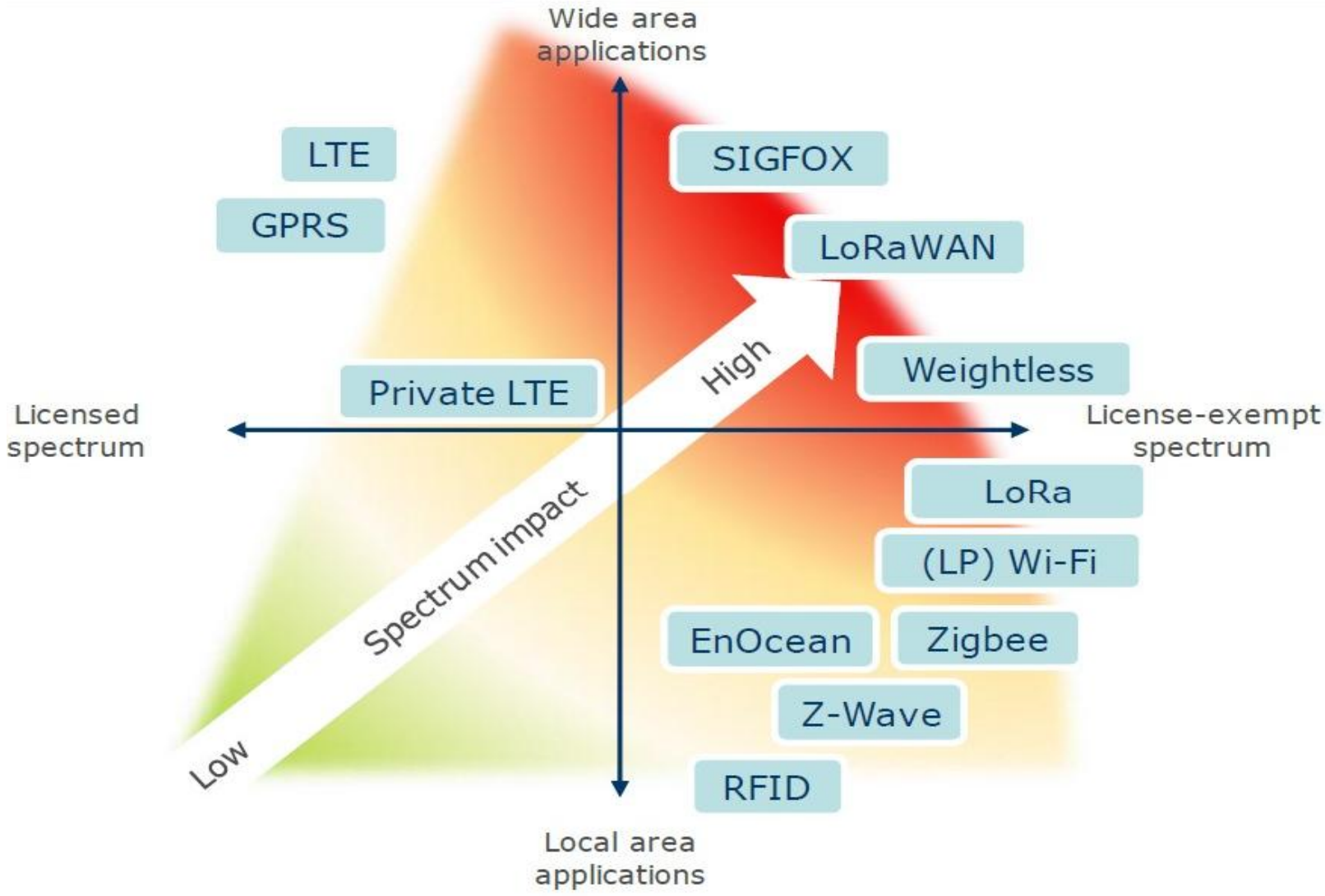
LPWSN


Smart Home
Smart City
Smart Transport
Smart health
Smart Education
Object Tracking
Connected Cars
Urban Areas
Deployment Opportunities
IoT Technologies
LoRaWAN
NB-IoT
LTE-M
Rural Areas
Deployment Opportunities
Smart Grid
Smart Utilities
Smart Manufacture
Smart Agriculture
Smart Broadcasting

Technology review



Licensed vs licensed-exempt



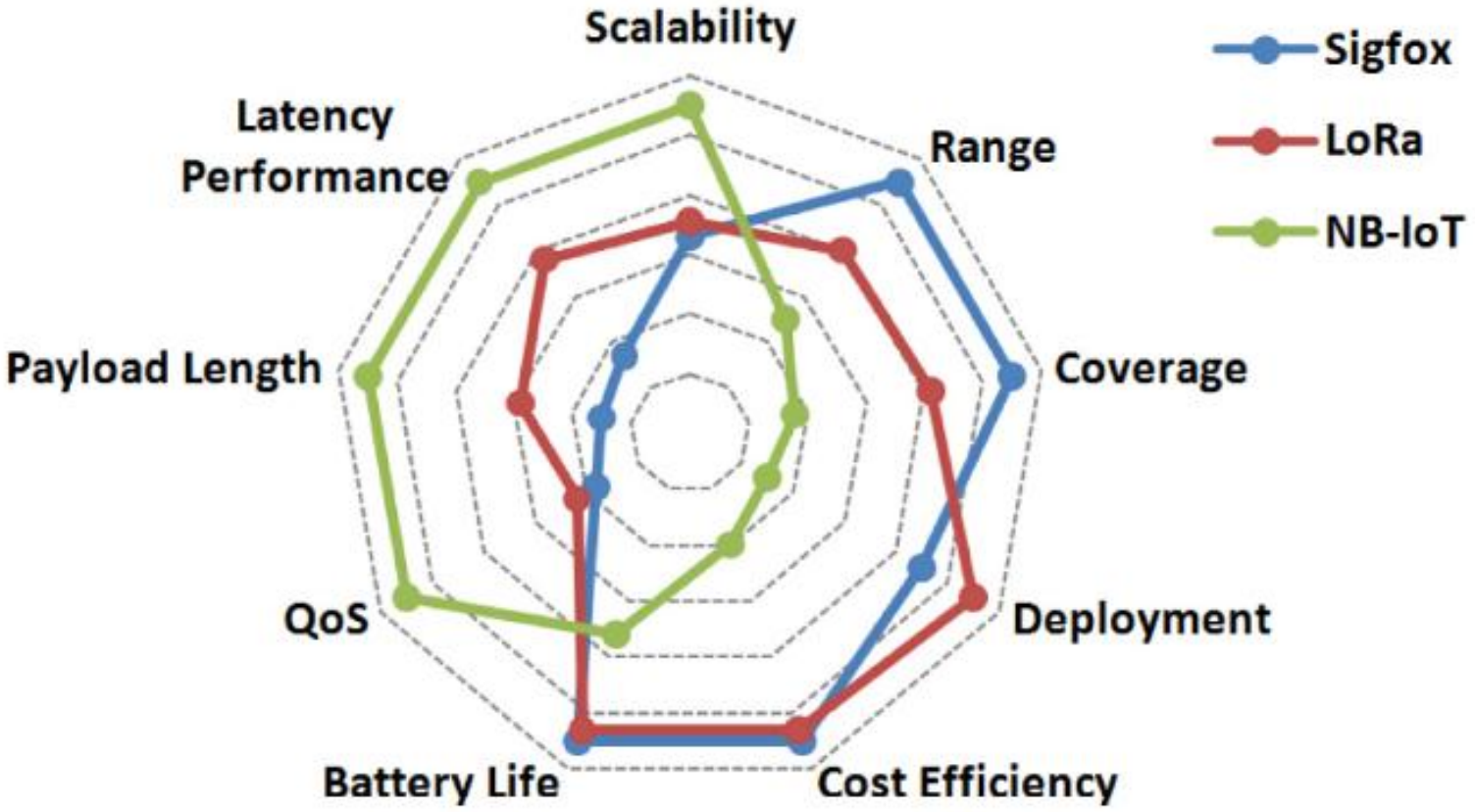
Overview of LPWAN

Overview of LPWAN technologies: Sigfox, LoRa, and NB-IoT.

	Sigfox	LoRaWAN	NB-IoT
Modulation	BPSK	CSS	QPSK
Frequency	Unlicensed ISM bands (868 MHz in Europe, 915 MHz in North America, and 433 MHz in Asia)	Unlicensed ISM bands (868 MHz in Europe, 915 MHz in North America, and 433 MHz in Asia)	Licensed LTE frequency bands
Bandwidth	100 Hz	250 kHz and 125 kHz	200 kHz
Maximum data rate	100 bps	50 kbps	200 kbps
Bidirectional	Limited / Half-duplex	Yes / Half-duplex	Yes / Half-duplex
Maximum messages/day	140 (UL), 4 (DL)	Unlimited	Unlimited
Maximum payload length	12 bytes (UL), 8 bytes (DL)	243 bytes	1600 bytes
Range	10 km (urban), 40 km (rural)	5 km (urban), 20 km (rural)	1 km (urban), 10 km (rural)
Interference immunity	Very high	Very high	Low
Authentication & encryption	Not supported	Yes (AES 128b)	Yes (LTE encryption)
Adaptive data rate	No	Yes	No
Handover	End-devices do not join a single base station	End-devices do not join a single base station	End-devices join a single base station
Localization	Yes (RSSI)	Yes (TDOA)	No (under specification)
Allow private network	No	Yes	No
Standardization	Sigfox company is collaborating with ETSI on the standardization of Sigfox-based network	LoRa-Alliance	3GPP

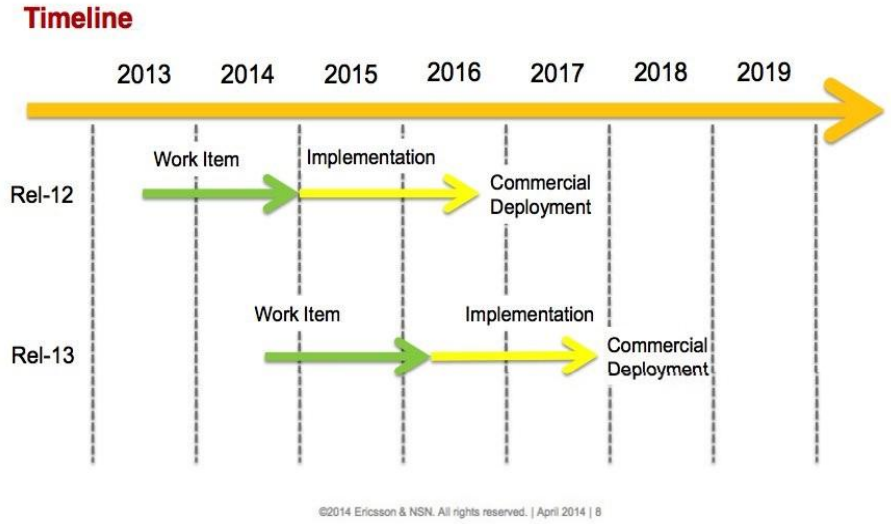
	Spectrum cost	Deployment cost	End-device cost
Sigfox	Free	>4000€/base station	<2€
LoRa	Free	>100€/gateway >1000€/base station	3–5€
NB-IoT	>500 M€ /MHz	>15 000€/base station	>20€

Comparison Radar



LTE-M - Overview

- Evolution of LTE optimized for IoT
- Low power consumption and autonomous
- Easy Deployment
- Interoperability with existing LTE networks
- Coverage up to 11 Km
- Max Throughput ≤ 1 Mbps



- ✓ First released in Rel.12 in 2 Q4 2014
- ✓ Optimization in Rel.13
- ✓ Specifications completed in Q1 2016
- ✓ Available since 2017

Evolution from LTE to LTE-M

3GPP Releases	8 (Cat.4)	8 (Cat. 1)	12 (Cat.0) LTE-M	13 (Cat. 1,4 MHz) LTE-M
Downlink peak rate (Mbps)	150	10	1	1
Uplink peak rate (Mbps)	50	5	1	1
Number of antennas (MIMO)	2	2	1	1
Duplex Mode	Full	Full	Half	Half
UE receive bandwidth (MHz)	20	20	20	1.4
UE Transmit power (dBm)	23	23	23	20

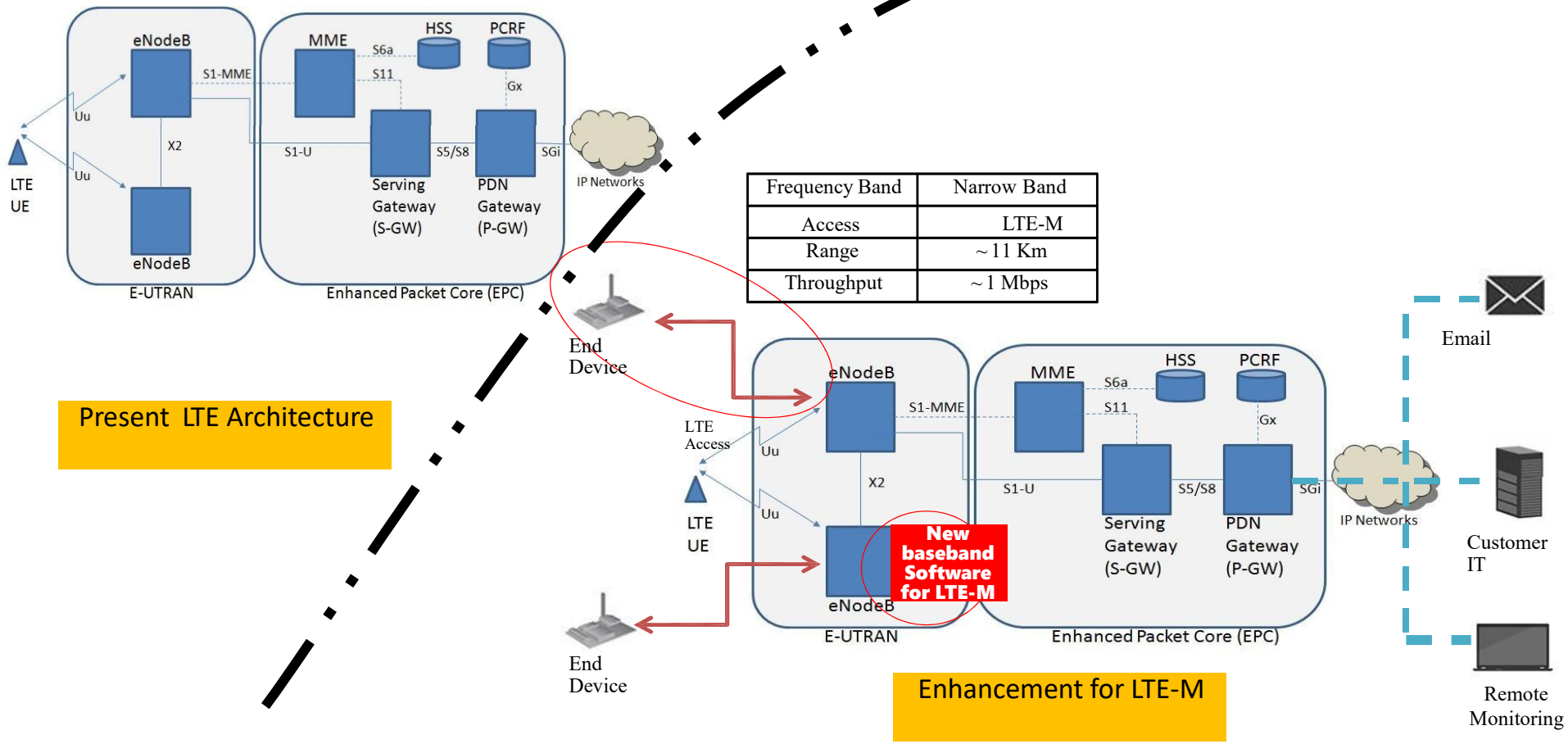
Release 12

- New category of UE ("Cat-0"): lower complexity and low cost devices
- Half duplex FDD operation allowed
- Single receiver
- Lower data rate requirement (Max: 1 Mbps)

Release 13

- Reduced receive bandwidth to 1.4 MHz
- Lower device power class of 20 dBm
- 15dB additional link budget: better coverage
- More energy efficient because of its extended discontinuous repetition cycle (eDRX)

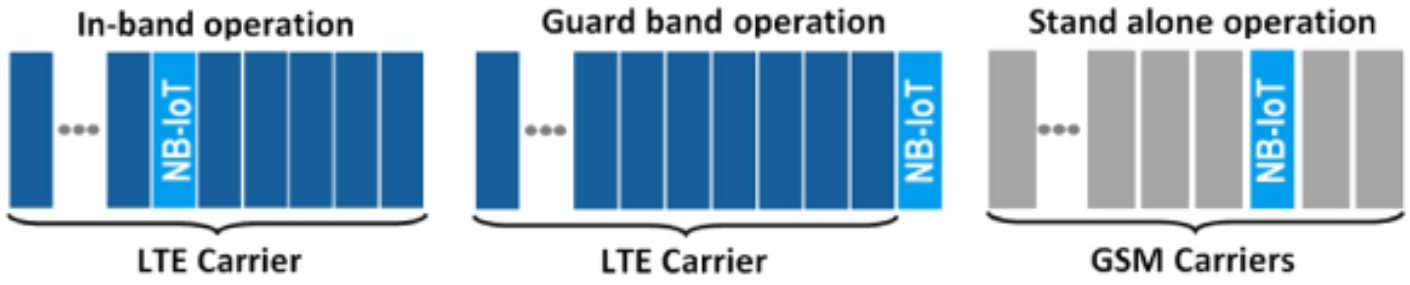
LTE to LTE-M - Architecture



Frequency Band	Narrow Band
Access	LTE-M
Range	~ 11 Km
Throughput	~ 1 Mbps

NB-IoT

- Defined in R13, another mode instead of LTE-M
- Bandwidth – 200KHz
  - One resource block in GSM/LTE
- Based in LTE protocol, stripped down
  - OFDMA(down)/FDMA(up), QPSK
  - 200kbps (down)/20kbps(up)
- Three modes of operation



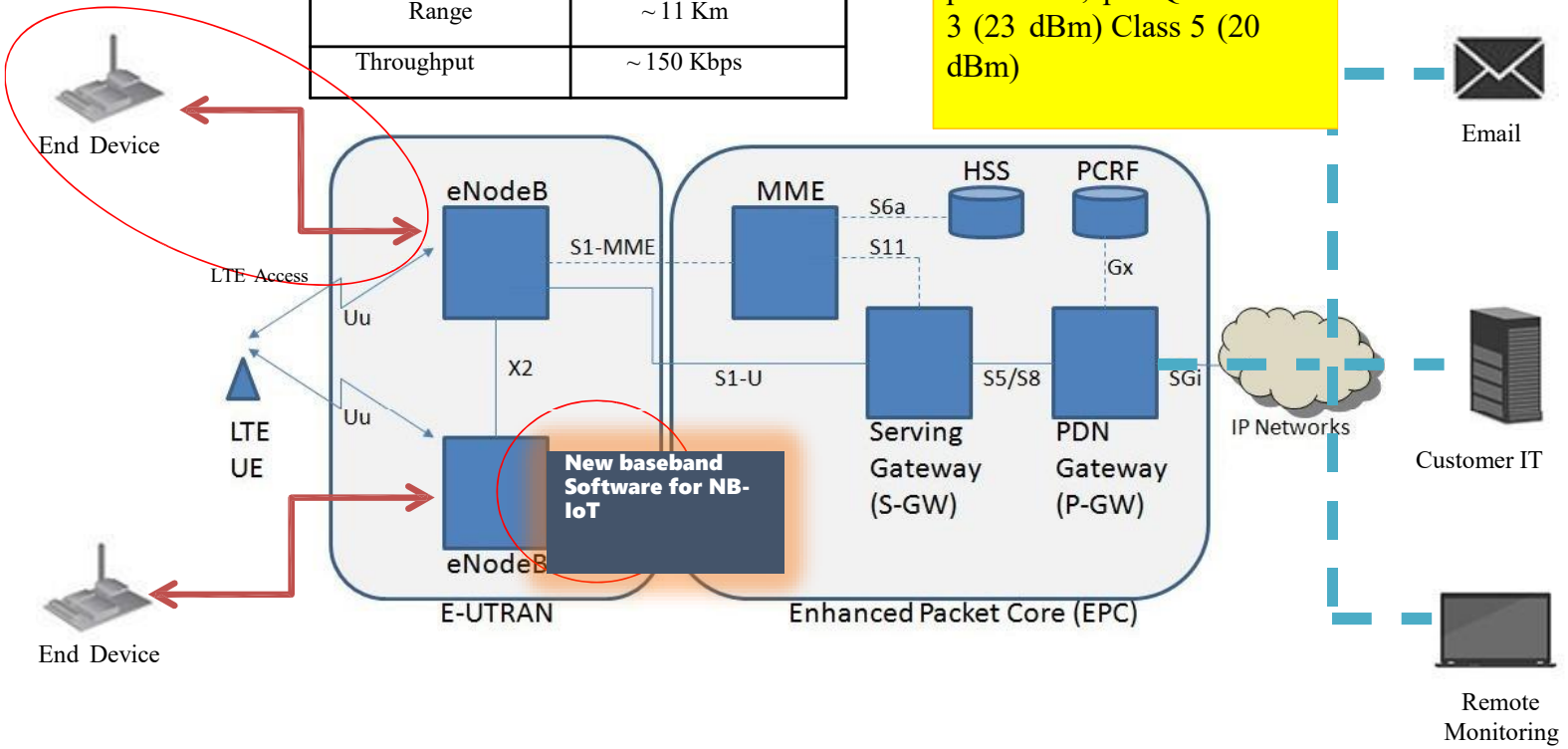
NB-IoT

- Uses LTE design extensively
- Lower cost in terms of channel utilization
- Extended coverage
- Low Receiver sensitivity = -141 dBm
- Long battery life: 10 years with 5 Watt Hour battery (depending on traffic and coverage needs)
- Support for massive number of devices: at least 50.000 per cell
- 3 modes of operation:
  - Stand-alone: *stand-alone carrier, e.g. spectrum currently used by GERAN (GSM Edge Radio Access Network) systems as a replacement of one or more GSM carriers*
  - Guard band: *unused resource blocks within a LTE carrier's guard-band*
  - In-band: *resource blocks within a normal LTE carrier*

NB-IoT - Architecture

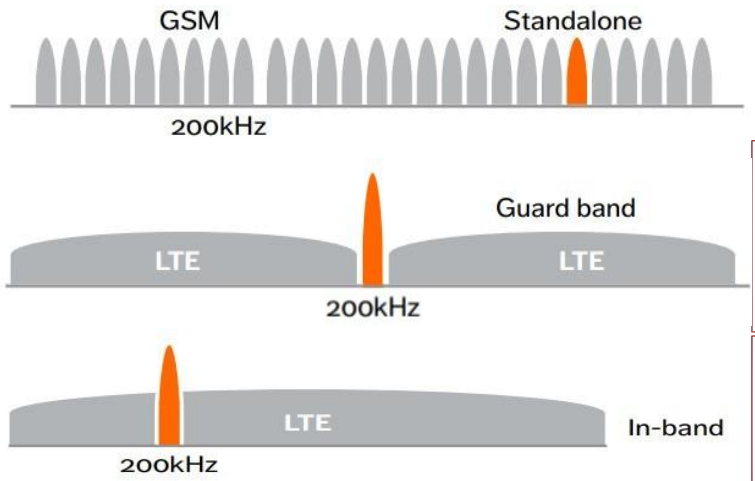
Frequency Band	Ultra Narrow Band
Range	~11 Km
Throughput	~150 Kbps

HD-FDD
p/2 BPSK, p/4 QPSK Class 3 (23 dBm) Class 5 (20 dBm)



NB-IoT – Spectrum & Access

Designed with a number of deployment options for licensed GSM , WCDMA or LTE spectrum to achieve efficiency



Stand-alone operation

Dedicated spectrum.

Ex.: By re-farming GSM channels

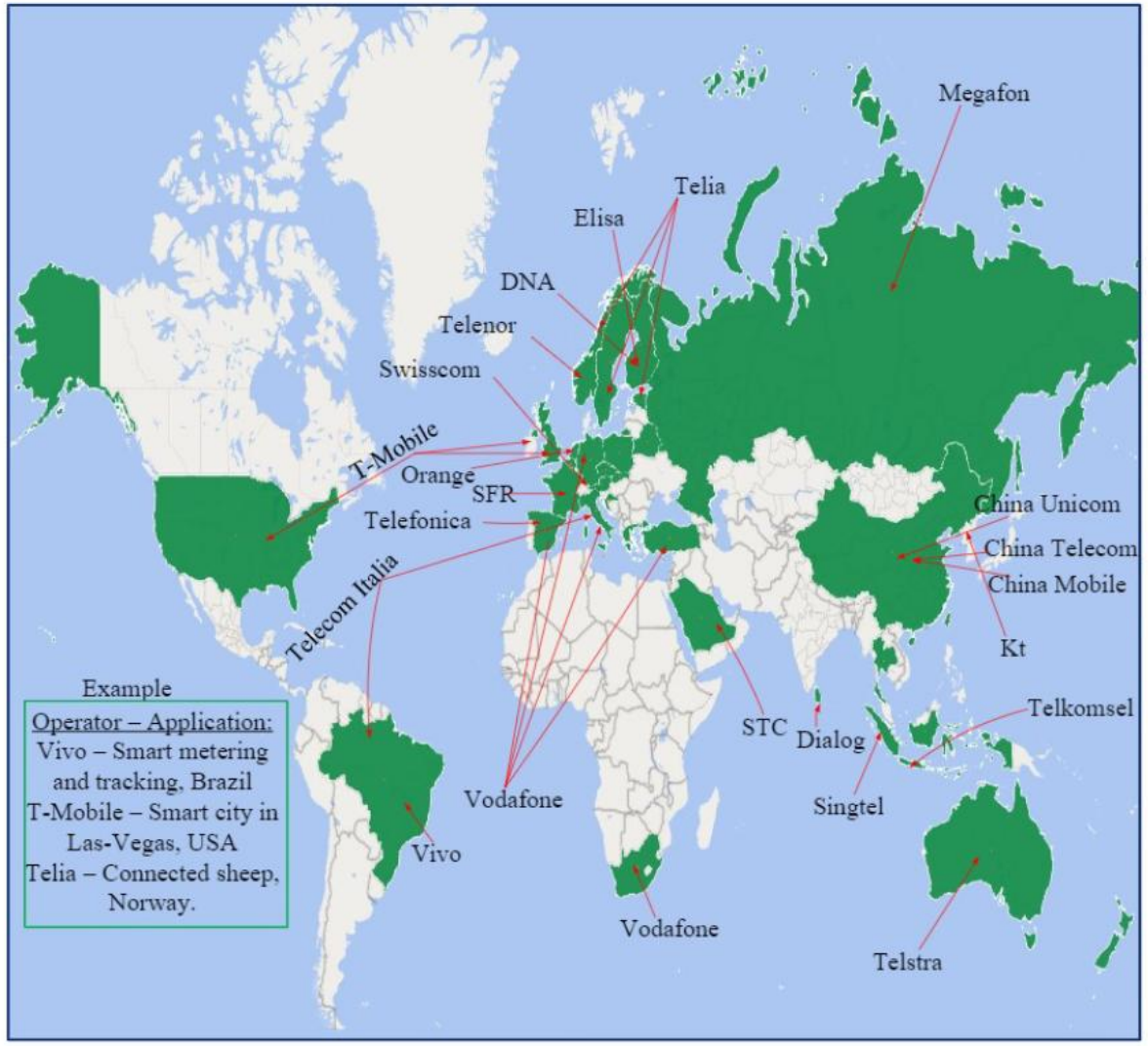
Guard band Operation:

Based on the unused RB within a LTE carrier's guard-band

In-band operation

Using resource blocks within a normal LTE carrier

NB-IoT (@2019)


Megafon
Telia
Elisa
DNA
Telenor
Swisscom
T-Mobile
Orange
SFR
Telefonica
Telecom Italia
China Unicom
China Telecom
China Mobile
Kt
Telkomsel
STC
Dialog
Singtel
Vodafone
Vivo
Vodafone
Telstra
Example
Operator – Application:
Vivo – Smart metering and tracking, Brazil
T-Mobile – Smart city in Las-Vegas, USA
Telia – Connected sheep, Norway.

Cellular technologies

- Two strategies, for different scenarios
  - No MIMO for lower end device energy.

	LTE-M	NB-IoT
Peak data rate	384 kbps	<100 kbps
Latency	50-100 ms	1.5-10 seconds
Power consumption	Best at medium data rates	Best at very low data rates
Mobility	Yes	No, stationary only
Voice	Yes	No
Antennas	1	1

SigFox



- Provide and maintain a PAID connectivity platform
  - Ultra Narrow Band: 100Hz per message
  - Ultra Low Bit rate: 12 byte messages, 140 messages per day (max!)
  - Long Range: ~50KM
  - Sensors lasting 10 years
  - Only provides connectivity, access control and a broker
- Business Model: connectivity service for alarms, smart meters, etc..

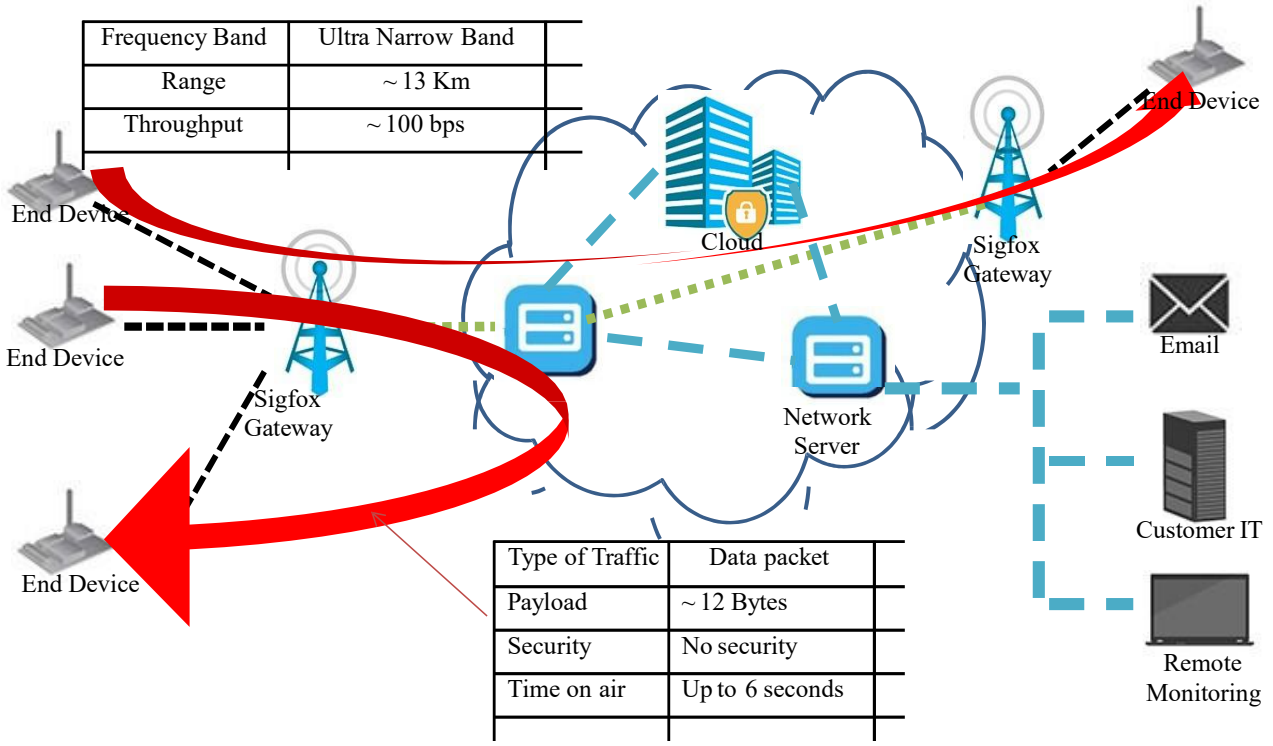
SigFox

- Low Power Wide Area Sensor Network (LPWASN)
- Thousands of millions ☺
  - A million per access point ;)
- Proprietary ☹ commercial
  - You have to use its access infrastructure (built with operators) and software
  - Open market for the endpoints
- 30-50km range in rural areas, and 3-10km range in urban
- Ultra narrow band, 868 (EU) or 902 (US) frequency (MHz)
- Low energy consumption
- Dedicated network

SigFox

- Each device can send up to 140 messages per day
  - Payload: 12 octets (~96 bytes)
  - Datarate: up to 100bps
- (**Duty cycle:** the time occupied by the operation of a device, which operates intermittently)
  - Common in the IoT
- Sigfox exploits this:
  - When a device has a message to be sent, the Sigfox interface wakes up, and the message is transmitted uplink
  - Then, the device listens for a short duration, if there is data to be sent to it
  - This is good for data acquisition scenarios
  - But not so good for command-and-control situations
- Use cases:
  - Smart meters, smoke detectors

Sigfox - Architecture



LoRa



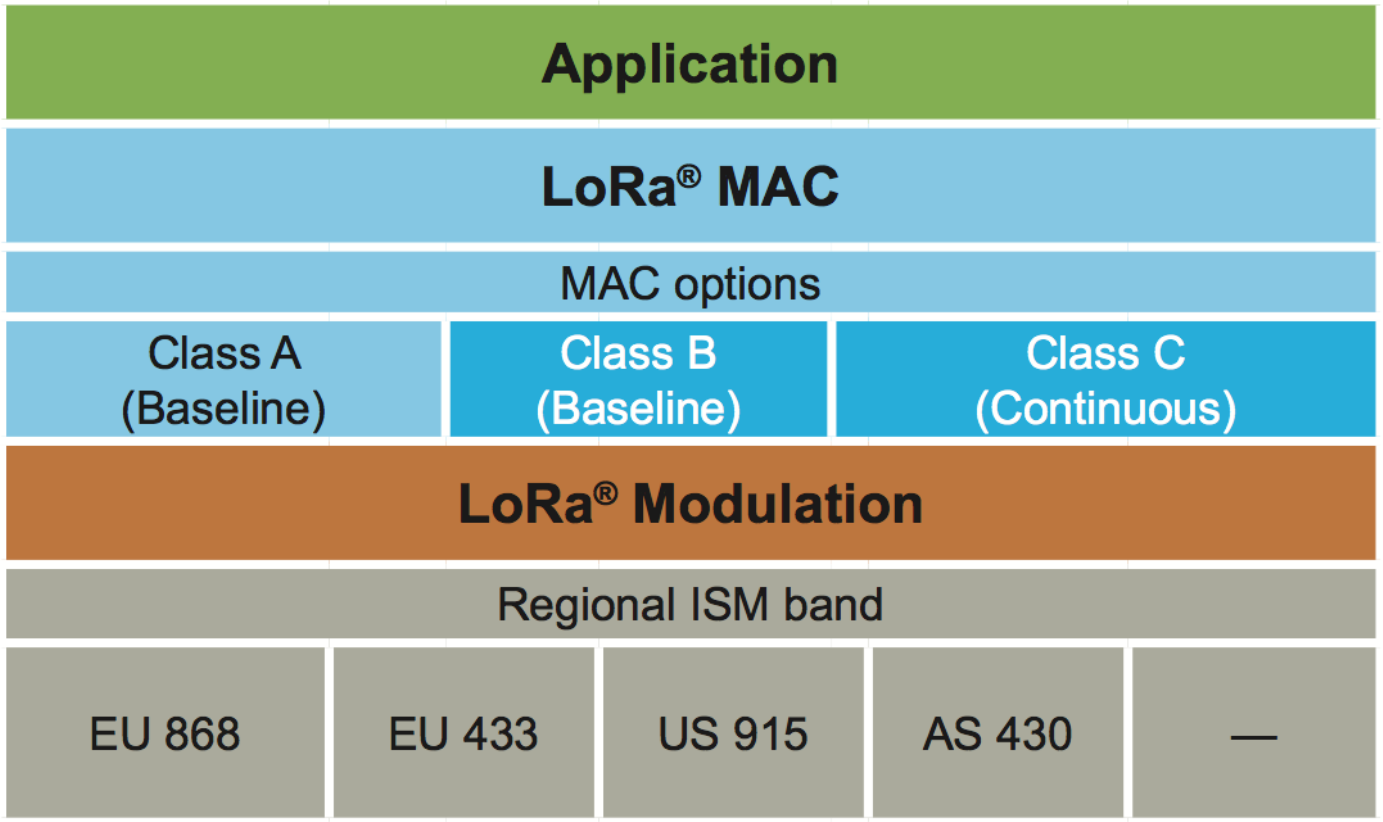
SEMTECH®

- Stands for “Long Range”
- To be used in long-lived battery-powered devices scenarios
- Semi-proprietary
  - Parts of the protocol are well documented, others not
  - In the beginning, they owned the radio part (Semtech)
  - But eventually they sublicensed it
  - You can install your own gateways



- LoRa usually means two different things:
  - LoRa: a physical layer that uses Chirp Spread Spectrum (CSS) modulation
  - LoRaWAN: a MAC layer protocol, specified by the LoRa Alliance

LoRa Stack


Application
LoRa® MAC
MAC options
Class A (Baseline)
Class B (Baseline)
Class C (Continuous)
LoRa® Modulation
Regional ISM band
EU 868
EU 433
US 915
AS 430
—

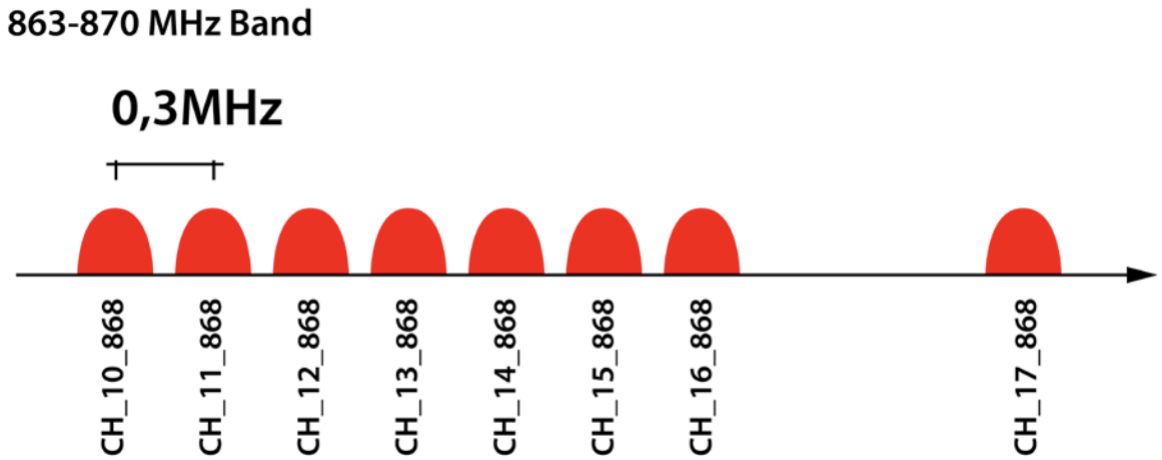
LoRa (the physical layer ☺)



- Developed by Semtech
- Low-range, low-power and low-throughput
- Operates on 433-, 868- (EU) or 915 (US) MHz bands
- Payload from 2 to 255 octets (2Kb)
  - Depends on configuration parameters
- Datarate: up to 50Kbps

LoRa (the physical layer ☺)

- In Europe, 8 channels with a bandwidth of 0.3MHz are used



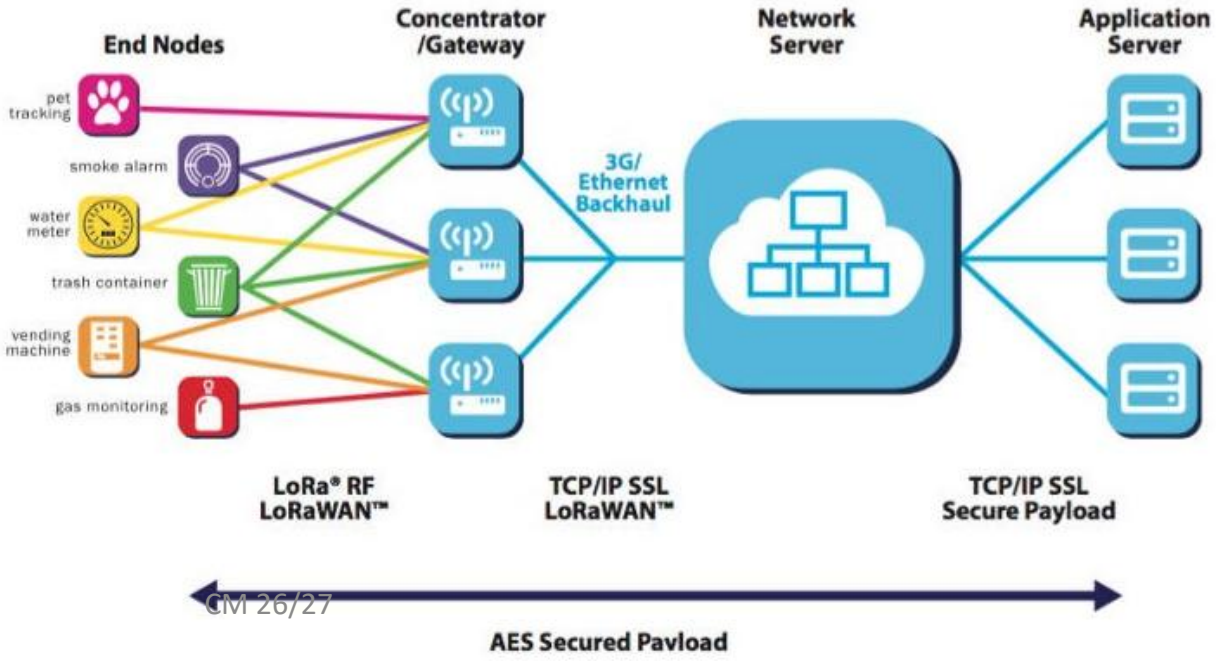
Source: Libelium

LoRaWAN

- MAC mechanism for controlling communications between end devices and LoRaWAN gateways. For all devices, it manages:
  - Communication frequencies
  - Data rate
  - Power
- Open Standard developed by the LoRa Alliance

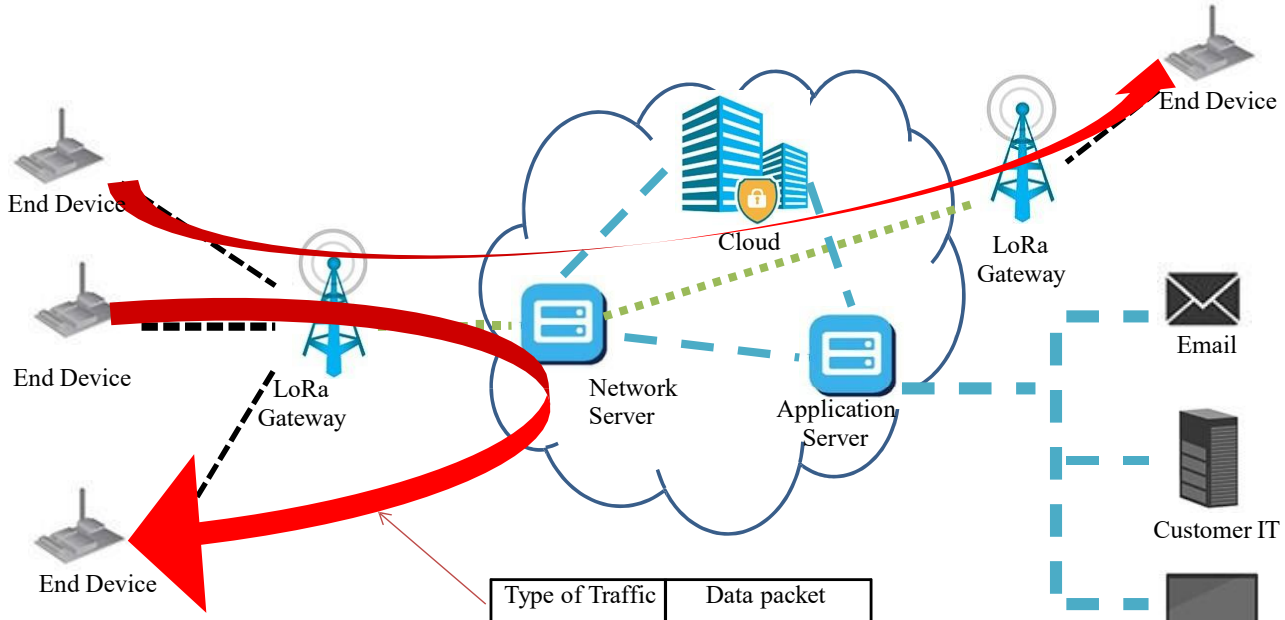
LoRA Network

- Star of stars topology
- Devices transmit data asynchronously
  - Data is received by multiple gateways
  - Each gateway forwards received data to a centralized network server, using a backhaul link (Ethernet or cellular)
  - The network server:
    - Filters duplicate packets
      - Packet with the strongest signal gets decoded
    - Realizes security checks
    - Manages the network



Source: semtech.com

LORA - Architecture



Type of Traffic	Data packet
Payload	~ 243 Bytes
Security	AES Encryption

Modulation	LoRa RF (Spread Spectrum)
Range	~ 15 Km
Throughput	~ 50 Kbps

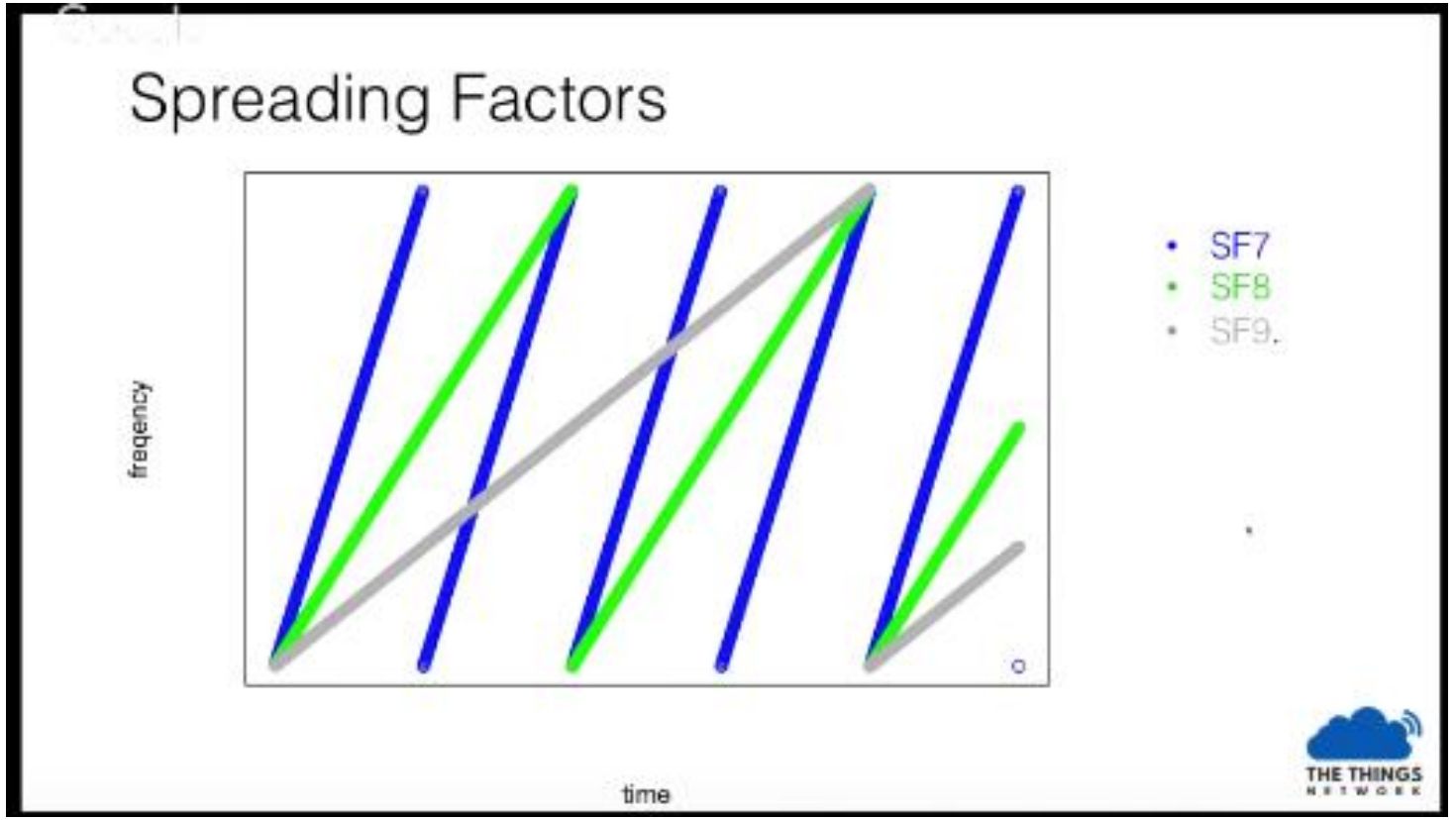
LoRa Physical Layer

- Modulation
  - (changing a signal, the carrier, in a way that allows it to contain information to be transmitted)
- LoRa uses a proprietary Spread-Spectrum modulation technique: Chirp Spread Spectrum (CSS)
    - (A chirp is a signal in which frequency raises or lowers with time)
  - Tries to increase range by:
    - Sending information with more power (within regulated values - <14dBm or 25mW)
    - Or by lowering the data rate
  - Increases link budget
    - Total sum of all gains and losses that a radio signal experiences from the transmitter, through the air, to the receiver
  - Increases immunity to in-band interference
- This, along with Forward Error Correction techniques, contribute to extend the range and robustness of radio communication links

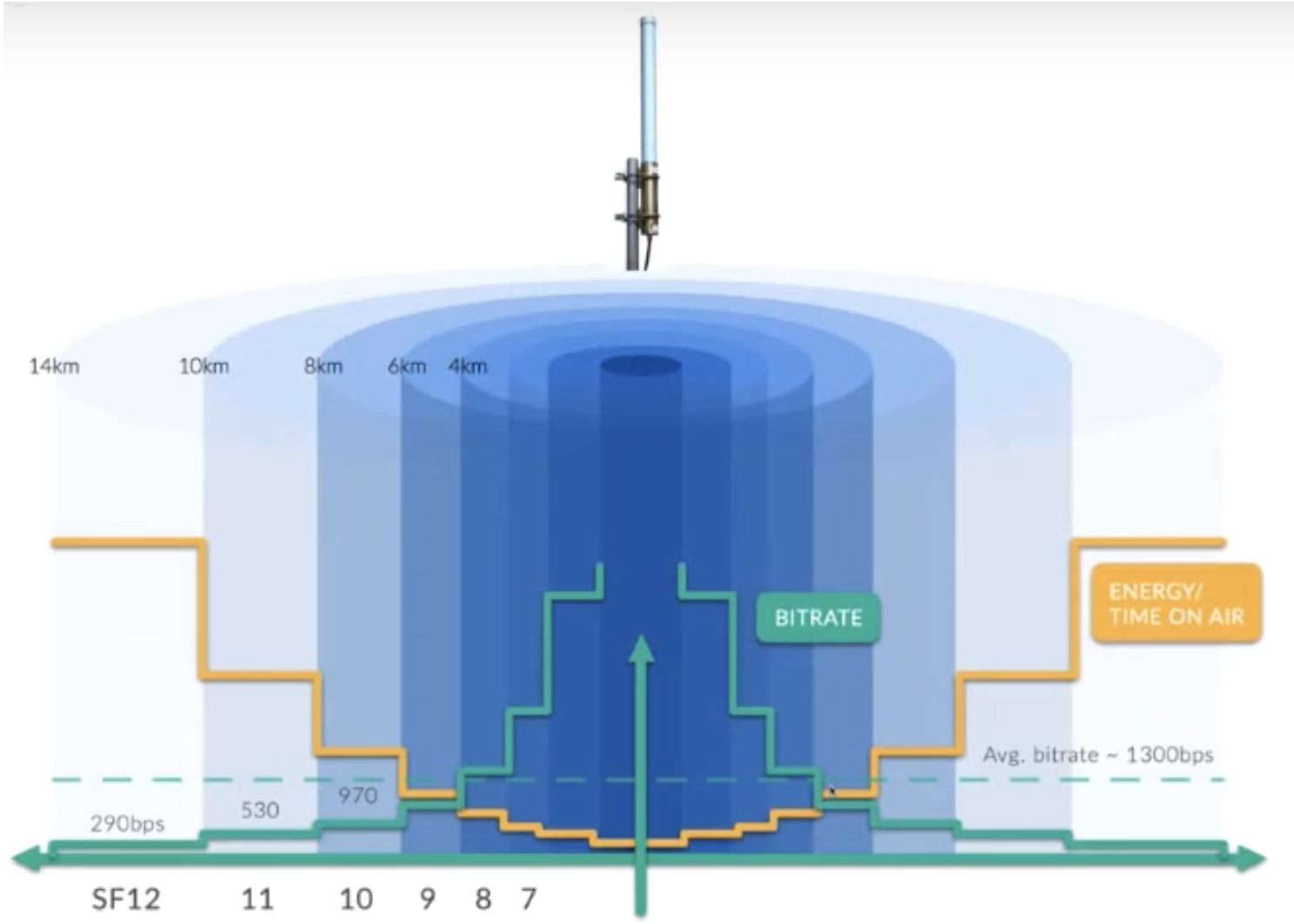
LoRa Physical Layer

- Has different Spread Factors (SF7 to SF12)
  - Spread factors can set the modulation rate and tune the distance
  - They indicate how fast or slow is the chirp (how many chirps you get per second) → **how much data you can encode per second**
    - The higher the SF, the lower the datarate
    - Each SF is 2x slower than the one before
    - The slower you send your data, the farther you can send it
    - The higher the SF, the more energy is required (time on air)
      - The interface has more time to decode and sensititviy is increased
- This helps on scaling the network
  - Closer nodes receive data much faster
  - Air is ”cleared” for other nodes to transmit
  - By adding more gateways, devices get nearer to them, applying the above

LoRa Physical Layer



Source: Thomas Telkamp


14km
10km
8km
6km
4km
BITRATE
ENERGY/
TIME ON AIR
Avg. bitrate ~ 1300bps
290bps
530
970
SF12
11
10
9
8
7

LoRa Physical Layer

- For a 125kHz bw (configurable by design)

Spreading Factor	Symbols/second	SNR limit	Time-on-air (10 byte packet) - ms	Bitrate - bps
7	976	-7.5	56	5469
8	488	-10	103	3125
9	244	-12.5	205	1758
10	122	-15	371	977
11	61	-17.5	741	537
12	30	-20	1483	293

LoRa Physical Layer

- The Bandwidth (kHz), Spreading Factor and Coding Rate are design variables that allow a system to optimize the trade-off between
  - Occupied bandwidth
  - Data rate
  - Link budget
  - Interference immunity
- By using software, it is possible to combine these values to define a transmission mode

LoRa Physical Layer

- Bandwidth
  - Show how wide is going to be the transmission signal
  - 3 options: 125 kHz, 250 kHz or 500 kHz
  - Greater reach: 125 kHz
  - Greater transmission speed: 500 kHz
  - Less bandwidth = more airtime = more sensitivity = more battery consumed

LoRa Physical Layer

- Coding Rate
  - 4 options: 4/5, 4/6, 4/7 and 4/8
  - Meaning:
    - Every 4 useful bytes are going to be encoded by 5, 6, 7 or 8 transmission bits
  - Smaller coding rate: 4/8
  - Lower coding rate = more airtime

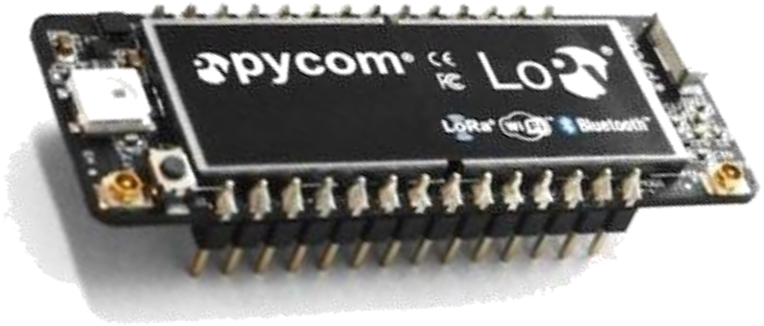
LoRa Physical Layer

- Spreading Factor
  - Number of chips per symbol used in data treatment before the transmission signal
  - 7 options: 6, 7, 8, 9, 10, 11 and 12
  - Greater Spreading Factor = Greater Range = more air time

LoRa Physical Layer

Mode	BW	CR	SF	Sensitivity (dB)	Transmission time (ms) for a 100-byte packet sent	Transmission time (ms) for a 100-byte packet sent and ACK received	Comments
1	125	4/5	12	-134	4245	5781	max range, slow data rate
2	250	4/5	12	-131	2193	3287	-
3	125	4/5	10	-129	1208	2120	-
4	500	4/5	12	-128	1167	2040	-
5	250	4/5	10	-126	674	1457	-
6	500	4/5	11	-125,5	715	1499	-
7	250	4/5	9	-123	428	1145	-
8	500	4/5	9	-120	284	970	-
9	500	4/5	8	-117	220	890	-
10	500	4/5	7	-114	186	848	min range, fast data rate, minimum battery impact

LoRa Physical Layer - Practical



- On the LoPy
  - Method
    - lora.init(mode, * , frequency=868000000, tx_power=14, **bandwidth=LoRa.BW_125KHZ**, **sf=7**, preamble=8, **coding_rate=LoRa.CODING_4_5**, power_mode=LoRa.ALWAYS_ON, tx_iq=False, rx_iq=False, adr=False, public=True, tx_retries=1, device_class=LoRa.CLASS_A)
- Bandwidth: **LoRa.BW_125KHZ / LoRa.BW_250KHZ / LoRa.BW_500KHZ**
- SF: **sf=6 / sf=7 / sf=8 / sf=9 / sf=10 / sf=11 / sf=12**
- Coding Rate: **LoRa.CODING_4_5 / LoRa.CODING_4_6 / LoRa.CODING_4_7 / LoRa.CODING_4_8**

LoRaWAN

- Components
  - End-Device
    - Devices (low-power) that communicate with the LoRa Gateway
    - They are not associated to a particular gateway.
    - They are, however, associated to a Network Server.
  - Gateway
    - Intermediate devices that relay packets between end-devices and a network server.
    - Linked to the Network server via a higher bandwidth backhaul network.
    - They add information about the quality of reception, when forwarding a packet from an end-device to a network server.
    - They are transparent to the end-devices.
    - There are multiple gateways in a network
    - Multiple gateways can receive the same packet transmitted from the same end-device
  - Network Server
    - Decodes and de-duplicates packets sent from devices.
    - Generates packets to be sent towards devices
    - Choses the appropriate gateway to send packets to a specific end-device

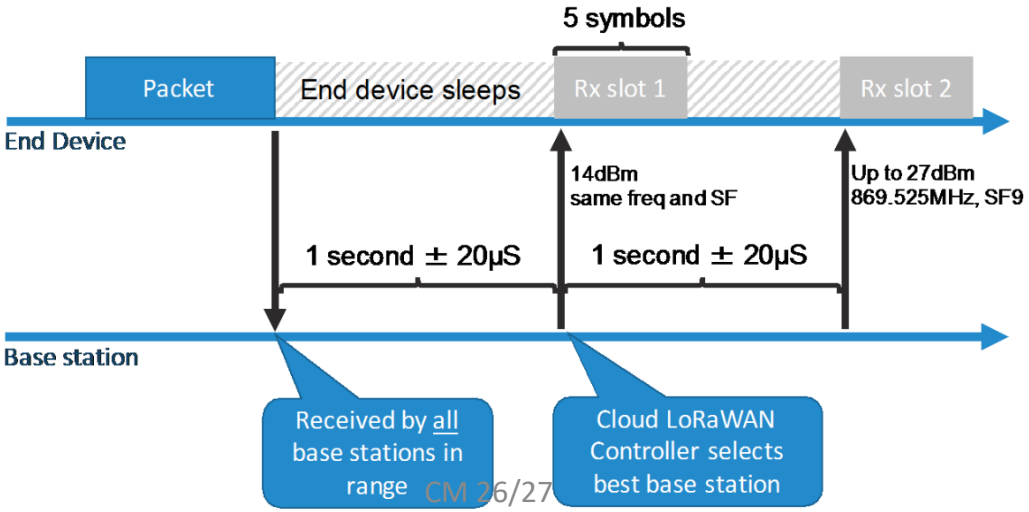
LORA – Device Classes

Classes	Description	Intended Use	Consumption	Examples of Services
A (« all »)	Listens only after end device transmission	Modules with no latency constraint	The most economic communication Class energetically.. Supported by all modules. Adapted to battery powered modules	• Fire Detection • Earthquake Early Detection
B (« beacon »)	The module listens at a regularly adjustable frequency	Modules with latency constraints for the reception of messages of a few seconds	Consumption optimized. Adapted to battery powered modules	• Smart metering • Temperature rise
C (« continuous »)	Module always listening	Modules with a strong reception latency constraint (less than one second)	Adapted to modules on the grid or with no power constraints	• Fleet management • Real Time Traffic Management

Any LoRa object can transmit and receive data

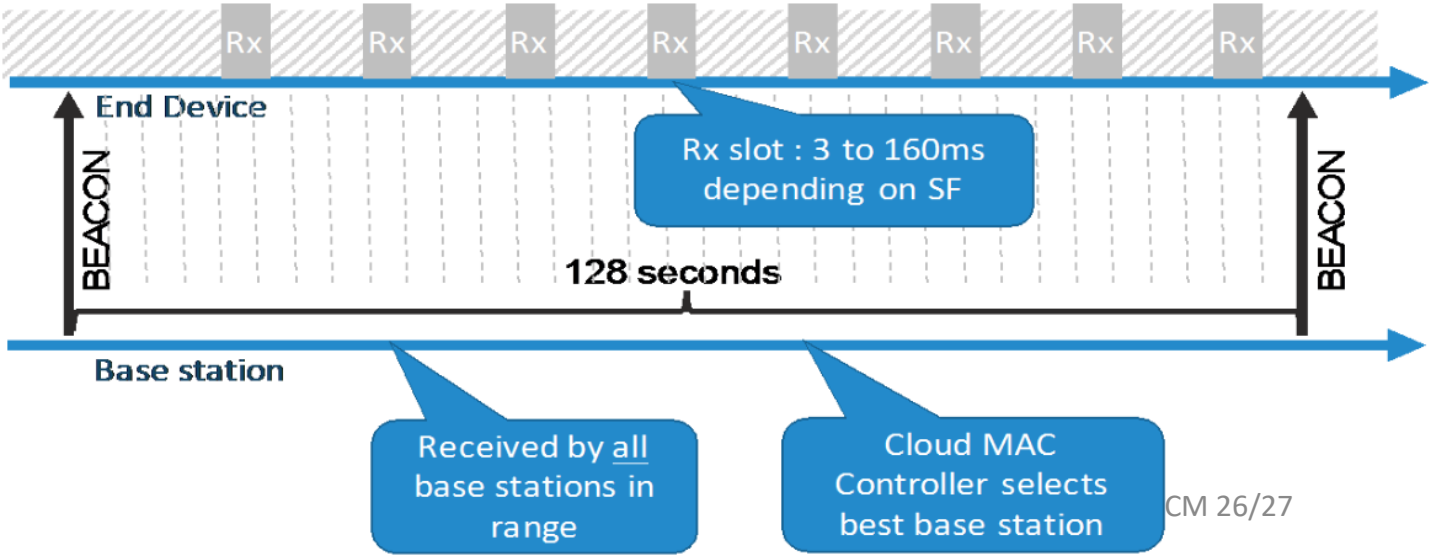
LoRaWAN

- End-devices classes
  - Class A – bi-directional
    - Lowest power consumption
    - Devices schedule uplink transmissions according to their requirements, with a small variation before transmission.
    - Each uplink transmission is followed by two short downlink receive windows
      - Downlink transmissions at any other time have to wait until the next uplink transmission
      - If no downlink frame is received during these windows, the radio goes back to sleep until the next scheduled or event-driven uplink.
      - Less flexibility for downlink
    - Low power drain
    - Common use cases: environmental monitoring, smart agriculture, asset tracking, smart parking sensors, smoke detectors



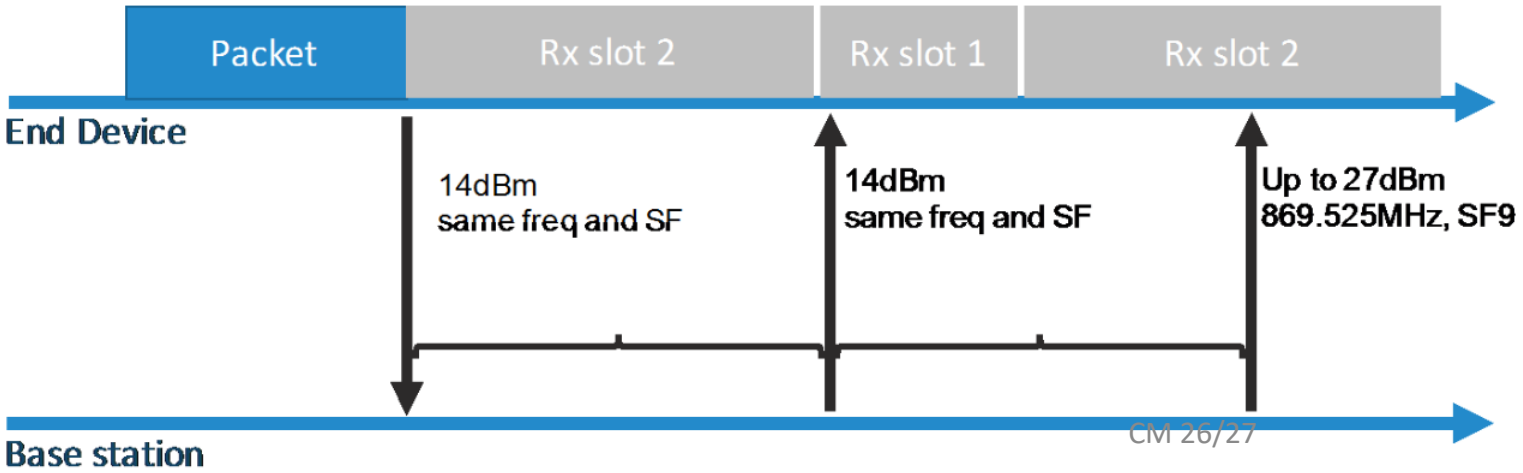
LoRaWAN

- End-devices classes
  - Class B – bi-directional with scheduled receive slots
    - Adds synchronized periodic receive ping slots
    - There is a synchronized beacon from the gateway to the network server, indicating when the device is listening
    - Medium power drain
  - Common use cases: Utility meters (water/gas), smart street lighting schedules, industrial actuators with non-critical timing



LoRaWAN

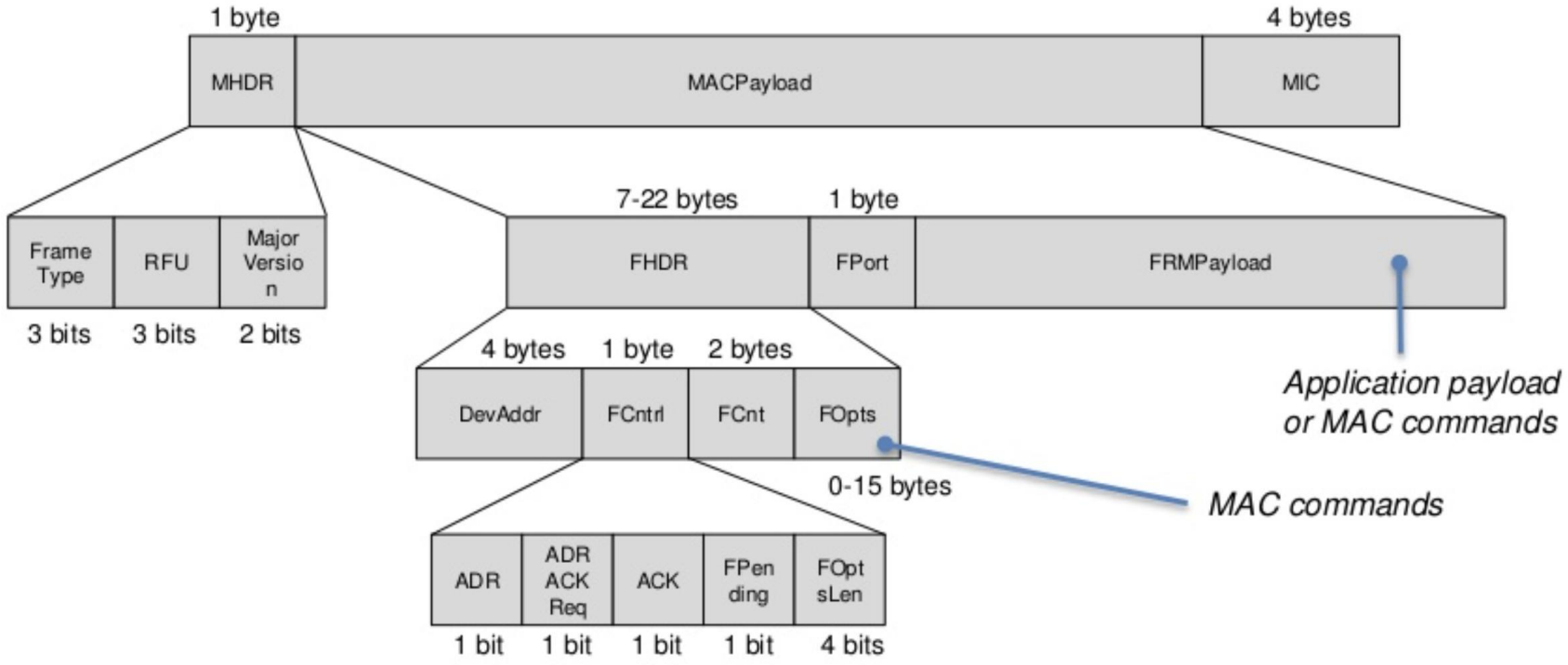
- End-devices classes
  - Class C – bi-directional with maximal receive slots
    - Greatest power consumption
    - Almost continuous receiving windows
      - Server can initiate transmission almost anytime
      - (when not transmitting)
    - Received remains tuned to RX2 frequency continuously and switches briefly to RX1 after an uplink.
  - Highest power drain
  - Downlink latency: lowest
  - Common use cases: smart grid power distributions, real-time industrial automation/actuation, smart traffic lights, connected streetlamp control systems, safety/alarm systems



LoRaWAN

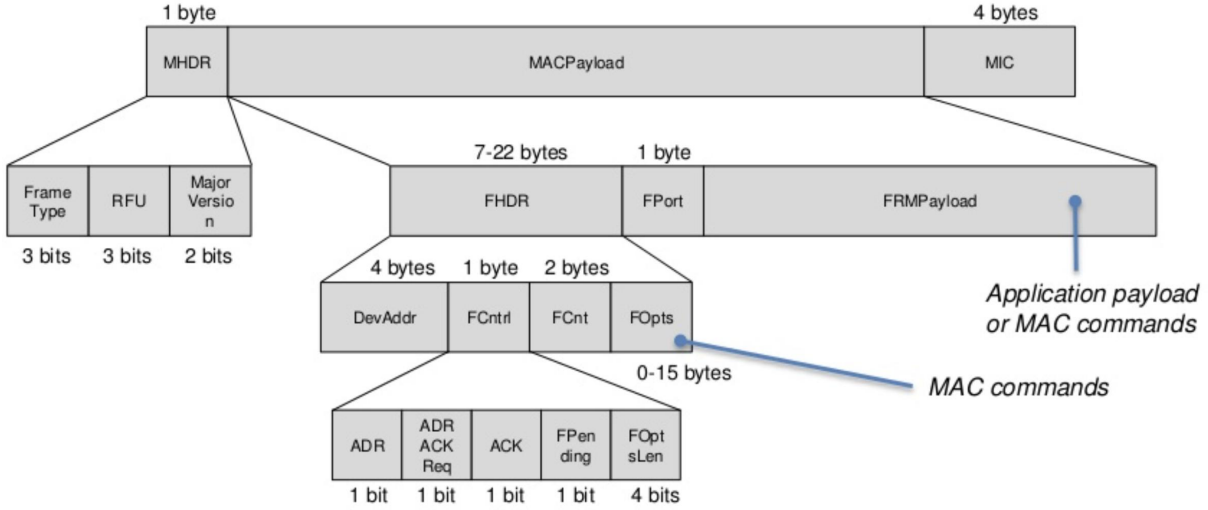
- End-Device Duty Cycle
  - Besides transmission frequency, duty cycle regulations apply
  - Delay between successive frames sent by a device
  - 1% limitation for end-devices
    - Device has to wait 100x the time it took for it to send the message, in order to be able to send again in the same channel
  - Gateways: 10%

LoRaWAN - Payload



Source: Stephen Pharrell

LoRaWAN



- *DevAddr* - short address of the device.
- *FPort* - multiplexing port field.
- *FCnt* - frame counter.
- *MIC* - cryptographic message integrity code
- *MType* - message type (uplink, downlink, confirmed (requires an ACK, …).
- *Major* - LoRaWAN version
- *ADR* and *ADRAckReq* - data rate control adaptation mechanism by the network server.
- *ACK* - acknowledges the last received frame.
- *Fpending* - indicates that there is still data to be sent by the network server (end-device is required to send another message to open a receive window).
- *FOptsLen* - length of the *FOpts* field in bytes.
- *FOpts* - contains MAC commands on a data message.
- *CID* - MAC command ID.
- *Args* -optional arguments of the command.
- *FRMPayload* - payload, encrypted using AES with a key length of 128 bits.

The minimal size of the MAC header is 13 bytes; its maximal size is 28 bytes.

There is no destination address on uplink packets, or source address on downlink packets.

LoRaWAN

- MAC Commands
  - Allows the network to customize end-device parameters
- Checks
  - Link status (this can be send by the end-device itself)
  - Device battery
  - Device margin (SNR)
- Settings
  - Datarate
  - TX power
  - TX and RX channels
  - RX timing
  - Repetition
  - Duty cycle
  - Dwell time

LoRaWAN

- End-Device Connection to a network
  - Also known as ***Activation***
- This process provides the end-device with:
  - End-device address (*DevAddr*): An identifier composed by the network identifier (7bit) and by the end-device's network address (25bit)
  - App identifier *(AppEUI):* Unique identification of the end-device owner
  - Network Session Key (*NwkSKey*): A key used by both the network server and end-device to verify and ensure message integrity
  - App Session Key (*AppSKey*): A key used by both the network server and end-device to encrypt the payload of received messages
- Note on security:
  - LoRaWAN protocol security is based on 802.15.4
    - AES-128

LoRaWAN

- To activate the device, there are two procedures:
  - Over-the-Air Activation (OTAA)
    - *Join-Request* and *Join-Response* messages are exchanged in each new session, allowing the end-devices to obtain the network and application session keys
  - Activation By Personalization (ABP)
    - The devices have both keys already stored internally

LoRaWAN

- Adaptive Data Rate
  - The network tells the node at which data rate it can send data
    - Manages the SF for each end-device
  - The aim is to:
    - Optimize for fastest data rate versus range
    - Maximize battery life
    - Maximize network capacity

LoRaWAN

- Typically, there is no node-to-node direct communication
  - LoRaWAN allows this by having 2 gateways and a network server in between the nodes
- However, most end-device vendors also include (for testing, mostly) a raw form of LoRa
  - Allows peer-to-peer communication between nodes
  - Contains only the link layer protocol
  - Only allows a very small number of nodes in a topology
    - There is no packet management

  (useful for a first try with LoRa)

LoRaWAN vs NB-IoT

Table 5. NB-IoT vs. LoRaWAN average power consumption, latency, and throughput.

Features	NB-IoT	LoRaWAN
Joining network	3 mAh	1 mAh
Uplink message (44 bytes)	1.8 mAh	100 µAh
UE class	Cat NB1	A
Data rate (20 bytes)	0.6–4 bps	
Frequency	28 Mhz	EU868 MHz

The Things Network

- Built in a crowdsourced manner by companies and enthusiasts
  - Low density coverage, mostly in larger cities (5 GW in Aveiro)
  - 1-2Km range in cities, 10km in open space
- Provides a Free connectivity service, and broker with APIs
- Composed by:
  - Nodes: owned by companies and citizens, send data to Gateways
  - Gateways: owned by companies and citizens, interface with TTN
  - TTN Servers: Hosted by TTN, routing data to/from user apps